Progress Report
Link Light Rail Program



Northgate Link—Installing walkway rebar in the Southbound tunnel

March | 2017

SOUNDTRANSIT

LINK LIGHT RAIL

CURRENT SERVICE AND APPROVED EXTENSIONS



SOUNDTRANSIT

October 2016

Projects

University Link Extension (U-Link): This project is a 3.15-mile light rail segment which is located entirely underground with tunnels traveling from Pine Street, under the I-5 freeway to an underground station at Capitol Hill, continuing north beneath SR 520 and the Montlake Cut to an underground station on the University of Washington campus, near Husky Stadium. The Sound Transit Board adopted the baseline capital budget of $1.76B in July 2008. Revenue Service began on March 19, 2016.

Northgate Link Extension: The University of Washington Station (UWS) to Northgate project is a 4.3-mile extension of light rail from the UWS north under campus via twin-bored tunnels to an underground station at NE 45th Street continuing to an underground station at NE 65th Street and via tunnel, retained cut and elevated sections to Northgate. The project is in Final Design and Construction. Revenue Service date is September 2021. The Sound Transit Board adopted the baseline capital budget of $2.13B originally in 2012 and revised the baseline budget to $1.89B in October 2015.

Lynnwood Link Extension: Lynnwood Link Extension extends light rail 8.5 miles to Snohomish County along I-5 from Northgate Mall in Seattle to the Lynnwood Transit Center, with four new light rail stations. The configuration includes at-grade, elevated, and retained cut/fill alignment. The proposed budget for this project is $488.4M.

East Link Extension: East Link Extension expands light rail to East King County via I-90 from Downtown Seattle to the Overlake Transit Center area of Redmond, with stations serving Rainier Avenue/I-90, Mercer Island, South Bellevue, Downtown Bellevue, Overlake Hospital, the Bel-Red corridor, Overlake Village and the Redmond Technology Center (formerly Overlake Transit Center). Revenue service to the Redmond Technology Center is forecast for early 2023 with a tunnel route in Downtown Bellevue. The Sound Transit Board adopted the baseline capital budget of $3.67B in April 2015.

South 200th Link Extension: S. 200th Link Extension consists of a 1.6-mile extension of the light rail from the SeaTac/Airport Station to an elevated station at South 200th Street. The Project was developed through a design-build (DB) delivery strategy. The Sound Transit Board adopted the baseline capital budget of $383M in 2011. Revenue Service began on September 24, 2016.

Federal Way Link Extension: The Federal Way Link Extension adds approximately 7.8 miles of light rail from the Angle Lake Station in the City of SeaTac to the Federal Way City Center. The extension generally parallels SR 99 and I-5 freeway with stations at Kent/Des Moines, South 272nd Star Lake park –and-ride and the Federal Way Transit Center. The proposed budget for this project is $412.6M.

Tacoma Link Expansion: The Tacoma Link Expansion is a 2.4-mile extension of the existing Tacoma Link system from the Theater District in downtown Tacoma to the Hilltop neighborhood. The extension will travel at-grade along Stadium Way, North 1st. Street, Division Avenue, and Martin Luther King Jr. Way. The Final Design budget is $34.6M.

Link Operations and Maintenance Facility East: This project located in the City of Bellevue, includes a 160,000 square foot operations and maintenance facility that will maintain, store and deploy and expand light rail fleet, including seven LRV service pays, a LRV wash facility and storage for up to 96 light rail vehicles. The ST Board adopted the baseline capital budget of $449M in July 2016.

ST2 Light Rail Vehicles (LRV) Fleet Expansion: This project includes the design, manufacturing, inspection, testing and delivery of 122 low floor LRVs for service requirements of Northgate, East Link and Lynnwood Link Extensions. The Sound Transit Board adopted the baseline capital budget of $733M in September 2015.

Downtown Redmond Link Extension: This project adds two new light rail stations beyond the future Redmond Technology Center Station being build as part of the East Link Extension. Preliminary Engineering for the Redmond light rail extension resumed in 2016 with a budget of $28.6M.

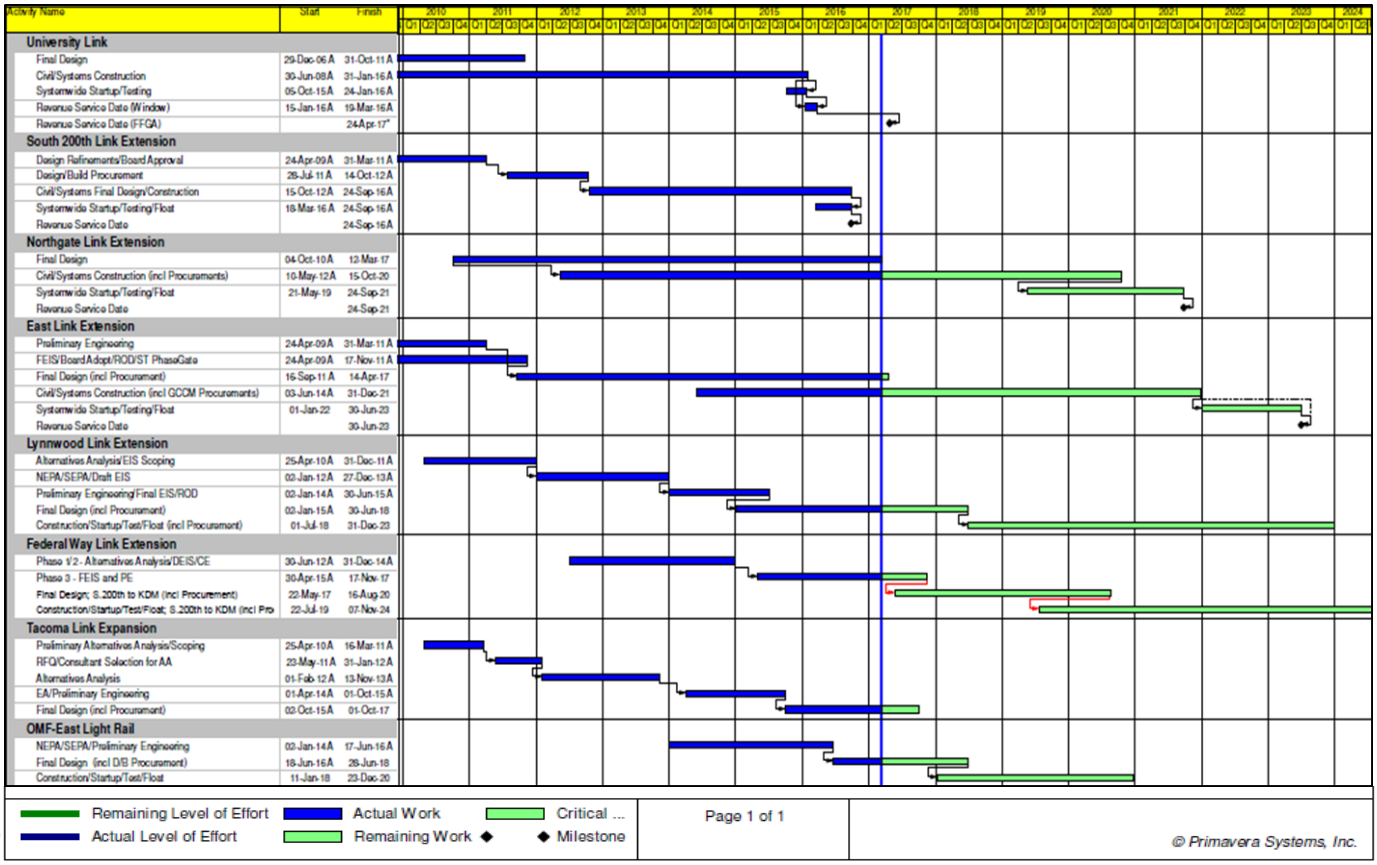
Program Budget

Project	Adopted Budget	Committed to Date	Incurred to Date	Forecast to Complete	Estimated Final Cost	Adopted Budget vs. EFC
University Link	$1,756.0	$1,514.1	$1,500.8	$40.5	$1,554.6	$201.4
Northgate Link Extension	$1,899.8	$1,297.5	$836.0	$602.3	$1,899.8	$0
Lynnwood Link Extension	$488.4	$200.2	$119.0	$288.2	$488.4	$0
East Link Extension	$3,677.2	$1,536.0	$639.7	$2,141.1	$3,677.1	$0
South 200th Link Extension	$383.2	$329.1	$323.4	$12.1	$341.2	$42.0
Federal Way Extension	$412.6	$40.9	$38.8	$371.7	$412.6	$0
Tacoma Link Expansion	$34.6	$20.3	$14.2	$14.3	$34.6	$0
Link O & M Facility: East	$449.2	$50.9	$49.5	$398.3	$449.2	$0
ST2 LRV Expansion	$733.0	$529.1	$9.9	$203.9	$733.0	$0
Downtown Redmond Link Ext.	$28.6	$13.6	$1.4	$15.0	$28.6	$0
Total Link	**$9,862.6**	**$5,531.7**	**$3,532.8**	**$4,087.5**	**$9,619.2**	**$243.4**

Table in millions.

Program Schedule

Schedules for active projects are summarized below.



Scope

Limits: 3.15-mile extension of the Initial Segment light rail transit (LRT) system from the Pine Street Stub Tunnel (PSST) in downtown Seattle to University of Washington (UW) campus near Husky Stadium.

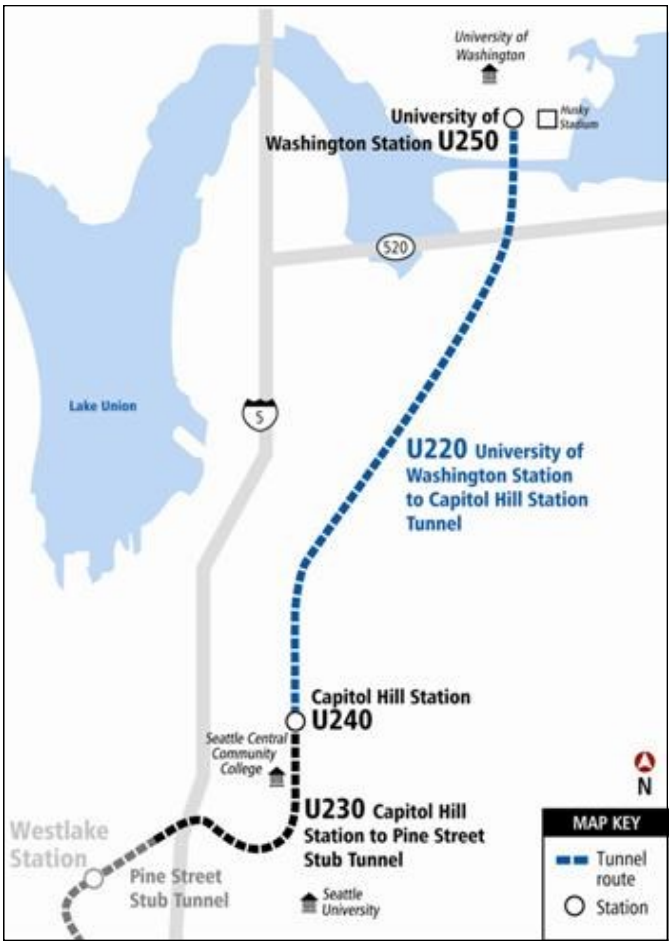
Tunnels: Two twin bored tunnels. Two contract segments: U220 starts at Husky Stadium, continues beneath SR 520 and the Montlake Cut south to the Capitol Hill underground station. U230 commences at Capitol Hill Station and proceeds south under the I-5 freeway to the PSST.

Stations: 2 underground center platform stations – Capitol Hill (U240) and University of Washington (U250). U240 is approximately 65' underground with three station entrances. U250 is about 100' below ground, with three station entrances and connections to the UW campus by a pedestrian bridge.

System: 27 LRV; direct fixation tracks, signals, traction electrification, and SCADA communications

Budget: $1.948 billion including finance cost (capital subtotal of $1.756 billion)

Schedule: Revenue Service began on March 19, 2016.



Key Project Issues

- Revenue Service commenced on March 19, 2016. Coordination between the Construction teams and the Operations group are ongoing to adjust the System for optimum operations. Contractors continue to perform punch list items but access to these activities poses to be challenging as they now have to work under Operations' rules and not Construction protocols.
- *U830 Systems*: Contractor is currently focusing on punch list on U-Link and systems refinement to achieve optimal operation status. U-Link Systems has encountered some issues related to this infancy period and is working for resolutions.
- *Certification of Occupancy*: University Link continues to operate under a temporary certification of occupancy until all permit requirements are fully met. The final items remaining are the utilities as-built to be submitted to City of Seattle (SPU) and a final review of the emergency ventilation systems by the Seattle Fire Department.
- Commercial issues with all prime Civil contractors (with the exception of Capitol Hill Station Contractor) have been negotiated as the project proceeds through the close-out process.
- Miscellaneous follow-on commitments and restoration work.

Project Cost Summary

The U-Link project cost is summarized in two types of cost classifications. In the first table, cost is classified in accordance with Sound Transit's Work Breakdown Structure (WBS); and in the second table, cost is summarized in accordance to the FTA's Standard Cost Categories (SCC) format. (Both tables' figures in millions.)

WBS Phase Elements	Baseline Budget	Current Adopted Budget	Commitment to Date*	Incurred to Date	Est. Final Cost (EFC)	Adopted Budget vs. EFC
ADMINISTRATION	$ 115.23	$ 113.55	$ 81.90	$ 80.89	$ 88.17	$ 25.38
PRELIMINARY ENGINEERING	$ 24.39	$ 24.26	$ 24.26	$ 24.26	$ 24.26	$ -
FINAL DESIGN	$ 77.94	$ 90.31	$ 87.98	$ 86.05	$ 88.66	$ 1.65
CONSTRUCTION SERVICES	$ 68.53	$ 94.81	$ 86.80	$ 86.32	$ 88.72	$ 6.10
3rd PARTY AGREEMENTS	$ 18.65	$ 18.65	$ 11.94	$ 11.31	$ 13.51	$ 5.13
CONSTRUCTION	$ 1,180.00	$ 1,158.18	$ 995.56	$ 986.35	$ 1,022.58	$ 135.61
VEHICLES	$ 103.91	$ 103.91	$ 99.20	$ 99.19	$ 101.91	$ 2.00
ROW	$ 167.33	$ 152.33	$ 126.43	$ 126.43	$ 126.79	$ 25.54
Capital Total	**$ 1,755.97**	**$ 1,756.01**	**$ 1,514.08**	**$ 1,500.80**	**$ 1,554.59**	**$ 201.41**
FINANCE COST	$ 191.71	$ 191.71	$ 191.71	$ 174.87	$ 191.71	$ -
Project Total	**$ 1,947.68**	**$ 1,947.72**	**$ 1,705.79**	**$ 1,675.67**	**$ 1,746.30**	**$ 201.41**

(*)Totals may not equal column sums due to rounding of line entries.

In March, the projected Estimated Final Cost (EFC) including Finance Cost remains at about $1.75B. As the project inches closer to close out, expenditure becomes less predictable in terms of its timing. This period, after some cost adjustments the project expenditure net approximately $160K and continues to inch the project's Incurred to Date amount above to $1.5B (Finance Cost excluded). Direct construction activities pertain to close out and miscellaneous follow on work. The construction EFC continues to be approximately $1B. The trend continues to hold as the project is essentially completed with only miscellaneous follow on scope and commercial issues remaining. Systems (U830) continues optimization process. The Total Incurred to Date for the Construction Phase is about $986M and LRV is at about $99.2M. Cost of repairs are excluded from this project and tracked independently.

Construction EFC under the SCC format remains relatively unchanged at about $1B. Construction SCC expenditures to date at about $974M an increase of about $0.3M—mostly attributed to Station punch lists and Systems' continuing optimization and resolutions to change order work and miscellaneous follow on work. Total capital cost incurred to date including non-construction cost (SCC 60 thru 90) now at $1.5B or about 85% of total project budget scoped (excluding Finance Cost). Total project cost incurred to date at the end of March 2017 close including Finance Cost is at $1.67B. The financing cost incurred to date is about $175M. University Link EFC continues to be projected to close out with at least $200M under budget excluding financing cost.

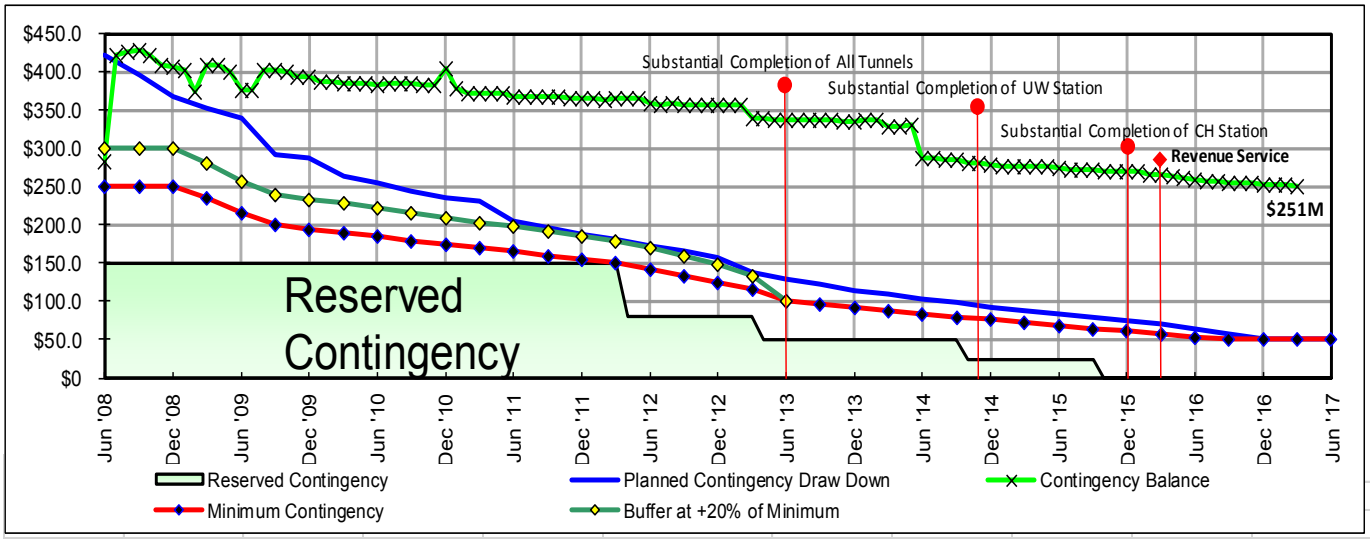
Project Elements by SCC	Baseline Budget	Current Adopted Budget	Commitment to Date*	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget vs. EFC
10 Guideway & Track Elements	$ 626.83	$ 450.46	$ 456.22	$ 455.80	$ 461.23	$ (10.77)
20 Stations	$ 366.33	$ 350.75	$ 344.23	$ 343.04	$ 353.09	$ (2.34)
30 Support Facilities: Yards, Shops	$ 7.01	$ 24.83	$ 23.40	$ 23.26	$ 24.50	$ 0.33
40 Sitework & Special Conditions	$ 59.03	$ 67.39	$ 57.09	$ 54.72	$ 56.92	$ 10.47
50 Systems	$ 69.63	$ 116.42	$ 101.63	$ 97.64	$ 102.89	$ 13.53
Construction Subtotal (SCC 10-50)	**$ 1,128.82**	**$ 1,009.85**	**$ 982.57**	**$ 974.46**	**$ 998.63**	**$ 11.22**
60 Row, Land, Existing Improvements	$ 167.33	$ 126.60	$ 126.43	$ 126.43	$ 125.77	$ 0.83
70 Vehicles	$ 99.76	$ 100.16	$ 99.87	$ 99.87	$ 100.16	$ 0.00
80 Professional Services	$ 306.41	$ 346.58	$ 305.20	$ 300.05	$ 318.55	$ 28.04
90 Unallocated Contingency	$ 53.65	$ 172.81	$ -	$ -	$ 11.48	$ 161.33
Capital Cost Total (SCC 10-90)	**$ 1,755.97**	**$ 1,756.01**	**$ 1,514.08**	**$ 1,500.80**	**$ 1,554.59**	**$ 201.41**
100 Finance Cost	$ 191.71	$ 191.71	$ 191.71	$ 174.87	$ 191.71	$ -
Project Total	**$ 1,947.68**	**$ 1,947.72**	**$ 1,705.79**	**$ 1,675.67**	**$ 1,746.30**	**$ 201.41**

(*)Totals may not equal column sums due to rounding of line entries.

Cost Contingency Management

Project contingencies remain at healthy at approximately $251M due to the favorable construction bidding climate, ROW acquisition cost trends, diligent project risk management practices, and excellent tunneling conditions. March's overall contingencies notched down by a little over $1M due to a global resolution to the MOW construction contract, miscellaneous change orders and follow on project commitments. While major construction activities are now done, there are still considerable follow-on scope to complete (pertaining to close-out of the project, systems adjustments to optimize the operating systems as well as settlement of commercial issues). Barring any catastrophic event, the likelihood that this contingency stays on trend is high. The forecast indicates that approximately 80% of these contingencies will be remain unused; thereby, generating at least $200M of budget savings excluding financing cost.

MINIMUM CONTINGENCY CURVES as of March 31, 2017 — $ in millions



U240 Contract – *Capitol Hill Station*

Close-out

The U240 Contractor achieved Substantial Completion on December 31, 2015.

Close-out Activities

Current Period

- Continued negotiating and finalizing outstanding commercial issues. The latest round of mediation was successful in resolving some subcontractor claims, but a final settlement with the general contractor has not been reached.

Next Period

- Continue negotiating and finalizing commercial issues.

Closely Monitored Issues

- Multiple commercial issues remain open including various notice of intents to claim and actual claims. These claims continue to be reviewed and assessed by Sound Transit and the Contractor.

Cost Summary

Present Financial Status	Amount
U240 Contractor - Turner Construction Co	
Original Contract Value	$104,850,276
Change Order Value	$11,254,629
Current Contract Value	$116,104,905
Total Actual Cost (Incurred to date)	$113,926,440
Financial Percent Complete:	98%
Physical Percent Complete:	99%
Authorized Contingency	$11,742,514
Contingency Drawdown	$11,254,629
Contingency Index	1.02

U250 Contract –*University of Washington Sta.*

Close-out

Milestone #7, Substantial Completion, was granted in November 2014. U250 achieved acceptance on April 26, 2016.

Close-out Activities

Current Period

- Continued closing out of all remaining commercial issues.

Next Period

- Continue closing out of all remaining commercial issues.

Closely Monitored Issues

- All outstanding claims have been verbally settled and will continue moving through the formal execution phase. A final settlement change order will be issued to the Contractor.

Cost Summary

Present Financial Status	Amount
U250 Contractor - Hoffman Construction Co.	
Original Contract Value	141,745,898
Change Order Value	8,568,882
Current Contract Value	150,314,781
Total Actual Cost (Incurred to date)	150,254,810
Financial Percent Complete:	99%
Physical Percent Complete:	99%
Authorized Contingency	9,152,295
Contingency Drawdown	8,568,882
Contingency Index	1.06

U810 Contract – *Maintenance of Way*

Close-out

Processing outstanding commercial issues.

Close-out Activities

Current Period

- Continued processing outstanding commercial issues.

Next Period

- Continue processing outstanding commercial issues and contract close-out.

Closely Monitored Issues

- ST and contractor have settled on a final contract amount. A final change order to the contract is pending formal processing.

Cost Summary

Present Financial Status	Amount
U810– Forma Construction (Design/ Build)	
Original Contract Value	$11,998,725
Change Order Value	$501,013
Current Contract Value	$12,499,738
Total Actual Cost (Incurred to date)	$12,438,547
Financial Percent Complete	99%
Physical Percent Complete:	99%
Authorized Contingency	$959,898
Contingency Drawdown	$501,013
Contingency Index	1.9

U830 Contract – *Track, Signal, Traction Power and Communications*

Close-out

Revenue Service on March 19, 2016. Punchlist and close-out work continue.

Close-out Activities

Current Period

- Ongoing work on Systems punch list items; delivery of spare parts; and finalizing O&M Manuals, As-built drawings, and other final project documentation.
- Ongoing work on commercial closure of the U830 project. ST working closely with the GC/CM and the EC/CMs to ensure rapid and fair commercial settlement of claims.

Next Period

- Continue Systems punch list, spare parts delivery, and finalizing O&M Manuals and documentation.
- Continue work on commercial closure of the project.

Closely Monitored Issues

- ST working closely with the GC/CM and the EC/CMs to ensure rapid and fair commercial settlement of claims.

Cost Summary

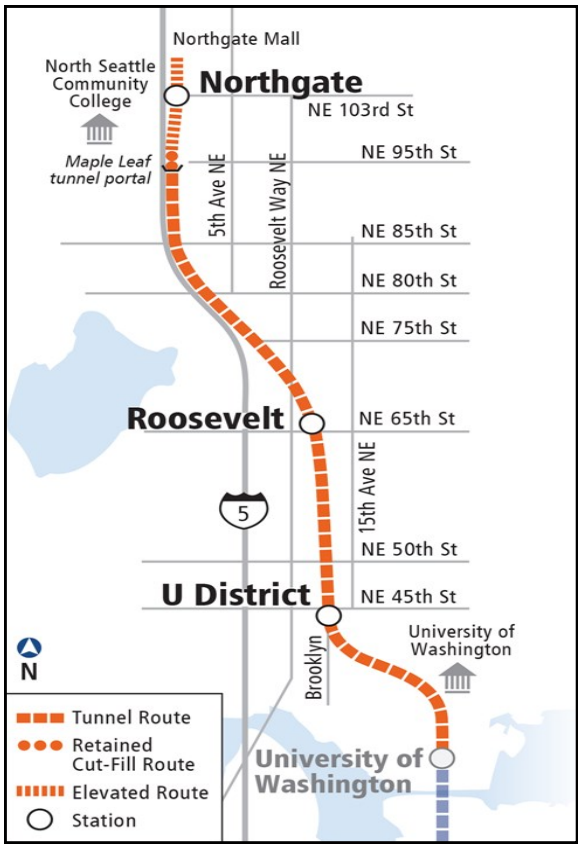
Present Financial Status	Amount
U830 GC/CM Contractor - Stacy & Witbeck	
Original Contract Value	$119,167,433
Change Order Value	$5,425,700
Current Contract Value	$124,593,133
Total Actual Cost (Incurred to date)	$121,323,510
Financial Percent Complete	97.4%
Physical Percent Complete:	99.0%
Authorized Contingency	$5,958,373
Contingency Drawdown	$5,425,700
Contingency Index	1.1

This page left intentionally blank.

Link Light Rail
Northgate Link Extension

Scope

Limits:	The Northgate Link Extension consists of 4.3 miles of light rail extending from the University of Washington to Northgate.
Alignment:	The extension begins at the UW Station, boring tunnels under campus then continues north to a portal located north of NE 94th Street on the east side of I-5, then transitioning to an aerial structure running north to the Northgate Mall.
Stations:	The *U District Station* is an underground station located on the west side of the UW campus near Brooklyn Ave. and NE 45th St. The *Roosevelt Station* is an underground station located near NE 65th St. and 12th Ave NE. The *Northgate Station* is an elevated station located at the southwest edge of the Northgate Mall property.
Systems:	Include Signals, track electrification, and SCADA communications.
Budget:	$1.899 Billion
Service:	September 2021
Phase:	Final Design and Construction



Map of Northgate Link Extension route and stations.

Key Project Activities

- Provided a project update at the ST- FTA Quarterly Meeting.
- Conducted a construction tour with FTA-PMOC.
- Briefed the Executive Leadership Team on Scott White and Joni Earl facility naming process; Board action planned next period.
- *For N125 TBM Tunnels,* Contractor and project team held an Executive Partnering Session; Contractor continued Stage 3 of the UW plaza restoration; Continued cross-passage work in the tunnels including freezing activities, dewatering, waterproofing and surface restoration.
- *For N140 University District Station*, ST Board approved the U District Station Finishes contract with Hoffman Construction.
- *For N150 Roosevelt Station,* Roosevelt Station Contractor, Hoffman Construction, continued shotcreting the station walls and finishing installation of temporary utilities.
- *For N160 Northgate Station,* Northgate Station Contractor, Absher is completing construction of the "C" series Station shafts.
- *For N180 Trackwork*, Contractor continues to focus on procurement, submittals, and manufacturing of the pre-cast floating slab and is working with a British supplier for the Ultra-Straight Rail.

Closely Monitored Issues

- Sound Transit issued work directive to N125 Contractor to expedite UW Plaza restoration work.

Project Cost Summary

The Northgate Link project cost is summarized below by two cost categories. In the first table, the cost is summarized in accordance with Sound Transit's Work Breakdown Structure (WBS). The second table summarizes the project in accordance with the FTA's Standard Cost Category (SCC) format. The figures in both tables are shown in millions.

Cost Summary by Phase

Project Phase	Baseline Budget	Current Adopted Budget	Commitment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget vs. EFC
ADMINISTRATION	$147.9	$147.9	$45.2	$46.0	$147.9	$0.0
PRELIMINARY ENGINEERING	$15.1	$15.1	$15.1	$15.1	$15.1	$0.0
FINAL DESIGN	$129.2	$129.2	$116.6	$102.9	$129.2	$0.0
CONSTRUCTION SERVICES	$118.3	$118.3	$84.8	$48.7	$118.3	$0.0
3rd PARTY AGREEMENTS	$11.8	$11.8	$9.9	$6.2	$11.8	$0.0
CONSTRUCTION	$1,343.0	$1,343.0	$932.2	$523.9	$1,343.0	$0.0
ROW	$112.3	$112.3	$93.7	$93.3	$112.3	$0.0
PROJECT CONTINGENCY	$22.2	$22.2	$0.0	$0.0	$22.2	$0.0
Total	**$1,899.8**	**$1,899.8**	**$1,297.5**	**$836.0**	**$1,899.8**	**$0.0**

The Estimated Final Cost (EFC) for the project as of this reporting period is $1,899.8M, which is equal to the current project budget. This period approximately $15.2M was incurred, of which $11.8M was for the N125 tunneling contract, the N150 Roosevelt Station Finishes contract, the N160 Northgate Station and Elevated Guideway contract, the N180 Trackwork contract and other miscellaneous construction; $0.6M was incurred for civil and systems final design and design support during construction; and $1.4M was for construction management. The remaining expenditures were for third party coordination, permits, staff, legal, right-of-way and other direct charges.

Cost Summary by SCC

SCC Element	Baseline Budget	Current Adopted Budget	Commitment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget vs. EFC
10 GUIDEWAY & TRACK	$595.6	$547.6	$491.9	$356.2	$548.8	($1.2)
20 STATIONS	$376.1	$429.1	$276.6	$53.9	$430.3	($1.3)
30 SUPPORT FACILITIES: YARD, SHOP	$5.3	$5.3	$6.4	$5.3	$6.4	($1.1)
40 SITEWORK & SPECIAL CONDITIONS	$140.8	$225.0	$143.2	$99.4	$219.2	$5.9
50 SYSTEMS	$110.9	$95.1	$4.0	$1.1	$97.4	($2.3)
Construction Subtotal (SCC 10 - 50)	**$1,228.7**	**$1,302.1**	**$922.1**	**$515.8**	**$1,302.1**	**$0.0**
60 ROW, LAND, EXISTING IMPROVEMENTS	$119.9	$110.9	$93.7	$93.3	$110.9	$0.0
80 PROFESSIONAL SERVICES	$420.7	$429.1	$281.7	$227.0	$429.0	$0.1
90 CONTINGENCY	$130.4	$57.8	$0.0	$0.0	$57.8	($0.1)
Capital Total (SCC 10 - 90)	**$1,899.8**	**$1,899.8**	**$1,297.5**	**$836.0**	**$1,899.8**	**$0.0**

The Estimated Final Cost (EFC) for some contract packages have been reassessed and revised as a result of updated design development estimates and the actual contract award amounts.

Cost Contingency Management

Compared to the baseline amount of $396.2M, the Total Contingency has decreased by $174.8M to $221.4M, which is 20.8% of project work remaining. During this reporting period, a decrease of $59.9M in the overall project contingency occurred. Detailed information is provided below.

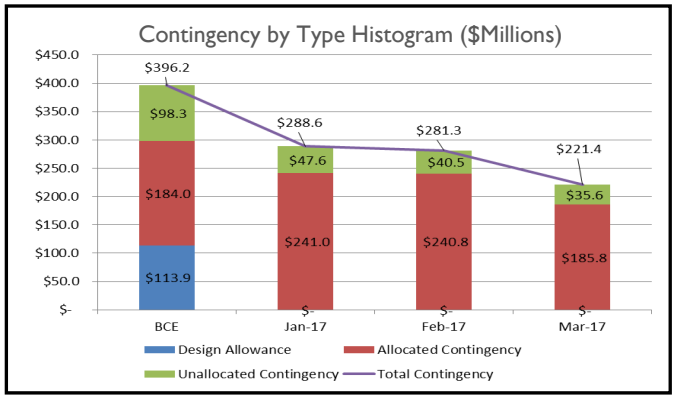
Design Allowance (DA) – The baseline Design Allowance of $113.9M has been fully depleted following the receipt of updated construction cost estimates on the N140, N150, and N160 Station Finishes contracts. DA has been utilized as scope development has progressed to 100% on N140, and the N150 and N160 contracts have been awarded. No changes to DA occurred during this period.

Cost Contingency Management, continued

Allocated Contingency (AC)– Compared to the baseline amount of $184.0M, Allocated Contingency has increased by $1.8M to $185.8M. During this reporting period, AC was decreased by $2.4M following execution of change orders on the N111, N125 and N160 construction contracts. In addition, administrative revisions were implemented to correct the Future AC in the Administration, Construction Services, and Construction Phases. These phases now show a decrease of $52.7M following reallocation of Future Contingency to Future Commitment in order to correct negative contingencies identified in the AC integrity report.

Unallocated Contingency (UAC) – Compared to the baseline amount of $98.3M, Unallocated Contingency has decreased by $62.7M to $35.6M. During this reporting period, UAC decreased by $4.8M following the transfer of funds from the Administration Phase UAC to cover additional Safety & Security and QA Consultant tasks ($1.9M), from the Final Design UAC to cover the Noise & Vibration and Systems FD requirements ($1.3M), and from Construction Phase UAC to the N830 contract to cover the total MACC award for the Systems contract ($1.6M).

Contingency Status	Baseline		Current	
	Amount	% of Total	Amount	% of Work Remaining
Design Allowance	$ 113.9	5.3%	$ -	0.0%
Allocated Contingency	$ 184.0	8.6%	$ 185.8	17.5%
Unallocated Contingency	$ 98.3	4.6%	$ 35.6	3.3%
Total	**$ 396.2**	**18.6%**	**$ 221.4**	**20.8%**



Project Schedule

The N113 SCL 115kV Relocation scope of work has been completed. However, WSDOT has requested a fence be erected along the MSE wall as an additional safety measure. ST has requested a proposal from Potelco.

The N125 Tunneling Contractor (JCM) is continuing work on cross passages and tunnel finishes. JCM's schedule update was returned Code 2— Exceptions Noted. JCM is continuing to forecast negative float in achievement of the following milestones: Milestone 5 - UDS Turnover (-159 days, due to Cross Passage 23 and claimed post-UDS Turnover access). Milestone 6 - Substantial Completion has slipped and is 14 days behind schedule. JCM has added a Milestone for Handover of the UDS Station Box with a constrained date of May 2, 2017 which has slipped to a forecast date of July 20, 2017. ST is currently working with JCM to achieve a date of July 1st.

The N140 U District Station construction contract received board approval this period, NTP is anticipated to be in early April.

The N150 Roosevelt Station contractor, Hoffman Construction, is currently shotcreting the station walls and finishing installation of temporary utilities. Waterproofing, wall, and invert rebar installation will begin in mid-April.

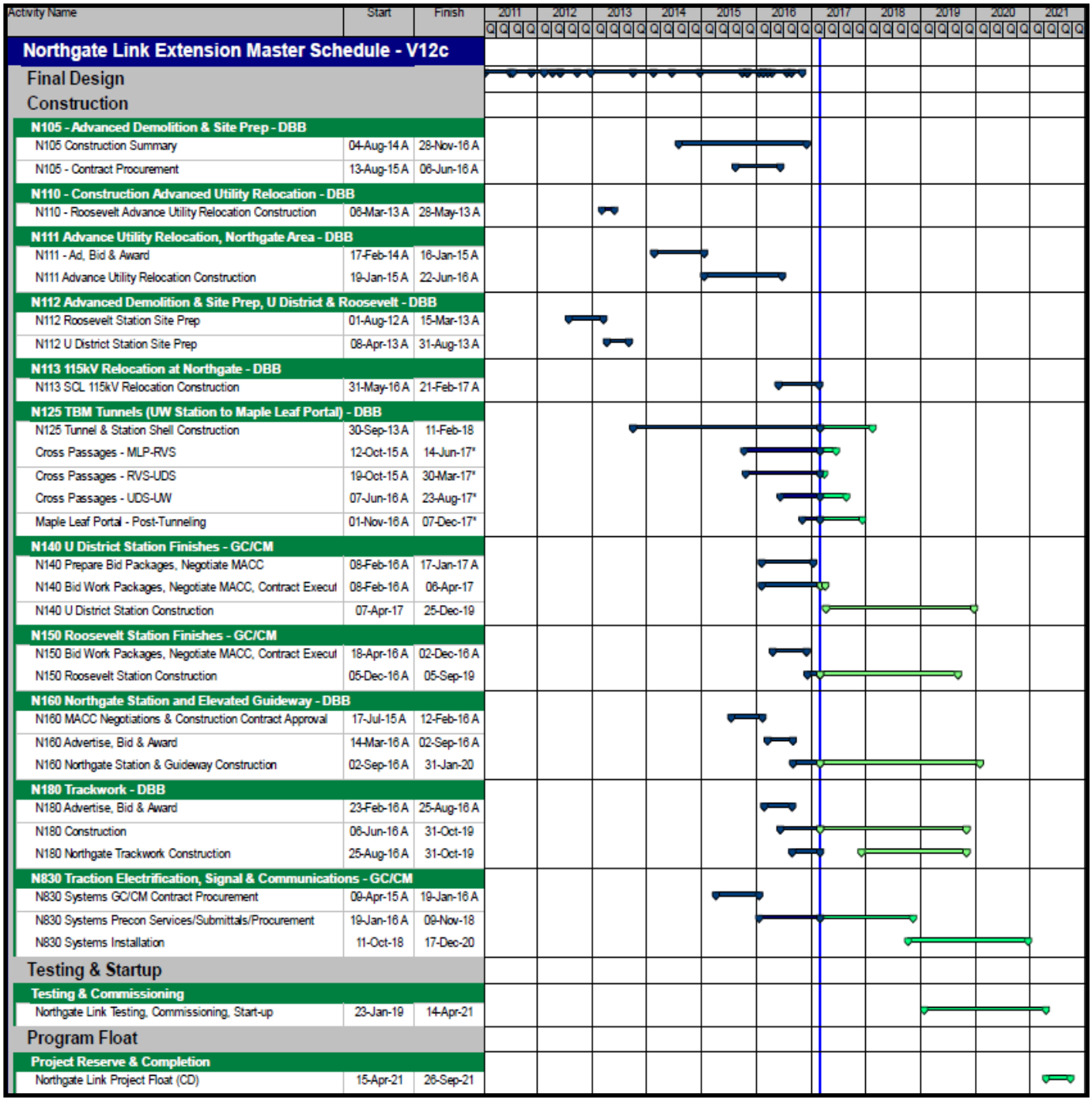
The N160 Contractor, is completing construction of the "C" series Station shafts. At the Parking Garage site, the Contractor is continuing installation of soldier piles and secant piles on the perimeter of the Garage. ST is providing revised plans for the north end realignment as they become available. Absher will likely be granted days for weather related delays that occurred in January and February.

The N180 Trackwork contract continues with submittal processing, and receipt of material samples. The material order for the isolations pads has been placed, and Columbia Precast is continues to set up for floating slab production. ST and SWI are currently working with the pre-caster on the test sections to be poured. The Contractor is working with British Steel on quotes for the Ultra-Straight Rail.

The N830/E750 Systems contract is on track for April board approval. ST has begun holding regular planning meetings with the Systems contractor/Operations/DECM regarding the testing and commissioning.

Revenue Service date for Northgate Link Extension is scheduled in September 2021. See schedule in the following page.

Project Schedule, continued

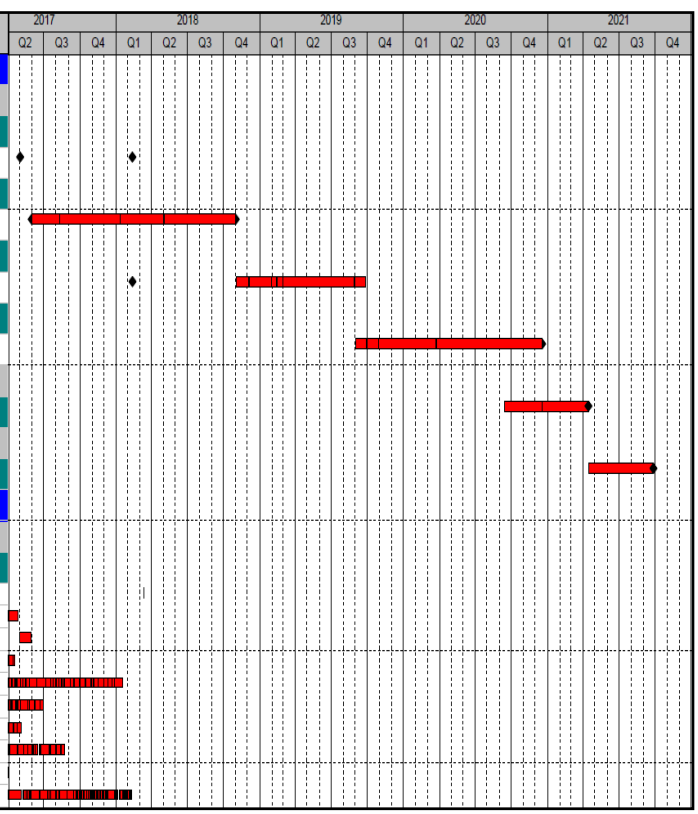


Activity Name	Start	Finish
Northgate Link Extension Master Schedule - V12c		
Final Design		
Construction		
N105 - Advanced Demolition & Site Prep - DBB		
N105 Construction Summary	04-Aug-14 A	28-Nov-16 A
N105 - Contract Procurement	13-Aug-15 A	06-Jun-16 A
N110 - Construction Advanced Utility Relocation - DBB		
N110 - Roosevelt Advance Utility Relocation Construction	06-Mar-13 A	28-May-13 A
N111 Advance Utility Relocation, Northgate Area - DBB		
N111 - Ad, Bid & Award	17-Feb-14 A	16-Jan-15 A
N111 Advance Utility Relocation Construction	19-Jan-15 A	22-Jun-16 A
N112 Advanced Demolition & Site Prep, U District & Roosevelt - DBB		
N112 Roosevelt Station Site Prep	01-Aug-12 A	15-Mar-13 A
N112 U District Station Site Prep	08-Apr-13 A	31-Aug-13 A
N113 115kV Relocation at Northgate - DBB		
N113 SCL 115kV Relocation Construction	31-May-16 A	21-Feb-17 A
N125 TBM Tunnels (UW Station to Maple Leaf Portal) - DBB		
N125 Tunnel & Station Shell Construction	30-Sep-13 A	11-Feb-18
Cross Passages - MLP-RVS	12-Oct-15 A	14-Jun-17*
Cross Passages - RVS-UDS	19-Oct-15 A	30-Mar-17*
Cross Passages - UDS-UW	07-Jun-16 A	23-Aug-17*
Maple Leaf Portal - Post-Tunneling	01-Nov-16 A	07-Dec-17*
N140 U District Station Finishes - GC/CM		
N140 Prepare Bid Packages, Negotiate MACC	08-Feb-16 A	17-Jan-17 A
N140 Bid Work Packages, Negotiate MACC, Contract Execut	08-Feb-16 A	06-Apr-17
N140 U District Station Construction	07-Apr-17	25-Dec-19
N150 Roosevelt Station Finishes - GC/CM		
N150 Bid Work Packages, Negotiate MACC, Contract Execut	18-Apr-16 A	02-Dec-16 A
N150 Roosevelt Station Construction	05-Dec-16 A	05-Sep-19
N160 Northgate Station and Elevated Guideway - DBB		
N160 MACC Negotiations & Construction Contract Approval	17-Jul-15 A	12-Feb-16 A
N160 Advertise, Bid & Award	14-Mar-16 A	02-Sep-16 A
N160 Northgate Station & Guideway Construction	02-Sep-16 A	31-Jan-20
N180 Trackwork - DBB		
N180 Advertise, Bid & Award	23-Feb-16 A	25-Aug-16 A
N180 Construction	06-Jun-16 A	31-Oct-19
N180 Northgate Trackwork Construction	25-Aug-16 A	31-Oct-19
N830 Traction Electrification, Signal & Communications - GC/CM		
N830 Systems GC/CM Contract Procurement	09-Apr-15 A	19-Jan-16 A
N830 Systems Precon Services/Submittals/Procurement	19-Jan-16 A	09-Nov-18
N830 Systems Installation	11-Oct-18	17-Dec-20
Testing & Startup		
Testing & Commissioning		
Northgate Link Testing, Commissioning, Start-up	23-Jan-19	14-Apr-21
Program Float		
Project Reserve & Completion		
Northgate Link Project Float (CD)	15-Apr-21	26-Sep-21

Critical Path Analysis

The longest path through the Northgate projects remains unchanged this period and continues to be driven by the N125 contract and turnover of the UDS box to N140, N140 turnover to N180 for trackwork, and subsequently to the N830 Systems contract. The revised schedule update from the Contractor indicates turnover of the U District Station at the beginning of June 2017 instead of the required May 2017 date. Turnover at the beginning of June 2017 will absorb the Contract Interface Float programmed in the Master Schedule between N125 and N140.

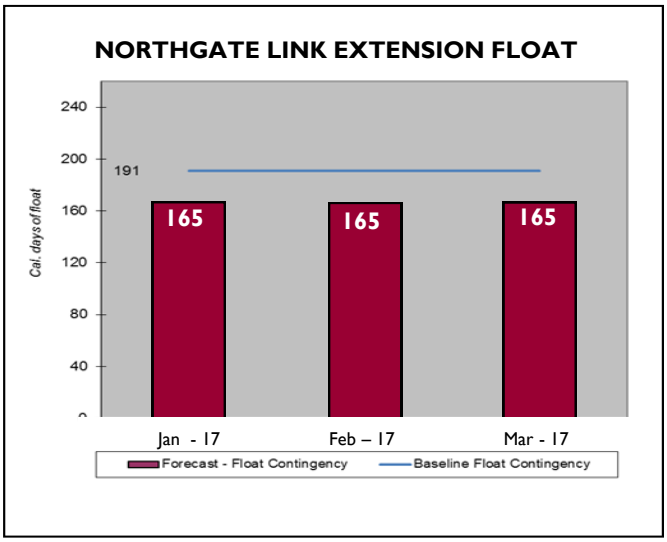
Activity ID	Activity Name	Start	Finish
	Northgate Link Extension Master Schedule - V12c	30-Sep-13 A	26-Sep-21
	Construction	30-Sep-13 A	17-Dec-20
	N125 TBM Tunnels (UW Station to Maple Leaf Portal) - DBB	30-Sep-13 A	11-Feb-18
	N125 Construction	30-Sep-13 A	11-Feb-18
	N140 U District Station Finishes - GC/CM	01-Jun-17	01-Nov-18
	N140 U- District Station Construction	01-Jun-17	01-Nov-18
	N180 Trackwork - DBB	28-Nov-17	27-Sep-19
	N180 Construction	28-Nov-17	27-Sep-19
	N830 Traction Electrification, Signal & Communications - GC/CM	11-Oct-18	17-Dec-20
	Construction - Systems Installation	11-Oct-18	17-Dec-20
	Testing & Startup	23-Jan-19	14-Apr-21
	Testing & Commissioning	23-Jan-19	14-Apr-21
	Program Float	15-Apr-21	26-Sep-21
	Project Reserve & Completion	15-Apr-21	26-Sep-21
	N125 Status Schedule Mar-17	01-Mar-16 A	12-Mar-18
	N125	01-Mar-16 A	12-Mar-18
	Construction	01-Mar-16 A	12-Mar-18
	Payment Triggers - Project Demobilization	12-Mar-18	12-Mar-18
	Submittals - Prepare	03-Apr-17	28-Apr-17
	Submittals - Review & Approve	20-May-16 A	28-May-17
	Station Work UW	03-Apr-17	18-Apr-17
	Tunneling	27-Sep-16 A	17-Jan-18
	Cross Passages - MLP - RVS	30-Mar-17 A	29-Jun-17
	Cross Passages - RVS - UDS	21-Dec-16 A	05-May-17
	Cross Passages - UDS - UW	25-Oct-16 A	22-Aug-17
	Cross Passages - Additional SEM Measures	01-Mar-16 A	03-Apr-17
	MLP Final Work	02-Feb-17 A	09-Feb-18



Critical Path Float

The Northgate Link Project currently retains 165 days of unallocated project float.

The Revenue Service date for the Northgate Link Extension remains September 2021.



Construction Safety

Data/ Measure	March 2017	Year to Date	Project to Date
Recordable Injury/Illness Cases	2	6	60
Days Away From Work Cases	0	0	4
Total Days Away From Work	0	0	219
Restricted or Modified Work Cases	0	1	24
Total Days Restricted or Modified Work	0	26	923
First Aid Cases	1	3	54
Reported Near Mishaps	0	1	55
Average Number of Employees on Worksite	456	-	-
Total # of Hours (GC & Subs)	76,762	205,526	2,178,398
OSHA Incident Rates	**Monthly Average**	**Year to Date**	**Project to Date**
Recordable Injury Rate	5.21	5.84	5.51
DART Rate	0.00	0.97	2.57
Recordable National Average	3.50	3.50	3.50
DART National Average	2.00	2.00	2.00
Recordable WA State Average	7.00	7.00	7.00
DART WA State Average	3.90	3.90	3.90

Right of Way

The U District and Roosevelt stations required the acquisition of a range of property interests including fee takings for stations and staging areas; tunnel easements are required for the running tunnel. These acquisitions resulted in owner and tenant, residential and commercial relocations.

Roosevelt Station – All parcels have been acquired. One parcel de-certified.

U District Station – All parcels have been acquired. One parcel de-certified.

Northgate Station - All parcels have been acquired.

Tunnel Easements:

RVS to North Portal - Two parcels are still in court.

UWS to U District Station - All parcels have been acquired.

Line Section	Total Parcels Certified	Offers Made	Signed Agree-ments	Admin. Settle-ments	Posses-sion and Use (P&U)	Closings to date	Baseline Relocations Required	Relocations Completed
U District Station	15	15	0	0	0	15	3	3
Roosevelt Station	19	19	0	0	0	19	26	26
Tunnel Easements	190	188	0	0	0	188	0	0
Northgate Station	11	11	0	0	0	9	13	13
Total	**235**	**233**	**0**	**0**	**0**	**231**	**42**	**42**

Community Outreach

- Distributed various construction alerts which includes the following:
  - Northgate Interim Lot B parking changes.
  - Nighttime striping on 1st Avenue NE.
  - Geotechnical borings along 12th Avenue NE.
- Met with Roosevelt businesses to discuss upcoming work for geotechnical boings.
- Gave project update to UW Tower Floor coordinator meeting.
- Emailed Roosevelt TOD Open House invite language to Roosevelt Neighborhood Association and City to distribute to their stakeholders.

Environmental

- None to report.

Sound Transit Board Actions

Board Action	Description	Date
M2017-31	*Execute a construction contract with Hoffman Construction Company to construct the U District Station Finishes within the Northgate Link Extension in the amount of $159,836,688 with a 5% contingency of $7,991,834 for a total authorized contract amount not to exceed $167,828,522.*	*Mar 23*
R2017-09	*Approving the chief executive officer's declaration that certain real properties acquired for the Northgate Link Extension will be surplus and no longer needed for a transit purpose upon completion of construction staging activities.*	*Mar 23*

Major Construction Contract Packages

Below are the major construction contract packages for the Northgate Link Extension with a brief scope description of each.

N105 Advance Demolition and Site Prep - Demolition of Key Bank Building, removal of underground storage tanks and remediation of contaminated soil at UDS. Demolition site and site prep at strip mall near the NG Station. (Complete)

N110 Utility Relocation at Roosevelt Station Area - Relocation of electrical and communication facilities at the Roosevelt Station site. (Complete)

N111 Advanced Utility Relocation at Northgate Area - Relocation of 26kV and communication lines at the Northgate Station vicinity. (Complete)

N112 U District and Roosevelt Station Site Preparation - Demolition of existing structures, grading and paving at the U District and Roosevelt Station sites. (Complete)

N113 115kV Relocation at Northgate Station Area - Relocation of overhead electrical facilities at the Northgate Station area. (Complete)

N125 TBM Tunnels UW Station to Maple Leaf Portal - Tunneling work and support of excavation of station boxes for the U District and Roosevelt stations.

N140 U District Station Finishes - Civil and architectural finishes work for the U District Station.

N150 Roosevelt Station Finishes– Civil and architectural finishes work for the Roosevelt Station.

N160 Northgate Station & Elevated Guideway & Parking Garage- Civil and architectural finishes work for the Maple Leaf Portal, the aerial guideway, and the Northgate Station and Tail Track. Construction of a 450 stall parking garage to be located at the south corner of the existing Northgate Mall surface parking lot at NE 103rd St. near 1st Avenue NE.

N180 Trackwork - Installation of all trackwork through two tunnels and three stations between UW Station and Northgate Station, including tail track.

N830 Track Electrification, Signals, Communication System - Installation of the major system elements – Traction Electrification, 26kV Distribution, Signals, and Communications.

Construction Schedule Highlights

Package	Bid Advertisement	NTP	Substantial Completion
N105-Advance Demolition & Site Prep	Mar 2016A	May 2016A	Nov 2016A
N111-Advanced Utility Relocation	Sept 2014A	Jan 2015A	Apr 2016A
N113-SCL 115kV Relocate	Feb 2016A	June 2016A	Feb 2017A
N125-Station Box Exc. & TBM Tunnels	Jan 2013A	Sep 2013A	Jan 2018
N140-U District Station Finishes	Oct 2013A (GC/CM-Precon)	Jun 2017 (construction)	Dec 2019
N150-Roosevelt Station Finishes	Apr 2013A (GC/CM-Precon)	Dec 2016A (construction)	Sep 2019
N160-Northgate Station & Guideway & Parking Garage	Apr 2016A	Sept 2016A	Jan 2020
N180-Trackwork to Northgate Station	Apr 2016A	Sept 2016A	Oct 2019
N830-Traction Power, Signals & Com	May 2015A (GC/CM-RFQ)	Jan 2018 (construction)	Dec 2020

A = Actual

Contract N105- Advanced Demolition and Site Prep

Close-out

The N105 Contractor, Pellco Construction, received Substantial Completion for the work at the Key Bank Site on November 23, 2016. Care and custody of the site has been returned to Sound Transit and will be transferred to the N140 Contractor in 2017.

This project is in the process of final payment and close-out.

No further updates will be made on this contract.

Cost Summary

Present Financial Status	Amount
N105 Contractor - Pellco Construction	
Original Contract Value	$2,691,500
Change Order Value	$20,709
Current Contract Value	$2,712,209
Total Actual Cost (Incurred to date)	$ 2,712,209
Financial Percent Complete:	100.0%
Physical Percent Complete:	100.0%
Authorized Contingency	$ 269,150
Contingency Drawdown	$ 20,709
Contingency Index	13.0

Contract N111 – Advanced Utility Relocation

Close-out

The N111 contract was Substantially Complete on April 26, 2016. The infrastructure has been turned over to Seattle City Light. Walsh Constructors have demobilized from the site.

No further updates will be made for this contract.

Cost Summary

Present Financial Status	Amount
N111 Contractor - Walsh Construction	
Original Contract Value	$9,370,000
Change Order Value	$ 802,471
Current Contract Value	$10,172,471
Total Actual Cost (Incurred to date)	$ 10,172,471
Financial Percent Complete:	100.0%
Physical Percent Complete:	100.0%
Authorized Contingency	$ 937,000
Contingency Drawdown	$ 802,471
Contingency Index	1.2

Contract N113– 115kv Relocation at Northgate Station

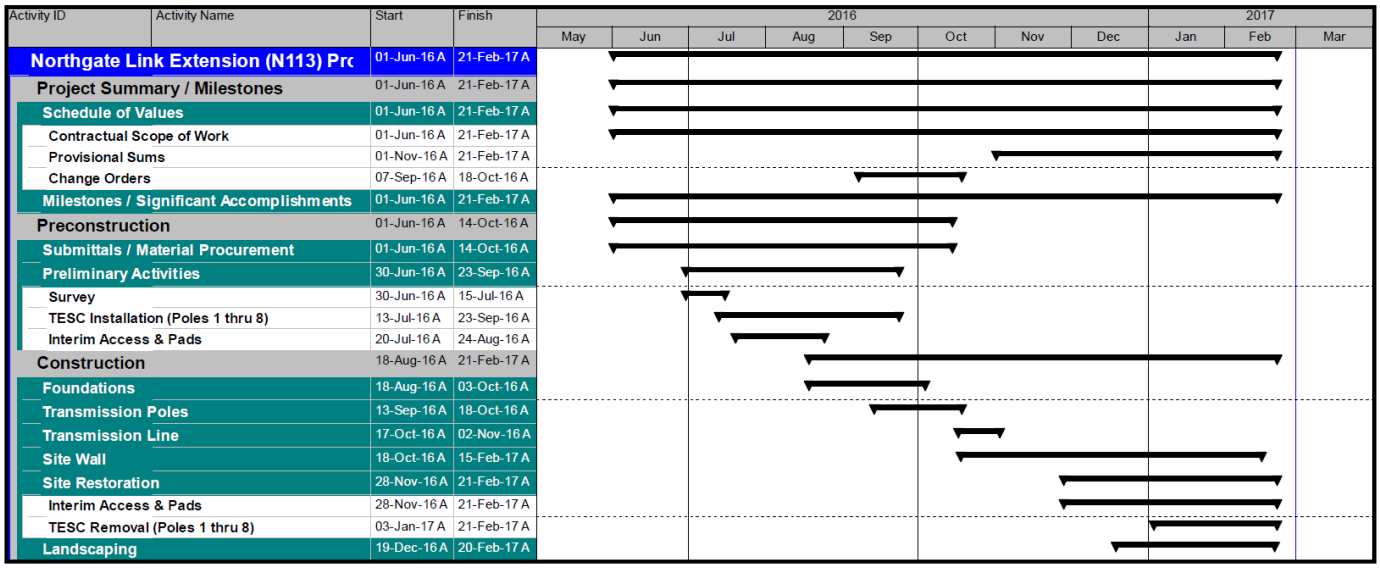
Current Progress

The N113 Contractor has completed the work. Substantial Completion has been was issued on February 21, 2017.

WSDOT has requested Sound Transit to design and install a protective fence along the MSE wall near Poles 1 and 2. ST has requested a proposal from Potelco.

Schedule Summary

The N113 contractor, Potelco, completed all work and achieved Substantial Completion on February 21. The team is currently working to complete the last punchlist item with WSDOT and begin the project close-out pending resolution of the late change order request by WSDOT.



Activity ID	Activity Name	Start	Finish
	Northgate Link Extension (N113) Prc	01-Jun-16 A	21-Feb-17 A
	Project Summary / Milestones	01-Jun-16 A	21-Feb-17 A
	Schedule of Values	01-Jun-16 A	21-Feb-17 A
	Contractual Scope of Work	01-Jun-16 A	21-Feb-17 A
	Provisional Sums	01-Nov-16 A	21-Feb-17 A
	Change Orders	07-Sep-16 A	18-Oct-16 A
	Milestones / Significant Accomplishments	01-Jun-16 A	21-Feb-17 A
	Preconstruction	01-Jun-16 A	14-Oct-16 A
	Submittals / Material Procurement	01-Jun-16 A	14-Oct-16 A
	Preliminary Activities	30-Jun-16 A	23-Sep-16 A
	Survey	30-Jun-16 A	15-Jul-16 A
	TESC Installation (Poles 1 thru 8)	13-Jul-16 A	23-Sep-16 A
	Interim Access & Pads	20-Jul-16 A	24-Aug-16 A
	Construction	18-Aug-16 A	21-Feb-17 A
	Foundations	18-Aug-16 A	03-Oct-16 A
	Transmission Poles	13-Sep-16 A	18-Oct-16 A
	Transmission Line	17-Oct-16 A	02-Nov-16 A
	Site Wall	18-Oct-16 A	15-Feb-17 A
	Site Restoration	28-Nov-16 A	21-Feb-17 A
	Interim Access & Pads	28-Nov-16 A	21-Feb-17 A
	TESC Removal (Poles 1 thru 8)	03-Jan-17 A	21-Feb-17 A
	Landscaping	19-Dec-16 A	20-Feb-17 A

Key Activities

Current Period

- Installed erosion control blankets in the jurisdictional ditch at Pole Site 1.
- Held landscape acceptance site review.
- Completed open punch list items.
- Reconciled Provisional Sums

Next Period

- Complete change orders.
- Construct Protective Fencing at MSE wall (change order requested by WSDOT)

Closely Monitored Issues

- WSDOT has expressed concern at the MSE wall indicating the lack of fence creates a potential fall hazard for maintenance crews at the top leading edge of the wall. ST currently reviewing the design and contract for determining merit.

Cost Summary

Present Financial Status	Amount
N113 Contractor - Potelco, Inc	
Original Contract Value	$2,811,926
Change Order Value	$ 63,500
Current Contract Value	$2,875,426
Total Actual Cost (Incurred to date)	$2,755,983
Financial Percent Complete:	95.8%,
Physical Percent Complete:	100.0%
Authorized Contingency	$281,194
Contingency Drawdown	$ 63,500
Contingency Index	4.43



Erosion blanket placed at Pole Site 1.

Contract N125 – TBM Tunnels

Current Progress

JCM is continuing work on the Cross Passages (CP), tunnel inverts, and troughs. Restoration of Husky Stadium Plaza is continuing.

Tunnel Finishes – Installation of the sidewalk, cable trough, drainage pipe, and conduit is continuing in both tunnels.

Cross Passage Construction—Work in progress at the end of this period included ground freezing and in-tunnel freezing at multiple cross passages; and waterproofing installation, crown excavation, invert rebar installation and finishing work at various other CPs along the alignment.

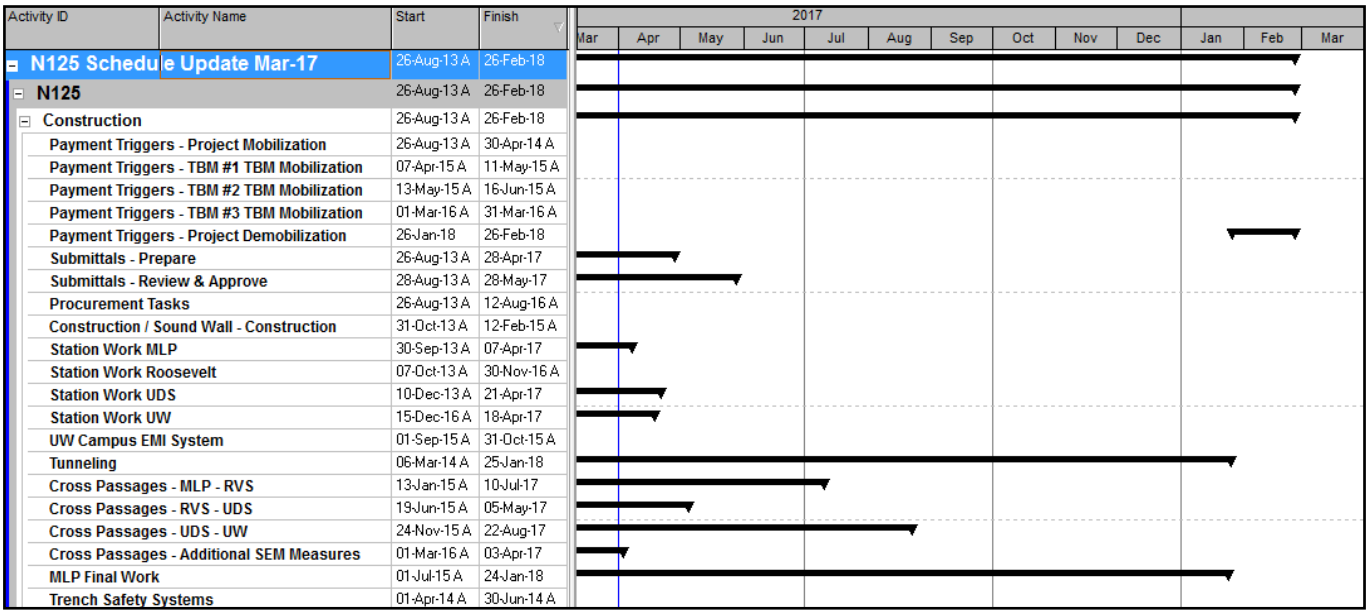
Schedule Summary

Sound Transit has received and reviewed the schedule update for March 2017.

The latest N125 schedule forecast a late turnover of the UDS site to the N140 contractor. ST is working with JCM to pull this date back to no later than July 1, 2017. JCM continues to forecast a completion of the tunnel finishes 158 days late. Milestone 6 - Substantial Completion has slipped from having 28 days of float to showing negative 14 days behind schedule.

The longest path through the schedule has shifted from the Cross Passages north of RVS to Cross Passage 21 and then into Cross Passage 38, following the crew responsible for grouting the spile holes and repairing the concrete. This in turn drives the block-outs, electrical, and mechanical in the southbound tunnel.

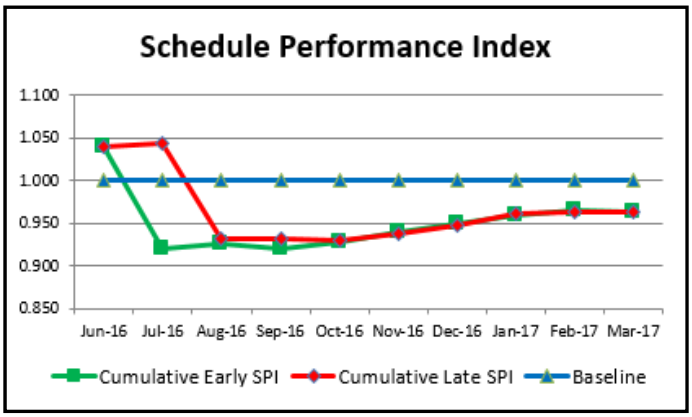
Due to the number of out-of-sequence logic ties in the schedule a recalculation of the schedule using progress override results in a longest path driven by the MSE Moment Slab Barrier and running through the MLP wall structures.



Activity ID	Activity Name	Start	Finish
N125 Schedule Update Mar-17		26-Aug-13 A	26-Feb-18
N125		26-Aug-13 A	26-Feb-18
Construction		26-Aug-13 A	26-Feb-18
Payment Triggers - Project Mobilization		26-Aug-13 A	30-Apr-14 A
Payment Triggers - TBM #1 TBM Mobilization		07-Apr-15 A	11-May-15 A
Payment Triggers - TBM #2 TBM Mobilization		13-May-15 A	16-Jun-15 A
Payment Triggers - TBM #3 TBM Mobilization		01-Mar-16 A	31-Mar-16 A
Payment Triggers - Project Demobilization		26-Jan-18	26-Feb-18
Submittals - Prepare		26-Aug-13 A	28-Apr-17
Submittals - Review & Approve		28-Aug-13 A	28-May-17
Procurement Tasks		26-Aug-13 A	12-Aug-16 A
Construction / Sound Wall - Construction		31-Oct-13 A	12-Feb-15 A
Station Work MLP		30-Sep-13 A	07-Apr-17
Station Work Roosevelt		07-Oct-13 A	30-Nov-16 A
Station Work UDS		10-Dec-13 A	21-Apr-17
Station Work UW		15-Dec-16 A	18-Apr-17
UW Campus EMI System		01-Sep-15 A	31-Oct-15 A
Tunneling		06-Mar-14 A	25-Jan-18
Cross Passages - MLP - RVS		13-Jan-15 A	10-Jul-17
Cross Passages - RVS - UDS		19-Jun-15 A	05-May-17
Cross Passages - UDS - UW		24-Nov-15 A	22-Aug-17
Cross Passages - Additional SEM Measures		01-Mar-16 A	03-Apr-17
MLP Final Work		01-Jul-15 A	24-Jan-18
Trench Safety Systems		01-Apr-14 A	30-Jun-14 A

Schedule Performance Index

The early SPI for this period remains 0.9, essentially unchanged from the 0.9 obtained in the previous period. The SPI continues to lag due to the pace of the cross passage work in comparison to the target dates for this work.



Key Activities

Current Period

- Continued formwork, rebar, and concrete pours at cross passages.
- Continued installation of fire standpipes in tunnels and cross passages.
- Continued forming and pouring cable troughs.
- Continued tunnel walkway pours.
- Continued installation of overhead electrical facilities in tunnel.
- Continued decommissioning dewatering wells, removal of discharge piping, and temporary street and surface restoration at cross passage locations.
- Continued UW Plaza restoration at Husky Stadium.
- Continued preparation of formwork for the MSE wall moment slab barrier.
- Continued preparation for TBM working slab removal and restoring permanent slab for site handover to N140 Contractor.

Next Period

- Continue cross passage finishes.
- Continue installation of fire standpipes and overhead electrical.
- Continue pouring cable troughs and tunnel walkways.
- Complete UW Plaza restoration.
- Begin preparation for MLP structural concrete work.
- Continue preparing UDS station box for handover to follow-on contractor.

Closely Monitored Issues

- Sound Transit has expressed concern to JCM regarding resourcing levels for ongoing concrete work south of UDS. Limited work is taking place in the NB tunnel. JCM is still devoting manpower to finishing the RVS-UDS tunnels and working in the SB tunnel north of RVS. These activities deprive the critical UDS-UWS section of much needed manpower.

Cost Summary

Present Financial Status	Amount
N125 Contractor - JCM Northlink	
Original Contract Value	$440,321,000
Change Order Value	$ 47,595,957
Current Contract Value	$487,916,957
Total Actual Cost (Incurred to date)	$451,585,407
Financial Percent Complete:	92%
Physical Percent Complete:	93%
Authorized Contingency	$66,048,150
Contingency Drawdown	$47,595,957
Contingency Index	1.29



Removing temporary walkway along eastside of MLP.

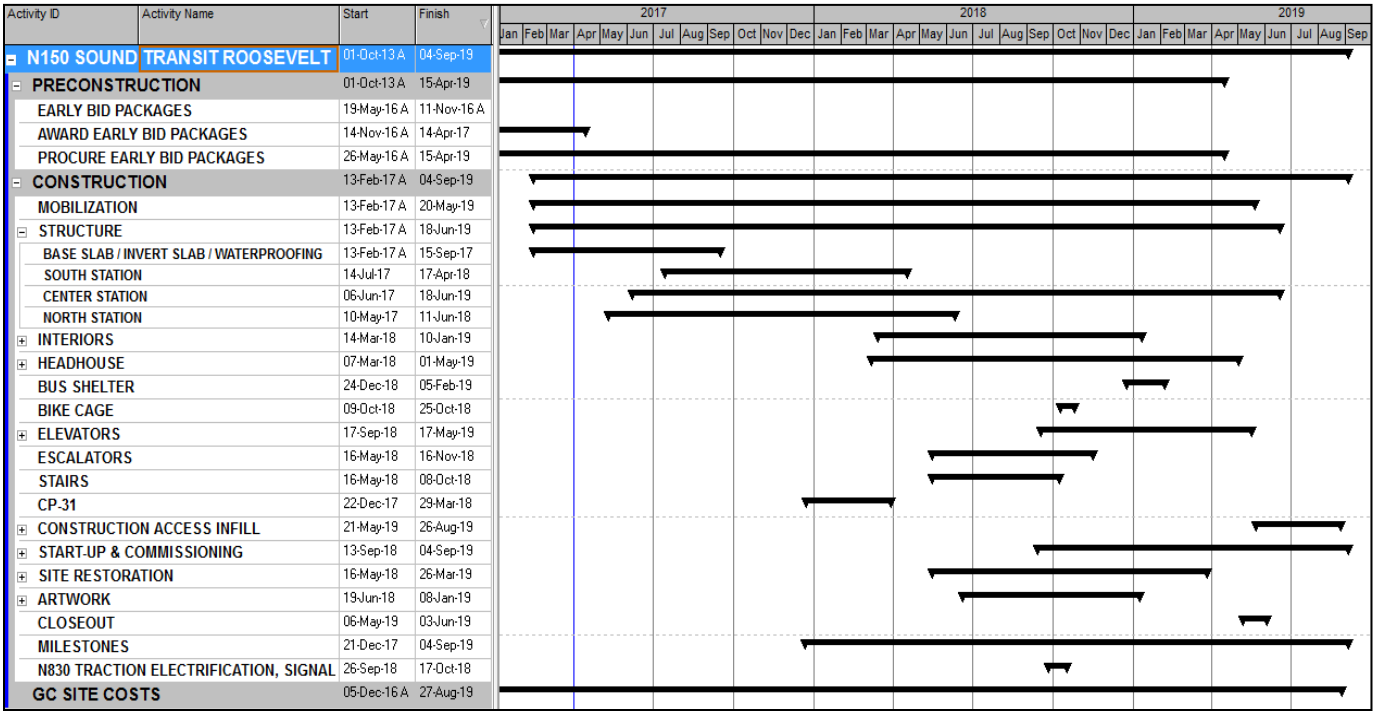
Contract N150 – Roosevelt Station Finishes

Current Progress

Hoffman Construction is shotcreting the station walls and installing the waterproofing system. Wall and invert slab rebar construction will begin in April.

Schedule

See latest schedule below.



Activity ID	Activity Name	Start	Finish
N150 SOUND	TRANSIT ROOSEVELT	01-Oct-13 A	04-Sep-19
PRECONSTRUCTION		01-Oct-13 A	15-Apr-19
EARLY BID PACKAGES		19-May-16 A	11-Nov-16 A
AWARD EARLY BID PACKAGES		14-Nov-16 A	14-Apr-17
PROCURE EARLY BID PACKAGES		26-May-16 A	15-Apr-19
CONSTRUCTION		13-Feb-17 A	04-Sep-19
MOBILIZATION		13-Feb-17 A	20-May-19
STRUCTURE		13-Feb-17 A	18-Jun-19
BASE SLAB / INVERT SLAB / WATERPROOFING		13-Feb-17 A	15-Sep-17
SOUTH STATION		14-Jul-17	17-Apr-18
CENTER STATION		06-Jun-17	18-Jun-19
NORTH STATION		10-May-17	11-Jun-18
INTERIORS		14-Mar-18	10-Jan-19
HEADHOUSE		07-Mar-18	01-May-19
BUS SHELTER		24-Dec-18	05-Feb-19
BIKE CAGE		09-Oct-18	25-Oct-18
ELEVATORS		17-Sep-18	17-May-19
ESCALATORS		16-May-18	16-Nov-18
STAIRS		16-May-18	08-Oct-18
CP-31		22-Dec-17	29-Mar-18
CONSTRUCTION ACCESS INFILL		21-May-19	26-Aug-19
START-UP & COMMISSIONING		13-Sep-18	04-Sep-19
SITE RESTORATION		16-May-18	26-Mar-19
ARTWORK		19-Jun-18	08-Jan-19
CLOSEOUT		06-May-19	03-Jun-19
MILESTONES		21-Dec-17	04-Sep-19
N830 TRACTION ELECTRIFICATION, SIGNAL		26-Sep-18	17-Oct-18
GC SITE COSTS		05-Dec-16 A	27-Aug-19

Key Activities

Current Period

- Continued preparing and processing submittals and RFIs.
- Completed installation of temporary man lift and scaffolding.
- Installed under-drain system drainage inlets.
- Completed installation of rebar curtains for shotcrete.
- Completed waterproofing system installation at north headwall and portions of the east and west walls.
- Completed topping slab at south end station box.
- Continued placement of shotcrete on east and west sides of station box.

Next Period

- Continue to prepare, review and respond to submittals and RFIs.
- Continue placement of shotcrete at east and west sides.
- Continue installation of waterproofing system on east and west side walls and invert slab.
- Continue installation of bottom rebar mats at invert slabs.

Closely Monitored Issues

- N150 continues to vet low bidder for the vertical transportation package including prior project references and ability to meet maintenance scope requirements.

Cost Summary

Present Financial Status	Amount
N150 Contractor - Hoffman Construction	
Original Contract Value	$152,291,184
Change Order Value	$ 26,359
Current Contract Value	$152,317,543
Total Actual Cost (Incurred to date)	$5,252,700
Financial Percent Complete:	3.4%
Physical Percent Complete:	5.0%
Authorized Contingency	$7,614,559
Contingency Drawdown	$26,359
Contingency Index	14.53



Waterproofing protection slab in SE corner station box.

Contract N160 – Northgate Station, Elevated Guideway, and Parking Garage

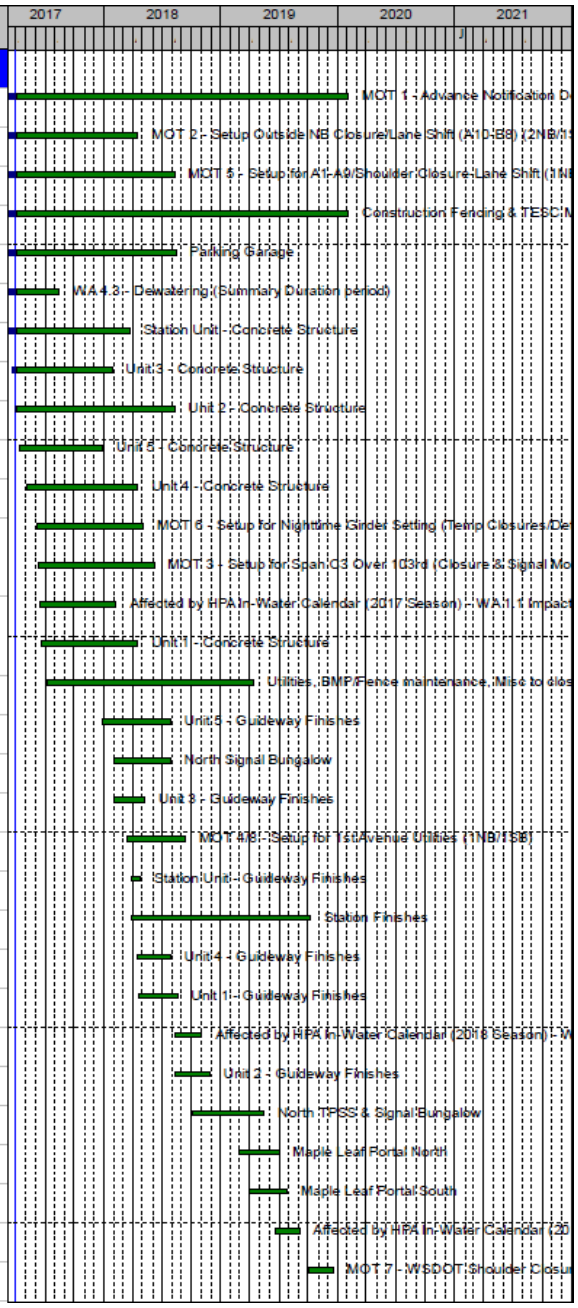
Current Progress

The N160 Contractor, is completing construction of the "C" series Station shafts.
At the Parking Garage site the Contractor is continuing installation of soldier piles and secant piles on the perimeter of the Garage.

Schedule

See latest schedule below.

Activity Name	OD	RD	Start	Finish
N160 - Northgate Station - 04/01/17	803	721	17-Oct-16 A	04-Feb-20
MOT 1 - Advance Notification Devices, Pedestrian Detour, Traffic Control, survey, sweeper	57	721	17-Oct-16 A	04-Feb-20
MOT 2 - Setup Outside NB Closure/Lane Shift (A10-B8) (2NB/1SB)	327	264	24-Oct-16 A	16-Apr-18
MOT 5 - Setup for A1-A9/Shoulder Closure-Lane Shift (1NB/1SB)	423	344	28-Oct-16 A	08-Aug-18
Construction Fencing & TESC Mainenance	5	721	07-Dec-16 A	04-Feb-20
Parking Garage	401	347	03-Jan-17 A	13-Aug-18
WA 4.3 - Dewatering (Summary Duration period)	128	93	04-Jan-17 A	11-Aug-17
Station Unit - Concrete Structure	297	248	12-Jan-17 A	23-Mar-18
Unit 3 - Concrete Structure	214	207	21-Mar-17 A	25-Jan-18
Unit 2 - Concrete Structure	346	344	30-Mar-17 A	08-Aug-18
Unit 5 - Concrete Structure	182	182	11-Apr-17	28-Dec-17
Unit 4 - Concrete Structure	240	240	05-May-17	16-Apr-18
MOT 6 - Setup for Nighttime Girder Setting (Temp Closures/Detours)	229	229	08-Jun-17	02-May-18
MOT 3 - Setup for Span C3 Over 103rd (Closure & Signal Mods)	252	252	09-Jun-17	06-Jun-18
Affected by HPA In-Water Calendar (2017 Season) - WA 1.1 Impacted Elevated Guideway; Bents A04-A07W	163	163	16-Jun-17	07-Feb-18
Unit 1 - Concrete Structure	211	211	20-Jun-17	18-Apr-18
Utilities, BMP/Fence maintenance, Misc to closeout activities	449	449	07-Jul-17	12-Apr-19
Unit 5 - Guideway Finishes	148	148	29-Dec-17	27-Jul-18
North Signal Bungalow	127	127	30-Jan-18	27-Jul-18
Unit 3 - Guideway Finishes	67	67	02-Feb-18	07-May-18
MOT 4/8 - Setup for 1st Avenue Utilities (1NB/1SB)	128	128	13-Mar-18	11-Sep-18
Station Unit - Guideway Finishes	23	23	26-Mar-18	25-Apr-18
Station Finishes	391	391	26-Mar-18	07-Oct-19
Unit 4 - Guideway Finishes	72	72	17-Apr-18	27-Jul-18
Unit 1 - Guideway Finishes	85	85	19-Apr-18	17-Aug-18
Affected by HPA In-Water Calendar (2018 Season) - WA 1.1 Impacted At-Grade Work; Utilities,	59	59	09-Aug-18	31-Oct-18
Unit 2 - Guideway Finishes	77	77	09-Aug-18	28-Nov-18
North TPSS & Signal Bungalow	157	157	02-Oct-18	14-May-19
Maple Leaf Portal North	87	87	01-Mar-19	02-Jul-19
Maple Leaf Portal South	83	83	01-Apr-19	26-Jul-19
Affected by HPA In-Water Calendar (2019 Season) - WA 1.1 Impacted At-Grade Work; JD & WC Habitat	53	53	20-Jun-19	04-Sep-19
MOT 7 - WSDOT Shoulder Closures for Landscaping	53	53	02-Oct-19	17-Dec-19



Key Activities

Current Period

- *Station:* Continued shaft activities; continued column stripping, rebar installation and post tensioning, and hillside removal of material.
- *Parking Garage:* Continued dewatering; installed soldier piles for excavation shoring; installed secant piles; and completed temporary stairway from NE 103rd into existing parking garage.
- *Offsite Fabrication:* Continued girder fabrication.

Next Period

- Continue shaft activities and continue formwork set up for columns.
- Continue soldier pile installation.
- Complete installation of secant piles.
- Continue mass excavation along west edge of future parking structure.

Closely Monitored Issues

- Interface coordination associated with the Lynnwood Link track re-alignment design changes continue.
- Elevator supply and installation subcontract remains under review.

Cost Summary

Present Financial Status	Amount
N160 Contractor - Absher Construction	
Original Contract Value	$174,000,000
Change Order Value	$ 636,327
Current Contract Value	$174,636,327
Total Actual Cost (Incurred to date)	$18,295,750
Financial Percent Complete:	10.5%
Physical Percent Complete:	10.1%
Authorized Contingency	$17,400,000
Contingency Drawdown	$ 636,327
Contingency Index	2.77



C2 columns with falsework supports installed.

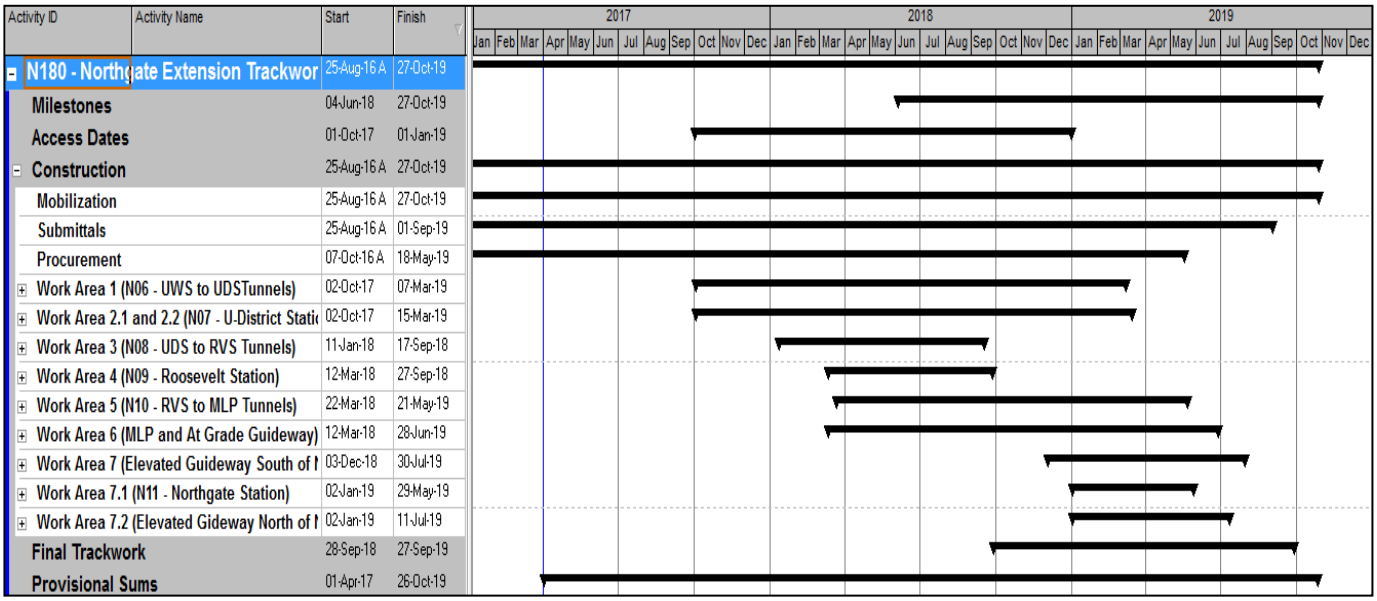
Contract N180 – Trackwork UWS to Northgate Station

Current Progress

The N180 Contractor continues to focus on procurement, submittals, and manufacturing of the pre-cast floating slabs. The Contractor is working with a British supplier for the Ultra-Straight Rail. The Pre-Cast yard in Woodland, WA is getting ready to cast the first test slabs. The rubber pad samples are being reviewed by ST and the Contractor.

Schedule Summary

Below is the latest schedule for N180.



Activity ID	Activity Name	Start	Finish
	N180 - Northgate Extension Trackwor	25-Aug-16 A	27-Oct-19
	Milestones	04-Jun-18	27-Oct-19
	Access Dates	01-Oct-17	01-Jan-19
	Construction	25-Aug-16 A	27-Oct-19
	Mobilization	25-Aug-16 A	27-Oct-19
	Submittals	25-Aug-16 A	01-Sep-19
	Procurement	07-Oct-16 A	18-May-19
	Work Area 1 (N06 - UWS to UDSTunnels)	02-Oct-17	07-Mar-19
	Work Area 2.1 and 2.2 (N07 - U-District Stati	02-Oct-17	15-Mar-19
	Work Area 3 (N08 - UDS to RVS Tunnels)	11-Jan-18	17-Sep-18
	Work Area 4 (N09 - Roosevelt Station)	12-Mar-18	27-Sep-18
	Work Area 5 (N10 - RVS to MLP Tunnels)	22-Mar-18	21-May-19
	Work Area 6 (MLP and At Grade Guideway)	12-Mar-18	28-Jun-19
	Work Area 7 (Elevated Guideway South of I	03-Dec-18	30-Jul-19
	Work Area 7.1 (N11 - Northgate Station)	02-Jan-19	29-May-19
	Work Area 7.2 (Elevated Gideway North of I	02-Jan-19	11-Jul-19
	Final Trackwork	28-Sep-18	27-Sep-19
	Provisional Sums	01-Apr-17	26-Oct-19

Key Activities

Current Period

- Finalized negotiations for procurement of Ultra-Straight Rail.
- All submittals required to begin fabrication of 5 Hz floating slabs and isolation pads have been submitted and approved.
- Columbia Precast Products (CPP) completed build out of QC office; continued build-out of 2nd batch plant, at Woodland, WA facility.
- Scougal continued fabrication of the 5Hz tapered side shim pads.

Next Period

- Execute Change Order for procurement of Ultra-Straight Rail.
- CPP to pour six floating slabs mock-ups for concrete final qualification testing.
- Helser will continue fabrication of the remaining steel forms for 5 Hz. 3' 10" slabs and begin fabrication of 7'-10" forms.

Closely Monitored Issues

- The thermite welding and electric flash-butt welding specifications do not account for welding Ultra-Straight Rail. ST Design is currently reviewing requirements.

Cost Summary

Present Financial Status	Amount
N180 Contractor - Stacy and Witbeck, Inc.	
Original Contract Value	$71,455,950
Change Order Value	$0
Current Contract Value	$71,455,950
Total Actual Cost (Incurred to date)	$4,727,639
Financial Percent Complete:	6.6%
Physical Percent Complete:	6.5%
Authorized Contingency	$10,718,393
Contingency Drawdown	$0
Contingency Index	N/A



Delivery of isolation pad rubber for floating slabs.

This page left blank intentionally.

Link Light Rail
Lynnwood Link Extension

Scope

Limits: North Seattle to Shoreline, Mountlake Terrace, and Lynnwood Transit Center

Alignment: Lynnwood Link extends light rail 8.5 miles to Snohomish County along I-5 from Northgate Mall in Seattle to the Lynnwood Transit Center, with four new light rail stations. The configuration includes at-grade, elevated, and retained cut/fill alignment.

Stations: NE 145th (Shoreline), NE185th (Shoreline), Mountlake Terrace Transit Center, Lynnwood Transit Center

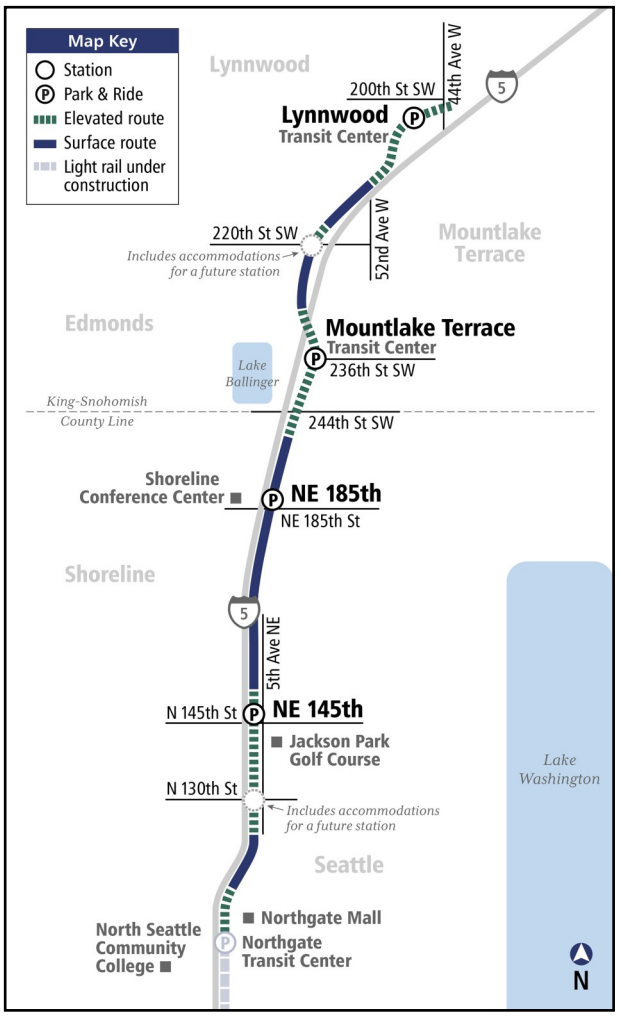
Systems: Signals, traction power, and communications (SCADA).

Budget: $488.4 Million Pre-baseline Budget; excludes Construction (Year of Expenditure Dollars)

Phase: Final Design

Const. Starts: 2018

Service: Late 2023



Map of Lynnwood Link Extension Alignment.

Key Project Activities

- L200 60% civil design submitted in February; reviews completed in March.
- L300 60% civil design submitted in March.
- L800 60% systems design submittal expected in June 2017.
- Anticipated NTP in July 2017 for the L800 Systems Construction Management Consultant.
- Continuing right-of-way acquisition and pre-construction planning.

Project Cost Summary

The Lynnwood Link project cost is summarized below in two cost categories. The first table summarizes costs in accordance with Sound Transit's budget Work Breakdown Structure (WBS). The second table summarizes costs in accordance with the FTA's Standard Cost Categories (SCC) format. The figures in both tables are shown in millions.

Commitment and Actuals under Construction Phase is related to Construction Permits for early demolition work associated with ROW Property Acquisitions.

Cost Summary by Phase

Project Phase	Current Adopted Budget	Commitment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget vs. EFC
Administration	$88.3	$14.4	$14.2	$88.3	$0.0
Preliminary Engineering	$42.0	$39.9	$39.0	$42.0	$0.0
Final Design	$111.5	$81.2	$27.2	$111.5	$0.0
Construction Services	$104.9	$18.0	$3.4	$104.9	$0.0
3rd Party Agreements	$17.4	$7.0	$1.5	$17.4	$0.0
Construction	$0.5	$0.1	$0.0	$0.5	$0.0
Light Rail Vehicles	$0.0	$0.0	$0.0	$0.0	$0.0
ROW	$123.8	$39.5	$33.7	$123.8	$0.0
Total	**$488.4**	**$200.2**	**$119.0**	**$488.4**	**$0.0**

Cost Summary by SCC

SCC Element	Current Adopted Budget	Commitment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget vs. EFC
10 Guideway & Track	$0.0	$0.0	$0.0	$0.0	$0.0
20 Stations	$0.0	$0.0	$0.0	$0.0	$0.0
30 Support Facilities	$0.0	$0.0	$0.0	$0.0	$0.0
40 Sitework & Special Conditions	$0.5	$0.1	$0.0	$0.5	$0.0
50 Systems	$0.0	$0.0	$0.0	$0.0	$0.0
Construction Subtotal (10 - 50)	**$0.5**	**$0.1**	**$0.0**	**$0.5**	**$0.0**
60 Row, Land	$123.8	$39.5	$33.7	$123.8	$0.0
80 Professional Services	$364.1	$160.5	$85.4	$364.1	$0.0
90 Unallocated Contingency	$0.0	$0.0	$0.0	$0.0	$0.0
Total (10 - 90)	**$488.4**	**$200.2**	**$119.0**	**$488.4**	**$0.0**

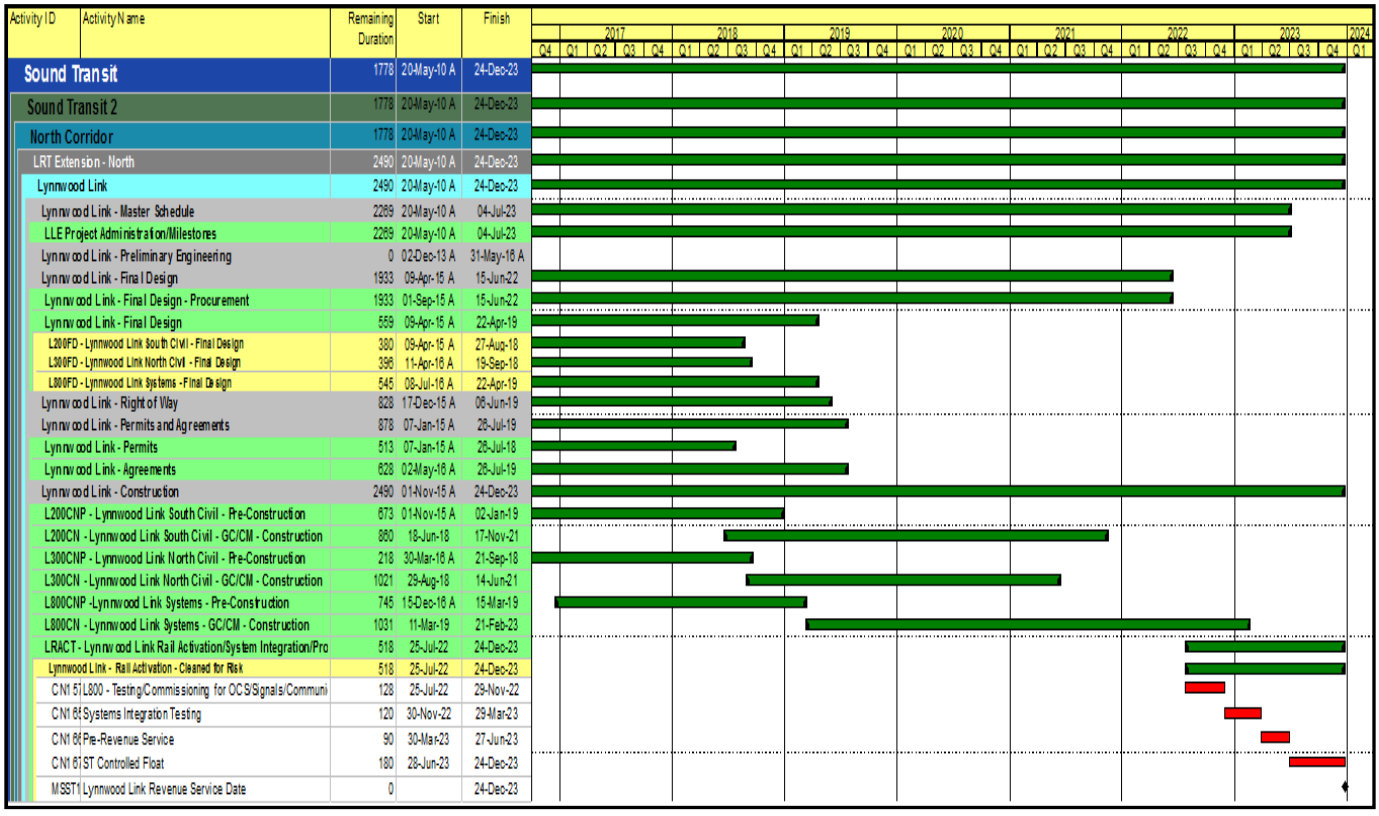
Risk Management

The Lynnwood Link Risk and Contingency Management Plan (RCMP) establishes a risk management and oversight process for identifying, assessing, and monitoring risks and develops risk reduction/mitigation plans. It provides a tool for the project team to proactively manage risks on the project. It is Sound Transit's responsibility for continuously monitoring project status and associated risks, taking mitigation actions for known risks identified on the risk register, and effectively responding in a timely manner to those risks that pose a significant impact to the project cost or schedule forecast.

The most recent Lynnwood Link Quarterly Risk Review Workshop was an FTA/PMOC led risk assessment workshop in February 2017.

Project Schedule

The project schedule is shown below. The schedule is at high risk; delays to permitting, right-of-way acquisition, or final design completion could push the anticipated revenue service date into 2024.



Activity ID	Activity Name	Remaining Duration	Start	Finish
	Sound Transit	1778	20-May-10 A	24-Dec-23
	Sound Transit 2	1778	20-May-10 A	24-Dec-23
	North Corridor	1778	20-May-10 A	24-Dec-23
	LRT Extension - North	2490	20-May-10 A	24-Dec-23
	Lynnwood Link	2490	20-May-10 A	24-Dec-23
	Lynnwood Link - Master Schedule	2269	20-May-10 A	04-Jul-23
	LLE Project Administration/Milestones	2269	20-May-10 A	04-Jul-23
	Lynnwood Link - Preliminary Engineering	0	02-Dec-13 A	31-May-16 A
	Lynnwood Link - Final Design	1933	09-Apr-15 A	15-Jun-22
	Lynnwood Link - Final Design - Procurement	1933	01-Sep-15 A	15-Jun-22
	Lynnwood Link - Final Design	559	09-Apr-15 A	22-Apr-19
	L200FD - Lynnwood Link South Civil - Final Design	380	09-Apr-15 A	27-Aug-18
	L300FD - Lynnwood Link North Civil - Final Design	396	11-Apr-16 A	19-Sep-18
	L800FD - Lynnwood Link Systems - Final Design	545	08-Jul-16 A	22-Apr-19
	Lynnwood Link - Right of Way	828	17-Dec-15 A	06-Jun-19
	Lynnwood Link - Permits and Agreements	878	07-Jan-15 A	26-Jul-19
	Lynnwood Link - Permits	513	07-Jan-15 A	26-Jul-18
	Lynnwood Link - Agreements	628	02-May-16 A	26-Jul-19
	Lynnwood Link - Construction	2490	01-Nov-15 A	24-Dec-23
	L200CNP - Lynnwood Link South Civil - Pre-Construction	673	01-Nov-15 A	02-Jan-19
	L200CN - Lynnwood Link South Civil - GC/CM - Construction	860	18-Jun-18	17-Nov-21
	L300CNP - Lynnwood Link North Civil - Pre-Construction	218	30-Mar-16 A	21-Sep-18
	L300CN - Lynnwood Link North Civil - GC/CM - Construction	1021	29-Aug-18	14-Jun-21
	L800CNP -Lynnwood Link Systems - Pre-Construction	745	15-Dec-16 A	15-Mar-19
	L800CN - Lynnwood Link Systems - GC/CM - Construction	1031	11-Mar-19	21-Feb-23
	LRACT - Lynnwood Link Rail Activation/System Integration/Pro	518	25-Jul-22	24-Dec-23
	Lynnwood Link - Rail Activation - Cleared for Risk	518	25-Jul-22	24-Dec-23
CN1 57	L800 - Testing/Commissioning for OCS/Signals/Communi	128	25-Jul-22	28-Nov-22
CN1 61	Systems Integration Testing	120	30-Nov-22	29-Mar-23
CN1 66	Pre-Revenue Service	90	30-Mar-23	27-Jun-23
CN1 67	ST Controlled Float	180	28-Jun-23	24-Dec-23
MSST1	Lynnwood Link Revenue Service Date	0		24-Dec-23

Right-of-Way

The Right-of Way project for Lynnwood Link involves the acquisition of a range of property interests which includes compensable (e.g. fee acquisitions, guideway easements, permanent and temporary construction easements) and non-compensable rights (e.g. rights of entry). These acquisitions will result in owner and tenant residential and commercial relocations. The right-of-way program status for this period is summarized in the following table.

Lynnwood Link Extension Property Acquisition Status								
Line Section	**Board Approved**	**Offers Made/In Negotiations**	**Signed Agreements**	**Possession and Use**	**Admin Settlement**	**Closings to date**	**Relocations Required**	**Parcels Vacated**
L200– South Segment	80	30	1	0	0	26	155	35
L300—North Segment	16	12	0	0	3	3	25	1
Total*	**96**	**42**	**1**	**0**	**3**	**29**	**180**	**36**

**Note– number of totals may differ from other reports due to the timing of reporting periods.*

L200 –Northgate to NE 200th Street Parcels

One additional parcel Board approved; 2 parcels closed; 17 more relocations required; 12 relocations vacated.

L300– NE 200th Street to Lynnwood Transit Center

Three admin settlements executed.

Sound Transit Board Actions

Board Action	Description	Date
R2017-10	*Authorizing the chief executive officer to acquire certain real property interests, including acquisition by condemnation to the extent authorized by law, and to reimburse eligible relocation and reestablishment expenses incurred by affected owners and tenants as necessary for the Lynnwood Link Extension.*	*Mar 23*

Community Outreach

- Distributed flyers to 27 residents of an apartment complex for upcoming geotech work.
- Began coordination for a community plant salvage event next period.
- Continued early planning for the 60% Station Design Open Houses in Shoreline, Mountlake Terrace and Lynnwood.
- Drafted and sent letters to property owners in Mountlake Terrace impacted by temporary parking plans and in Shoreline impacted by the Ridgecrest Park mitigation.
- Getting public input through the ST website for Station Naming—for top two names.
- Attended a public meeting on March 15, sponsored by the City of Shoreline to publicly introduce and solicit feedback on the proposal to locate a bike and pedestrian trail adjacent to the light rail alignment.

Civil Final Design Overview

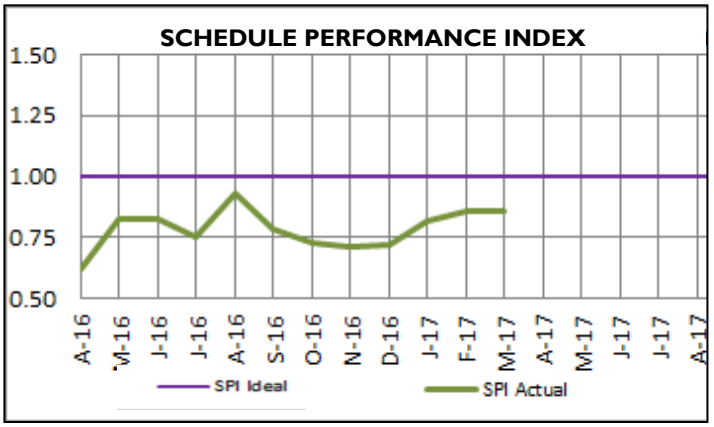
Sound Transit executed a professional services contract with HNTB Jacobs in April 2016 to perform Civil Final Design Services.

Civil Final Design Activities

- L200 60% design deliverable submitted in February; completed reviews in March.
- L200 60% design deliverable submitted in March.
- Continuing design refinements to 185th garage and Lynnwood Transit Center area, and advancing design to 90%.

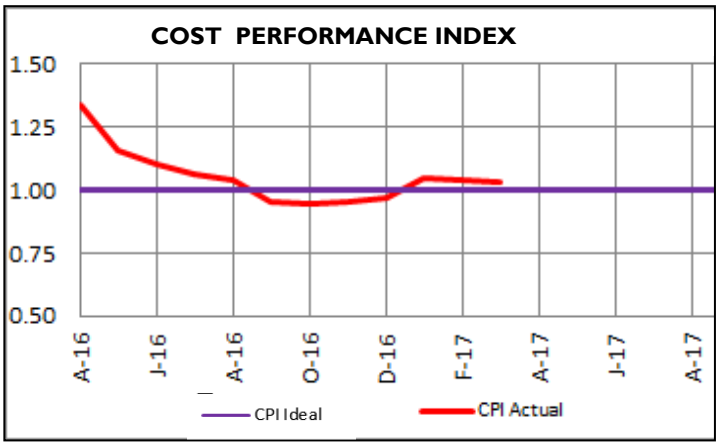
Civil Final Design SPI

The cumulative Schedule Performance Index (SPI) trends at 0.85 through March 2017, which means that cumulative work accomplished is less than work originally planned. Currently the Civil Final Design is behind schedule due to modifications to the design between 30% and 60%, and delay in finalizing configuration at 185th and Lynnwood Transit Center.



Civil Final Design Cost Performance

$25.9M of the total contract amount, 36%, has been spent through March 2017. The Civil Final Design percent complete is 37%, with an earned value of $26.7M. The cumulative Cost Performance Index (CPI) through March is 1.03 indicating that expenditures are slightly less than the earned value of work performed.



Final Design Performance	Cumulative To-date
Amount Invoiced	$25.9M
% Spent	36%
Earned Value	$26.7M
% Complete	37%
SPI	**0.85**
CPI	**1.03**

Systems Final Design Overview

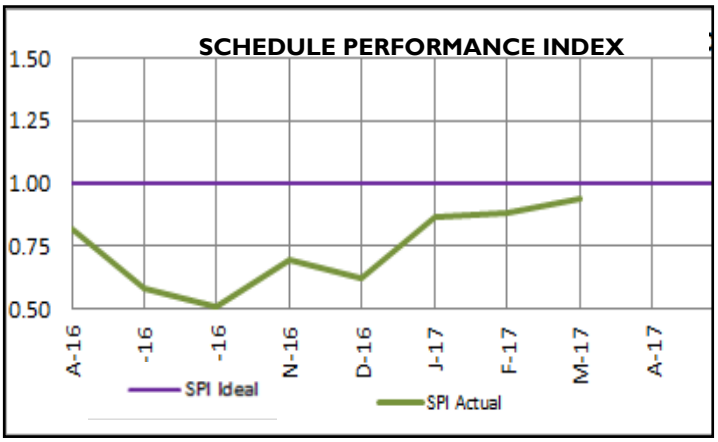
Sound Transit executed a professional services contract with LTK Engineering in July 2016 to perform Systems Final Design Services.

Systems Final Design Activities

- Working on 60% design deliverable based on civil 60% design.
- L800 60% final design submitted expected in June 2017.

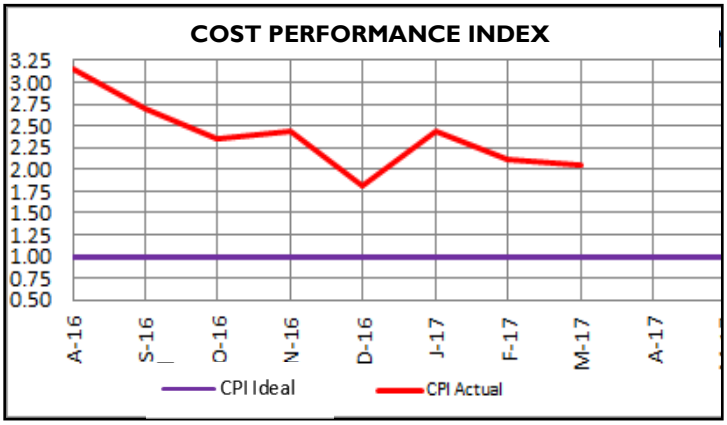
Systems Final Design SPI

The cumulative Schedule Performance Index (SPI) trends at 0.94 through March 2017, which means that cumulative work accomplished is slightly less than work planned. The monthly SPI for March was 1.23, indicating recovery to the original plan.



Systems Final Design Cost Performance

$1.2M of the total contract amount, 13%, has been spent through March 2017. The Systems Final Design percent complete is 26%, with an earned value of $2.4M. The cumulative Cost Performance Index (CPI) through March is 2.04 indicating that expenditures are significantly lower than the earned value of work performed.



Final Design Performance	Cumulative To Date
Amount Invoiced	$1.2M
% Spent	13%
Earned Value	$2.4M
% Complete	26%
SPI	0.94
CPI	2.04

Civil Construction Management Overview

Sound Transit executed a professional services contract with PGH Wong in May 2016 to perform Civil Construction Management (CMC) Services

Civil Construction Management Activities

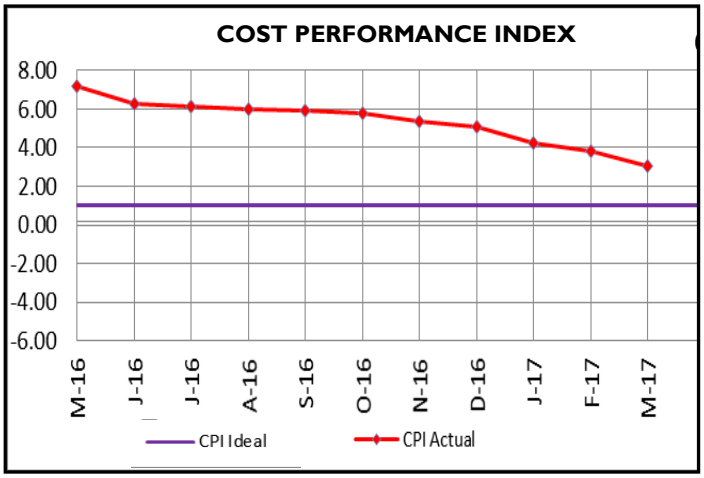
- Coordinating meetings with design, GC/CM, and ST teams.
- Coordination on various preconstruction activities including cost estimates, schedule, and constructability.
- Coordinating GC/CM review of 60% civil plans.

Civil Construction Management SPI

Work by the CMC is on a level-of-effort basis to support Sound Transit during the final design/pre-construction period. Earned value for level-of-effort is calculated as equal to the planned value, so SPI = 1.00.

Civil CM Cost Performance

$1.0M of the total contract amount, 14%, is spent through March 2017, with an earned value of $3.2 M. The cumulative Cost Performance Index (CPI) through March is 3.03; indicating significantly lower costs than expected on the contract. The consultant assumed more support staff would be required early in the schedule and that has not been the case, leading to a high CPI. Staffing has increased with the 60% design deliverables, and CPI is trending closer to an expected value of 1.0.



Civil CM Performance	Cumulative To-date
Amount Invoiced	$1.0M
% Spent	14%
Earned Value	$3.2M
CPI	**3.03**

GC/CM Pre-Con Overview - L200 Northgate to NE 200th Street

Sound Transit executed a professional services contract with Stacy & Witbeck Kiewit Hoffman in June 2016 to perform GC/CM Pre-Construction Services for the L200 segment.

L200 GC/CM Pre-Con Activities

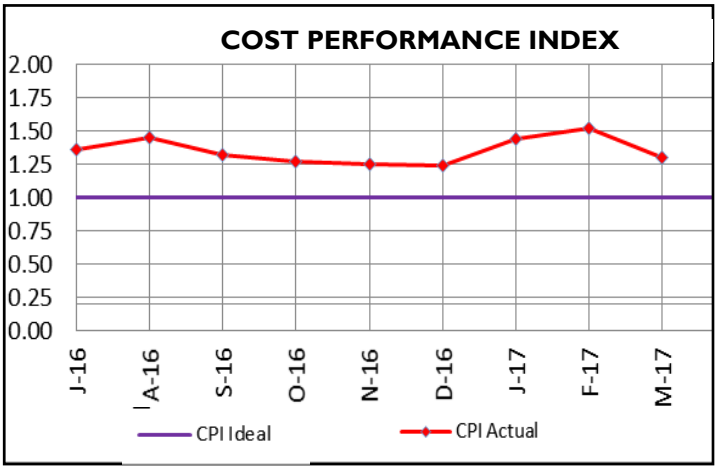
- Coordinating meetings with design, CM, and ST teams.
- Progress on various preconstruction activities including cost estimates, schedule, and constructability.
- Review of 60% civil plans.

L200 GC/CM Pre-Con Schedule Performance

Driving the L200 GC/CM Contract is the LLE Final Design contract; GC/CM Pre-Construction work is on a level-of-effort basis to support Sound Transit during the final design/pre-construction period. Earned value for level-of-effort is calculated as equal to the planned value, so SPI = 1.00.

L200 GC/CM Pre-Con Cost Performance

$1.7M of the total contract amount, 32%, is spent through March 2017, with an earned value of $2.2M. The cumulative Cost Performance Index (CPI) through March is 1.30; indicating actual costs are lower than planned.



L200 GC/CM Performance	Cumulative To-date
Amount Invoiced	$1.7M
% Spent	32%
Earned Value	$2.2M
CPI	**1.30**

GC/CM Pre-Con Overview - L300 NE 200th St. to Lynnwood Transit Center

Sound Transit executed a professional services contract with Skanska in October 2016 to perform GC/CM Pre-Construction Services for the L300 segment.

L300 GC/CM Pre-Con Activities

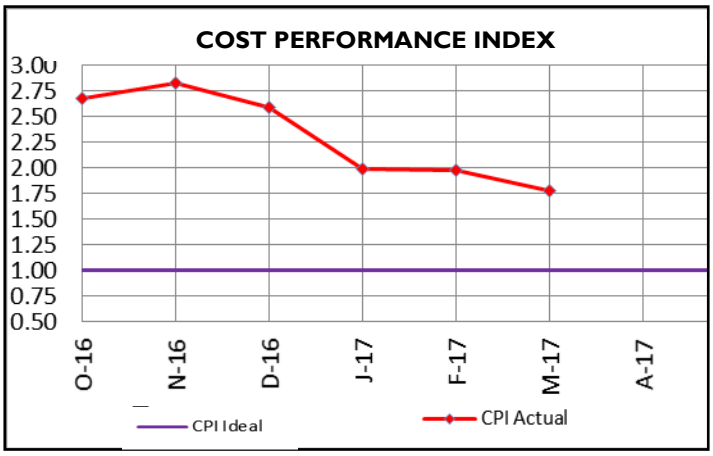
- Coordinating meetings with design, CM, and ST teams.
- Progress on various preconstruction activities including cost estimates, schedule, and constructability.
- Review of 60% civil plans.

L300 GC/CM Pre-Con Schedule Performance

Driving the L300 GC/CM Contract is the LLE Final Design contract; GC/CM Pre-Construction work is on a level-of-effort basis to support Sound Transit during the final design/pre-construction period. Earned value for level-of-effort is calculated as equal to the planned value, so SPI = 1.00.

L300 GC/CM Pre-Con Cost Performance

$1.0M of the total contract amount, 19%, has been spent through March 2017, with an earned value of $1.8M. The cumulative Cost Performance Index (CPI) through March is 1.78; indicating actual costs are significantly lower than planned.



L300 GC/CM Performance	Cumulative To-date
Amount Invoiced	$1.0M
% Spent	19%
Earned Value	$1.8M
CPI	**1.78**

This page intentionally left blank.

Scope: The I-90 Two-Way Transit and HOV Operations (Stage 3) project will provide approximately four miles of the HOV lanes in each direction of the outer roadway between 80th Avenue South on Mercer Island and Rainier Avenue in Seattle. The project includes upgrades and retrofits to the fire, life, safety, and other systems in the Mount Baker and Mercer Island/ First Hill tunnels. The project also includes remaining dowel bar retrofit work on the west side of Lake Washington, the installation of screening on the shared–use pathway on the I-90 floating bridge. Following completion of this project, the center roadway will be closed to allow the construction and operation of East Link Extension.



The project, when completed, will enable 24/7 HOV access on I-90 between Seattle, Mercer Island and Bellevue.

Budget: $225.6 Million

Phase: Construction

Construction Start: January 2015

Construction Completes: June 2017

Major Contracts

	Scope	Agreement/Contract
WSDOT	Final Design	$ 20,942,000
WSDOT	CM Services	$ 16,358,000
IMCO	Construction	$122,021,000

Key Project Activities

- *Work outside the tunnels* – Ongoing work on Island Crest Way interchange: prep grade at Island Crest Way HOV exit, pour barrier; splice fiber for intelligent traffic systems; install emergency lighting switch gear; install exterior fuel piping to generators.
- *Mercer Island Tunnel (MIT)* – Start-up and test supply fan 9, continue generator and switch gear commissioning. Westbound live fire tests conducted in March; Eastbound live fire tests scheduled for weekend of April 8th.
- *Mount Baker Ridge Tunnel (MBRT)* – Continuation of work on preliminary punch list items; resolution on exhaust fan cavitation issue; Eastbound live fire tests scheduled for weekend of April 8th.
- *SCADA* – ITS system coordination and programming; continue work on punch list items and programming issues discovered during live fire tests.
- *Simplex*– Continued work on SCADA interface; testing new software; added system tamper switches; installed battery at MIT/MBRT and connection of flame detection cameras & linear heat detectors; powered up LH panels in MIT control room; existing operation test at MBRT; punch list review, training in progress.

Closely Monitored Issues

- *SCADA/Simplex implementation* – Challenges with network programming tasks for Phase 3 commissioning and Simplex / SCADA video interface and integration; weekly meeting with SCADA team & WSDOT commissioning expert to expedite implementation progress.
- *High Temperature Fabric* – Material has been accepted, could be a cost issue; researching what information was available at time of bid (ongoing).
- *Weekend closure scheduling* – Proactively working to prioritize Spring/Summer closures and schedule within limitations of other major area projects.

Project Cost Summary

Total Project Cost expended this period is $5.5M. The following tables summarize the cost information for the I-90 Two - way Transit and HOV Operations (Stage 3) project. Tables in millions.

Cost Summary by Phase

Project Elements by Phase	Baseline Budget	Current Adopted Budget	Commit-ment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget vs. EFC
Administration	$3.6	$3.6	$1.6	$1.6	$3.6	$0.0
Preliminary Engineering	$1.5	$1.5	$1.5	$1.5	$1.5	$0.0
Final Design	$22.1	$22.1	$18.6	$17.9	$22.1	$0.0
Construction Services	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0
3rd Party Services	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0
Construction	$198.3	$198.3	$170.9	$148.4	$198.3	$0.0
Total	**$225.6**	**$225.6**	**$192.6**	**$169.5**	**$225.6**	**$0.0**

Cost Summary by SCC

Project Elements by SCC	Baseline Budget	Current Adopted Budget	Commitment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget vs. EFC
40 Sitework & Special Condi-tions	$180.6	$180.6	$170.8	$148.4	$180.6	$0.0
80 Professional Services	$26.3	$26.3	$21.8	$21.1	$26.3	$0.0
90 Unallocated Contingency	$18.7	$18.7	$0.0	$0.0	$18.7	$0.0
Project Total (SCC 10-90)	**$225.6**	**$225.6**	**$192.6**	**$169.5**	**$225.6**	**$0.0**

Cost Contingency Management

In the table below, the Allocated Contingency includes both WSDOT-controlled construction contingency & the total ST-controlled allocated contingencies. Changes to the contingency level since last period are due to WSDOT executing change orders to the construction contract.

Contingency Status	Baseline		Current	
	Amount	% of Total	Amount	% of Work Remaining
Design Allowance	$.0.0	0.0%	$0.0	0.0%
Allocated Contingency	$17.0	7.5%	$10.8	19.1%
Unallocated Contingency	$18.7	8.3%	$18.7	33.3%
Total	**$35.7**	**15.8%**	**$29.5**	**52.5%**

Note: Table in millions. Contract does not carry Design Allowance.
Contingency % of Work Remaining based on Contingency Remaining as a % Budget Remaining.
Allocated contingency includes WSDOT-controlled contingency & ST-controlled Contingency.

Project Schedule

WSDOT has executed a change order implementing a schedule recovery plan that preserves the commitment to turn over the center roadway to Sound Transit in June. Although Physical Completion is now anticipated for July 13, 2017, the Center Roadway Turnover is scheduled for June 14th; this 2-week adjustment to the access date of the Center Roadway will not impact the E130 Construction contractor's anticipated completion of the East Link I-90 Segment. The Substantial Completion date of April 23 is tentative; negotiations are currently under way between WSDOT and the Prime Contractor to finalize a date.

The remaining Project Milestones for construction are indicated below:

Contract	Final Tunnel Commissioning Complete		Substantial Completion		Center Roadway Turnover		Physical Completion	
	Baseline	Forecast / Actual	Baseline	Forecast / Actual	Baseline	Forecast / Actual	Baseline	Forecast / Actual
I-90 Two-Way Transit & HOV Operations Stage 3	2/16/2017	7/13/2017	2/20/2017	4/23/2017	5/31/2017	6/14/2017	5/31/2017	7/13/2017

Changes from previous update are indicated in **RED**; A indicates Actual

This page left intentionally blank.

Scope

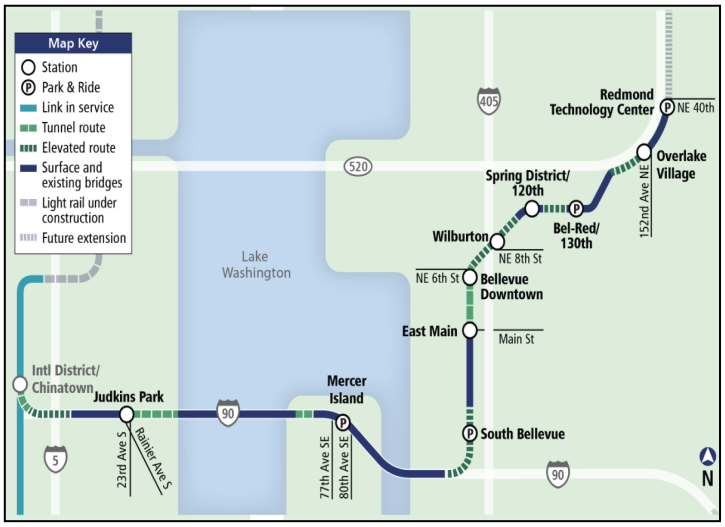
Limits/Alignment: Fourteen-mile extension from Downtown Seattle to Mercer Island, Bellevue and the Overlake area of Redmond. The extension includes 10 stations along the alignment that includes at-grade, elevated, bridges, new and retrofitted tunnels.

Stations: Judkins Park (formerly Rainier), Mercer Island, South Bellevue, East Main, Bellevue Downtown, Wilburton (formerly Hospital), Spring District/120th, Bel-Red/130th, Overlake Village and Redmond Technology Center (RTC).

Systems: Signals, traction electrification, and communications (SCADA).

Budget: $3.677 Billion Year of Expenditure Dollars

Schedule: Revenue Service - June 2023



Map of East Link Extension Alignment.

Key Project Activities

Seattle to South Bellevue

- Reviewed VA/CR alternatives developed from International District Value Assessment workshops.
- I-90 ASL being reviewed by WSDOT and FHWA.

South Bellevue to Redmond

- *E320 South Bellevue:* Special Use Agreement is being processed. I-90 ASL being reviewed by WSDOT and FHWA. Preparations for the closure of the South Bellevue Park-Ride on May 30.
- *E330 Downtown Bellevue Tunnel:* Refined spilling installation criteria based on ground stability. Set up of concrete batch plant and preparing for final shotcrete application in Stage 3 of the Skyline Bldg. garage.
- *E335 Downtown Bellevue to Spring Dist.:* Continued working with Wright Runstad and City of Bellevue to address project interface issues; I-405 TCAL undergoing final review by WSDOT.
- *E340 Bel-Red:* NTP issued to Max J. Kuney; ST and Kuney working on schedule for utility disconnects. City of Bellevue issued permits for Verizon/Integra relocation work at NE 20th/136th NE
- *E360 SR520 to Redmond Technology Center:* Development of 60% and 100% design packages. Preparations for May 1 closure of the parking facilities at OTC. Received Governance team approval from Redmond to include Artwork in the OVS pedestrian bridge.

Closely Monitored Issues

- Meeting Civil and Systems construction procurement schedules
- Timely property acquisitions including execution of Air Space Leases.
- Timely submissions and issuance of permits (City of Bellevue and City of Redmond) and resolution of appeals.
- Resolutions of issues related to Mercer Island.
- E330 Downtown Bellevue Tunnel slower than anticipated tunneling production rates .

Project Cost Summary

The East Link project cost is summarized below in two cost categories. In the first table, the cost is summarized in accordance with Sound Transit's Work Breakdown Structure (WBS). The second table summarizes the project in accordance with the FTA's Standard Cost Category (SCC) format. The figures in both tables are shown in millions. The Estimated Final Cost (EFC) for this project is $3,677.2M in year of expenditure dollars. Approximately $16.8M was incurred this period. The ($2.4M) Baseline Budget vs. EFC in construction SCCs was due to a slight difference in actual cost allocation of E360 DB contract compared to Baseline Budget. This was offset by a positive variance in non-construction SCCs.

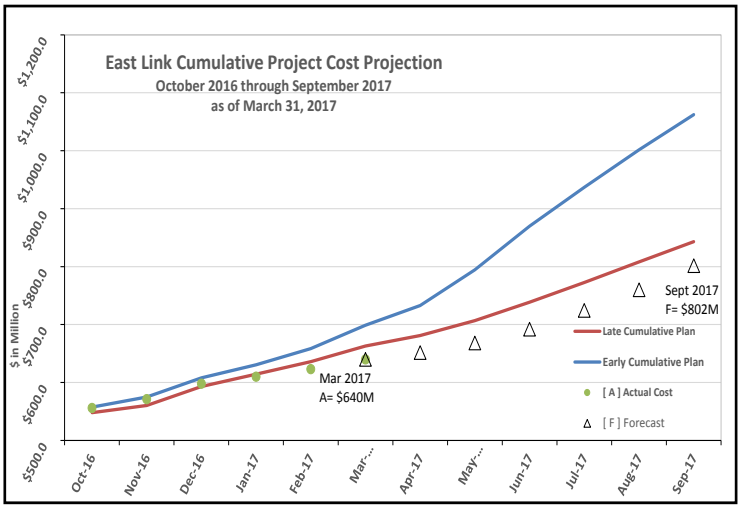
Cost Summary by Phase

Project Phase	Baseline Budget	Current Budget	Commitment to	Incurred to Date	Estimated Final Cost	Current Budget
Administration	186.2	186.2	$56.8	$56.1	186.2	$0.0
Preliminary Engineering	55.9	55.9	$54.8	$54.5	55.9	$0.0
Final Design	283.0	283.0	$231.5	$190.5	283.0	$0.0
Construction Services	257.5	257.5	$137.5	$26.2	257.5	$0.0
3rd Party Agreements	52.2	52.2	$34.7	$10.0	52.1	$0.0
Construction	2,544.3	2,544.3	$805.1	$92.8	2,544.3	$0.0
Light Rail Vehicles	0.0	0.0	$0.0	$0.0	0.0	$0.0
ROW	298.2	298.2	$215.5	$209.7	298.2	$0.0
Total	**3,677.2**	**3,677.2**	**$1,536.0**	**$639.7**	**3,677.2**	**$0**

Cost Summary by SCC

SCC Element	Baseline Budget	Current Budget	Commitment to Date	Incurred to Date	Estimated Final Cost (EFC)	Current Budget vs. EFC
10 Guideway & Track	$744.6	$747.6	$328.6	$42.5	857.7	($110.2)
20 Stations	$397.7	$397.7	$174.7	$8.5	439.4	($41.7)
30 Support Facilities	$0.0	$0.0	$0.0	$0.0	0.0	$0.0
40 Sitework & Special Conditions	$808.5	$806.0	$243.7	$37.6	658.8	$147.2
50 Systems	$353.8	$353.8	$36.9	$0.9	359.4	($5.6)
Construction Subtotal (10 - 50)	**$2,304.6**	**$2,305.1**	**$783.9**	**$89.5**	**2,315.5**	**($10.4)**
60 Row, Land	$288.5	$288.5	$215.5	$209.7	288.5	($0.0)
70 Vehicles (non-revenue)	$2.8	$2.8	$0.0	$0.0	2.8	$0.0
80 Professional Services	$898.4	$899.7	$536.6	$340.5	889.2	$10.6
90 Unallocated Contingency	$182.9	$181.0	$0.0	$0.0	181.2	($0.2)
Total (10 - 90)	**$3,677.2**	**$3,677.2**	**$1,536.0**	**$639.7**	**3,677.2**	**$0.0**

The cost projection on the right shows East Link cost is experiencing some tapering because planned construction activities have been slowed. The forecast continues to trend a near term lag and anticipates to pick up by Summer. East Link's total cost incurred to date is a little under $640M of which about $92.7M were within the Construction Phase. The projects continues to transition from into Construction. E330 tunneling; E360 design build progressing to 100% submittals. E320 and E340 commenced mobilization. E130 contract awarded but progress is uncertain due to Mercer Island litigation. E335 contract awarded, contracts awaits execution. E750/ N830 MACC negotiated, combined system contract with Northgate Link Extension.



Risk and Contingency Management

Risk Management

The RCMP established a risk management and oversight process for identifying, assessing, and monitoring risks and develops risk reduction/mitigation plans. It provides a tool for the project team to proactively manage risks on the project. Sound Transit (ST) continuously monitor project status and associated risks, taking mitigation actions for known risks identified on the risk register, and effectively responding in a timely manner to those risks that pose a significant impact to the project cost or schedule forecast.

This period, Sound Transit began the quantitative risks update and risk assessment workshops. ST included participation of construction management consultants and construction contractors who are already under contract (E130, E330, E335, E360 and E750) in its risks management program.

Contingency Status

The project's baseline budget was approved by the Sound Transit Board in April 2015, contained a total of $795.9M contingency. This period, total contingency decreased by $320.8M primarily due to the procurement of E130 I-90 Heavy Civil GC/CM. This procurement effectively reduced the current total contingency balance to $506.3M, which is slightly above the planned contingency drawdown.

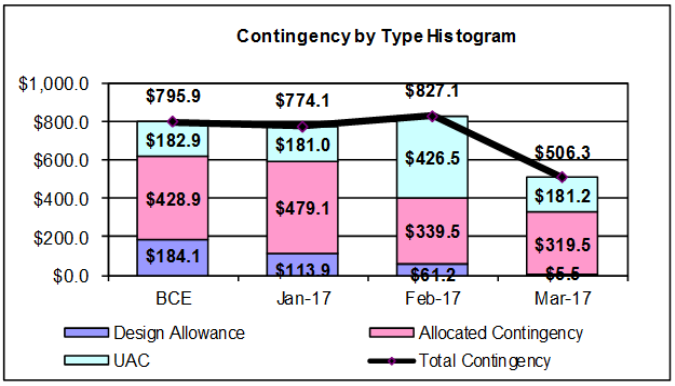
Design Allowance (DA) was set up to account for unquantified scope at the time of the cost estimate. This period, DA balance was drawn down by $55.6M associated with contracts E130 to a new balance of $5.5M. DA balance is expected to be fully drawn when all construction scope are executed.

Allocated Contingency (AC) is used to address risks, complexity and unforeseen conditions during the execution of work, including procurement. To finalize the procurement of E130, approximately $20M was drawn from its budgeted AC. The total AC remain relatively stable at $319.5M.

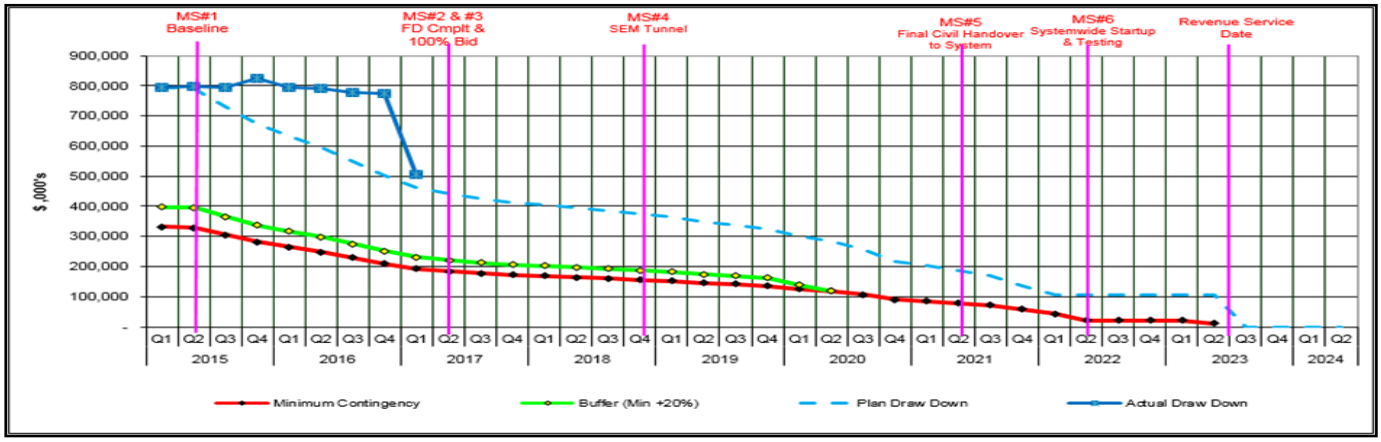
Unallocated Contingency (UAC) is used to address general project-wide cost risks and uncertainties. This period, UAC balance decreased by $245.2M to $181.2M due to a draw down in anticipation of E130 budget shortfall. However, the balance UAC continues to be stable compared to the initial balance of $181M at Baseline Cost Estimate (BCE).

Contingency Status	BCE		Current Status	
	Amount	% of Total Budget	Remaining Amount	% of Work Remaining
Design Allowance	$184.1	5.0%	$5.5	0.2%
Allocated Contingency	$428.9	11.7%	$319.5	10.5%
Unallocated Contingency	$182.9	5.0%	$181.2	6.0%
Total:	**$795.9**	**21.6%**	**$506.3**	**16.7%**

Dollar figures on this page are displayed in millions.



Contingency Drawdown



Project Schedule

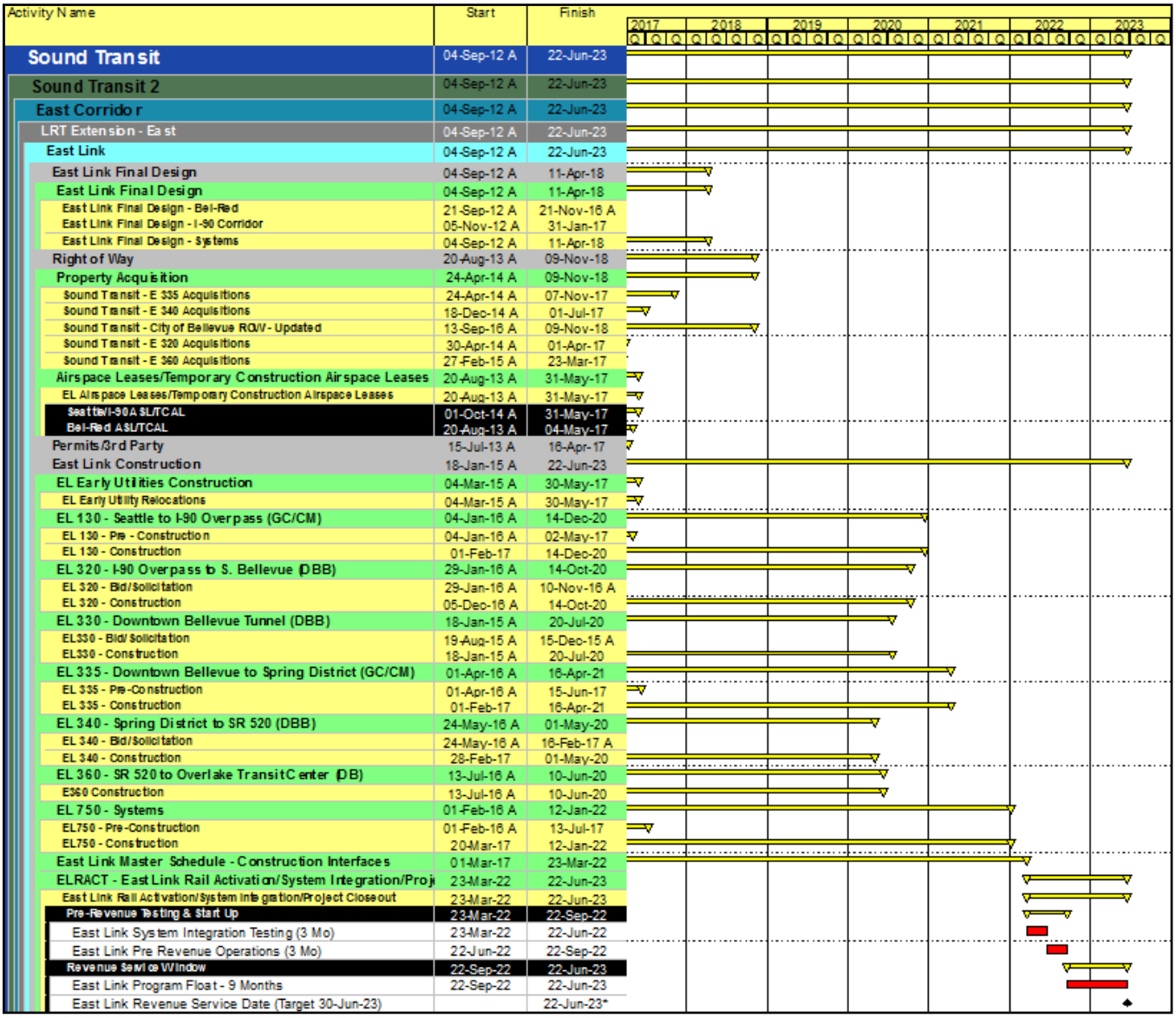
All GC/CM contract negotiations have concluded and all contracts are currently being executed. NTP for E335 Downtown Bellevue to Spring District is anticipated in April, while the stations work is being re-packaged for future procurement. NTP for E130 I-90 is anticipated by mid-May. The Contractor is already proceeding with a limited scope to prepare critical submittals and work plans and to design and fabricate steel for the floating bridge post-tensioning retrofit. NTP for E750 Systems is expected in June 2017.

NTP was issued for E340 Bel-Red. Contractor is preparing work plans and key submittals; construction activity is anticipated to begin mid-May. E320 South Bellevue continued preparing key submittals and work plans. Construction is anticipated to start in late May.

E330 Downtown Bellevue Tunnel Contractor continued with tunnel excavation. Work continued on the Skyline building retrofit .

E360 SR 520 to RTC design is progressing toward 100%. The first construction tasks are anticipated in late Spring 2017 with the removal of trees. Civil construction is scheduled to start in earnest later in the Summer.

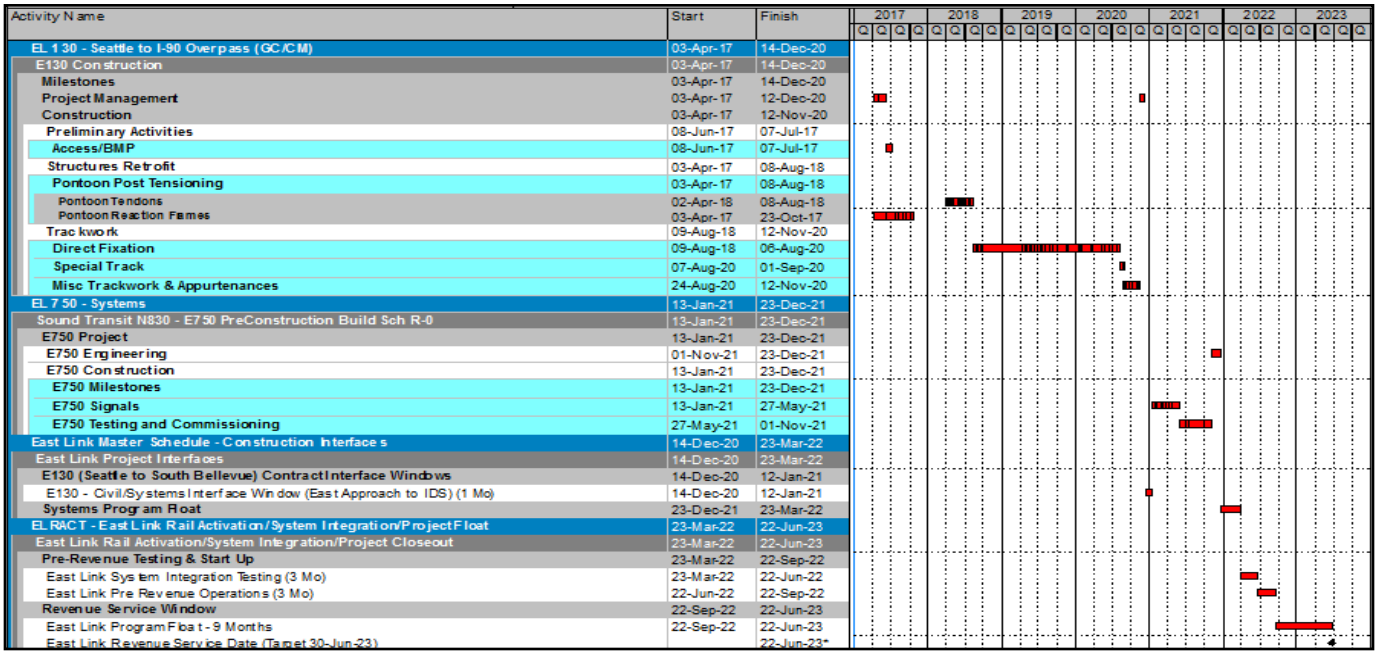
All contracts are forecast to complete on or before target. Revenue Service is forecasted on June 30, 2023.



Activity Name	Start	Finish
Sound Transit	04-Sep-12 A	22-Jun-23
Sound Transit 2	04-Sep-12 A	22-Jun-23
East Corridor	04-Sep-12 A	22-Jun-23
LRT Extension - East	04-Sep-12 A	22-Jun-23
East Link	04-Sep-12 A	22-Jun-23
East Link Final Design	04-Sep-12 A	11-Apr-18
East Link Final Design	04-Sep-12 A	11-Apr-18
East Link Final Design - Bel-Red	21-Sep-12 A	21-Nov-16 A
East Link Final Design - I-90 Corridor	05-Nov-12 A	31-Jan-17
East Link Final Design - Systems	04-Sep-12 A	11-Apr-18
Right of Way	20-Aug-13 A	09-Nov-18
Property Acquisition	24-Apr-14 A	09-Nov-18
Sound Transit - E 335 Acquisitions	24-Apr-14 A	07-Nov-17
Sound Transit - E 340 Acquisitions	18-Dec-14 A	01-Jul-17
Sound Transit - City of Bellevue ROW - Updated	13-Sep-16 A	09-Nov-18
Sound Transit - E 320 Acquisitions	30-Apr-14 A	01-Apr-17
Sound Transit - E 360 Acquisitions	27-Feb-15 A	23-Mar-17
Airspace Leases/Temporary Construction Airspace Leases	20-Aug-13 A	31-May-17
EL Airspace Leases/Temporary Construction Airspace Leases	20-Aug-13 A	31-May-17
Seattle/I-90 ASL/TCAL	01-Oct-14 A	31-May-17
Bel-Red ASL/TCAL	20-Aug-13 A	04-May-17
Permits/3rd Party	15-Jul-13 A	16-Apr-17
East Link Construction	18-Jan-15 A	22-Jun-23
EL Early Utilities Construction	04-Mar-15 A	30-May-17
EL Early Utility Relocations	04-Mar-15 A	30-May-17
EL 130 - Seattle to I-90 Overpass (GC/CM)	04-Jan-16 A	14-Dec-20
EL 130 - Pre - Construction	04-Jan-16 A	02-May-17
EL 130 - Construction	01-Feb-17	14-Dec-20
EL 320 - I-90 Overpass to S. Bellevue (DBB)	29-Jan-16 A	14-Oct-20
EL 320 - Bid/Solicitation	29-Jan-16 A	10-Nov-16 A
EL 320 - Construction	05-Dec-16 A	14-Oct-20
EL 330 - Downtown Bellevue Tunnel (DBB)	18-Jan-15 A	20-Jul-20
EL330 - Bid/ Solicitation	19-Aug-15 A	15-Dec-15 A
EL330 - Construction	18-Jan-15 A	20-Jul-20
EL 335 - Downtown Bellevue to Spring District (GC/CM)	01-Apr-16 A	16-Apr-21
EL 335 - Pre-Construction	01-Apr-16 A	15-Jun-17
EL 335 - Construction	01-Feb-17	16-Apr-21
EL 340 - Spring District to SR 520 (DBB)	24-May-16 A	01-May-20
EL 340 - Bid/Solicitation	24-May-16 A	16-Feb-17 A
EL 340 - Construction	28-Feb-17	01-May-20
EL 360 - SR 520 to Overlake Transit Center (DB)	13-Jul-16 A	10-Jun-20
E360 Construction	13-Jul-16 A	10-Jun-20
EL 750 - Systems	01-Feb-16 A	12-Jan-22
EL750 - Pre-Construction	01-Feb-16 A	13-Jul-17
EL750 - Construction	20-Mar-17	12-Jan-22
East Link Master Schedule - Construction Interfaces	01-Mar-17	23-Mar-22
ELRACT - East Link Rail Activation/System Integration/Proj	23-Mar-22	22-Jun-23
East Link Rail Activation/System Integration/Project Closeout	23-Mar-22	22-Jun-23
Pre-Revenue Testing & Start Up	23-Mar-22	22-Sep-22
East Link System Integration Testing (3 Mo)	23-Mar-22	22-Jun-22
East Link Pre Revenue Operations (3 Mo)	22-Jun-22	22-Sep-22
Revenue Service Window	22-Sep-22	22-Jun-23
East Link Program Float - 9 Months	22-Sep-22	22-Jun-23
East Link Revenue Service Date (Target 30-Jun-23)		22-Jun-23*

Critical Path Analysis

The East Link critical path is currently running through the handover of the I-90 corridor from the E130 contractor to the E750 Systems contractor. The critical path for E130 is driven by the pontoon post-tensioning retrofit.



Activity Name	Start	Finish
EL 130 - Seattle to I-90 Overpass (GC/CM)	03-Apr-17	14-Dec-20
E130 Construction	03-Apr-17	14-Dec-20
Milestones	03-Apr-17	14-Dec-20
Project Management	03-Apr-17	12-Dec-20
Construction	03-Apr-17	12-Nov-20
Preliminary Activities	08-Jun-17	07-Jul-17
Access/BMP	08-Jun-17	07-Jul-17
Structures Retrofit	03-Apr-17	08-Aug-18
Pontoon Post Tensioning	03-Apr-17	08-Aug-18
Pontoon Tendons	02-Apr-18	08-Aug-18
Pontoon Reaction Frames	03-Apr-17	23-Oct-17
Trackwork	09-Aug-18	12-Nov-20
Direct Fixation	09-Aug-18	06-Aug-20
Special Track	07-Aug-20	01-Sep-20
Misc Trackwork & Appurtenances	24-Aug-20	12-Nov-20
EL 750 - Systems	13-Jan-21	23-Dec-21
Sound Transit N830 - E750 PreConstruction Build Sch R-0	13-Jan-21	23-Dec-21
E750 Project	13-Jan-21	23-Dec-21
E750 Engineering	01-Nov-21	23-Dec-21
E750 Construction	13-Jan-21	23-Dec-21
E750 Milestones	13-Jan-21	23-Dec-21
E750 Signals	13-Jan-21	27-May-21
E750 Testing and Commissioning	27-May-21	01-Nov-21
East Link Master Schedule - Construction Interfaces	14-Dec-20	23-Mar-22
East Link Project Interfaces	14-Dec-20	23-Mar-22
E130 (Seattle to South Bellevue) Contract Interface Windows	14-Dec-20	12-Jan-21
E130 - Civil/Systems Interface Window (East Approach to IDS) (1 Mo)	14-Dec-20	12-Jan-21
Systems Program Float	23-Dec-21	23-Mar-22
ELRACT - East Link Rail Activation/System Integration/Project Float	23-Mar-22	22-Jun-23
East Link Rail Activation/System Integration/Project Closeout	23-Mar-22	22-Jun-23
Pre-Revenue Testing & Start Up	23-Mar-22	22-Sep-22
East Link System Integration Testing (3 Mo)	23-Mar-22	22-Jun-22
East Link Pre Revenue Operations (3 Mo)	22-Jun-22	22-Sep-22
Revenue Service Window	22-Sep-22	22-Jun-23
East Link Program Float - 9 Months	22-Sep-22	22-Jun-23
East Link Revenue Service Date (Target 30-Jun-23)		22-Jun-23*

Right of Way

The East Link Extension involves the acquisition of a range of property interests including fee acquisitions for stations, guideway, staging areas, support facilities, and tunnel and guideway easements as well as permanent and temporary easements for construction. These acquisitions will result in owner and tenant residential and commercial relocations. Market value trends in both residential and commercial properties are exhibiting strong upward pressure. The overall impact on the Right-of-Way (ROW) budget is being closely monitored. Efforts are also underway to prioritize acquisitions by contract package to help ensure timely availability of property. Property acquisition is recognized as a high priority and the management team has formed an ad hoc ROW team consisting of internally involved departments to prioritize and manage the process. See right-of-way program status in the table below.

Several of the remaining properties needed for construction are City of Bellevue Properties committed in the Memorandum of Understanding. The ST/City of Bellevue Steering Committee consisting of senior staff are collaborating to ensure timely transfer of these property rights consistent with the project schedule and priorities.

East Link Extension Property Acquisition Status								
Line Section	**Board Approved**	**Offers Made/In Negotiations**	**Signed Agreements**	**Possession and Use**	**Admin Settlement**	**Closings to date**	**Relocations Required**	**Parcels Vacated**
Total	**242**	**232**	**3**	**31**	**4**	**170**	**243**	**197**

Note: Excluded from the table above are 9 parcels that were approved under a separate ST Board authorized agreement. These parcels are tracked in a working file and will be added to the table above in future reports.

Seattle to S. Bellevue

E130 IDS Rail Connection & IDS & I-90 Upgrades - 3 new relocatees identified. Air Space Lease discussions with WSDOT continue.

S. Bellevue to RTC parcels

E320 S. Bellevue– Air Space Lease discussions continue.

E330 Downtown Bellevue – No new activity.

E335 Downtown Bellevue to Spring District - 1 parcel closed;

E340 Bel-Red - No new activity.

E360 SR 520 to RTC – Air Space Lease discussions with WSDOT continue.

Construction Safety

Data/ Measure	March 2017	Year to Date	Project to Date
Recordable Injury/Illness Cases	0	0	0
Days Away From Work Cases	0	0	0
Total Days Away From Work	0	0	0
Restricted or Modified Work Cases	0	0	0
Total Days Restricted or Modified Work	0	0	0
First Aid Cases	*1*	*6*	*10*
Reported Near Mishaps	*2*	*6*	*19*
Average Number of Employees on Worksite	*80*	-	-
Total # of Hours (GC & Subs)	*12,500*	*31,650*	*72,231*
OSHA Incident Rates	**Monthly Average**	**Year to Date**	**Project to Date**
Recordable Injury Rate	0.00	0.00	0.00
DART Rate	0.00	0.00	0.00
Recordable National Average	*3.50*	*3.50*	*3.50*
DART National Average	*2.00*	*2.00*	*2.00*
Recordable WA State Average	*7.00*	*7.00*	*7.00*
DART WA State Average	*3.90*	*3.90*	*3.90*

Sound Transit Board Actions

Board Action	Description	Date
M2017-30	*Execute a contract amendment with Kiewit-Hoffman, AJV to provide Heavy Civil GC/CM construction services for the Seattle to South Bellevue construction within the East Link Extension in the amount of $661,315,300, plus a 7% contingency, for a new total authorized contract amount not to exceed $711,660,541. Further authorizes Sound Transit to issue a notice-to-proceed under the executed contract when appropriate.*	*Mar 23*
M2017-35	*Execute an Operations and Maintenance Agreement with the Washington state Department of Transportation to define maintenance responsibilities, purchase specialized vehicles, and provide payment for certain maintenance services during light rail construction in the I-90 corridor as part of the East Link Extension for a total authorized agreement amount not to exceed $2,390,000.*	*Mar 23*

Board Action	Description	Date
M2017-38	*Authorizing the Board Chair to appoint up to three board members to meet with City of Mercer Island elected officials to discuss issues of concern regarding the East Link Project.*	*Mar 23*

Environmental

- Continued developing environmental permit application packages. Environmental commitments are being incorporated into the design, program, and specifications of the project.

Community Outreach

- Social media launch of the I-90 track bridge video which had 20,500 views in over two days.
- Visited and updated various Bel– Red businesses interested in the project .
- Hosted an outreach kickoff meeting with the E320 CM team, contractor and project partners.
- Sent out notifications to multiple businesses on NE Spring Blvd. of upcoming PSE 115kV utility pole relocation on the MGI site beginning early April.
- Coordinated postcard delivery for the SEPA evaluation of the FEIS for the I-90 segment of East Link.
- Provided E335 briefing to Meydenbauer Center.
- Staffed SDOT Rainier Avenue Forum and approximately 50 people attended.



Video launch of the I-90 track bridge had 20.5k views in two days.

Major East Link Construction Contract Packages

Below are the major construction contract packages for the East Link Extension with a brief scope description and status of each.

R8A Stage 3A – Modification of HOV Ramp between I-90 and Bellevue Way SE. [Design-bid-build]
Status: Completed.

E130 Seattle to South Bellevue – International District Station (IDS) modification, Retrofit of existing WSDOT structures, I-90 Center Roadway and two light rail stations. [Heavy Civil GC/CM]
Status: MACC negotiated. Proceed with the Contractor on long lead procurement and planning for early work. The balance of the scope authorized by the ST Board. Execution of change order for balance of work is pending.

E160 Track Bridge System – Fabrication, delivery and installation of eight track bridges that will span over fixed structures and the floating bridge. [Furnish and Install]
Status: Incorporated as part of E130.

E320 South Bellevue – One light rail station and a parking garage at South Bellevue P&R; aerial, at-grade and retained-cut guideway along Bellevue Way SE and 112th Avenue SE. [Design-bid-build]
Status: ST Board approved. NTP was issued.

E330 Downtown Bellevue Tunnel – SEM Tunneling along 110th Ave. NE in Downtown Bellevue. [Design-bid-build]
Status: See following pages.

E335 Downtown Bellevue to Spring District – Four light rail stations; aerial, at-grade and retained-cut guideway from Downtown Bellevue, over I-405 to 124th Ave. NE. [Heavy Civil GC/CM]
Status: Contract with the exception of the Station package scope awarded, pending execution. The station scope will be repackage and procured later in the year.

E340 Bel-Red – One light rail station, aerial and at-grade guideway from 124th Ave. NE to NE 20th. [Design-bid-build]
Status: Contract Award in January 2017. NTP issued in March 2017.

E360 SR520 to Redmond Technology Center – Two light rail stations; aerial and retained-cut guideway along SR-520 in the Overlake area of Bellevue and Redmond. [Design-Build]
Status: Preparation of 60% design submittals.

E750 Systems – Light rail systems elements, including Traction Power Electrification, Overhead Catenary System, Train Signals and Communication System. Combined with Northgate Link N830. [Heavy Civil GC/CM]
Status: MACC negotiated, Contract pending award in April and execution to follow in May.

Contract E330 – Downtown Bellevue Tunnel

Current Progress

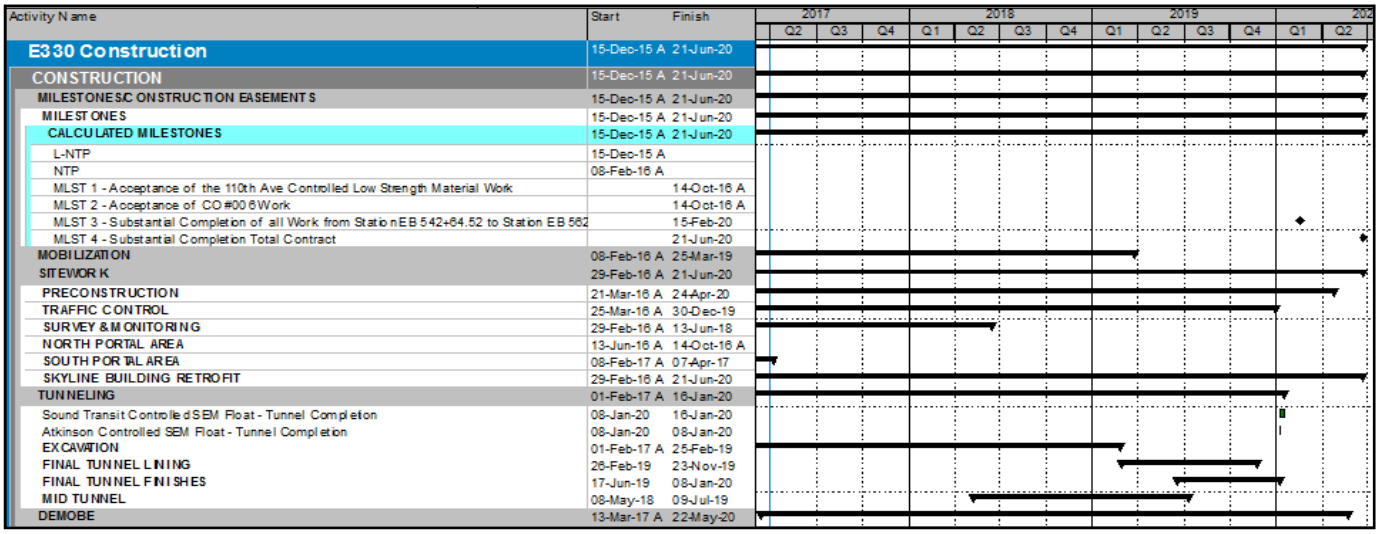
South Portal: *The* Contractor continued tunneling throughout the month and completed reach 1 type 1 excavation through the length of the pipe canopy. Continued setting up facilities for concrete batch plant operations to begin next month. Continued to maintain TESC and traffic control measures.

North Portal: Completed Phase II of the Skyline Building scope of work, and began Phase III. Skyline work will run primarily during the night shift for the next 2-3 months.

Between Portals: Completed water main replacement at intersection of 110th Ave SE and NE 2nd Ave.

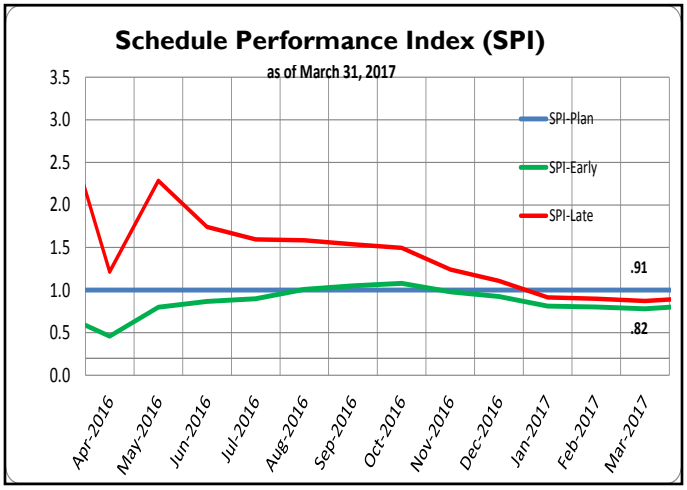
Schedule Summary

The critical path for this contract follows the tunnel excavation and the completion of the South Portal structure. Tunneling has been slow to ramp up and the Contractor has not yet met their anticipated production rate. They attribute this to a learning curve and anticipate production increasing in the months ahead. The Contractor was issued a Change Order to use a "three-drift method" of tunnel excavation. The terms of the CO also allow E335 to access the tunnel two months ahead of substantial completion which will relax the interface with the E750 contractor.



Schedule Performance Index

This period, the SPI early is at 0.82 and the SPI late is at 0.91. Both the early and late indexes indicate that the Contractor continues to be behind schedule when compared to the baseline schedule, but has slightly improved this month. This trend is supported by the fact that the start of tunnel excavation began later than originally planned, and the Contractor experienced a much steeper learning curve than anticipated. The Contractor has stated that they expect to increase tunneling progression in the near future with the use of the onsite batch plant, an improved shotcrete robot being used, and the potential decrease in the number of rebar spiling required moving forward.



Next Period's Activities

- Continue batch plant installation and begin testing.
- Continue tunnel excavation. Continue improvements to shotcrete production, delivery and application.
- Continue Skyline Building work phase III shotcrete and rebar dowels.
- Complete 110th Avenue waterline replacement work and finalize street paving.
- Dewatering consultant to develop dewatering plan and identify additional test wells are required for northernmost end of tunnel.

Closely Monitored Issues

- Obtaining right of entry agreements and temporary construction easements for properties along alignment.
- Contractor has yet to meet their anticipated tunneling production rate.
- Contractor provided notification of added cost and time for additional shotcrete at face for work stoppages under 24 hours placed during the weekend. ST responded with direction to follow the specification requirement of applying 5" on the face when crews are not actively advancing the tunnel. Contractor has submitted a Notice of Intent to Claim.
- Contractor working to resolve issues with the quality and efficiency of shotcrete application.

Cost Summary

Present Financial Status	Amount
E330 Contractor– Guy F Atkinson Construction, LLC.	
Original Contract Value	$121,446,551
Change Order Value	$1,835,434
Current Contract Value	$123,281,985
Total Actual Cost (Incurred to date)	$29,837,919
Financial Percent Complete:	24.2%
Physical Percent Complete:	24.2%
Authorized Contingency	$12,144,655
Contingency Drawdown	$1,835,434
Contingency Index	1.61



Tunneling advancing shotcrete robot in operation.

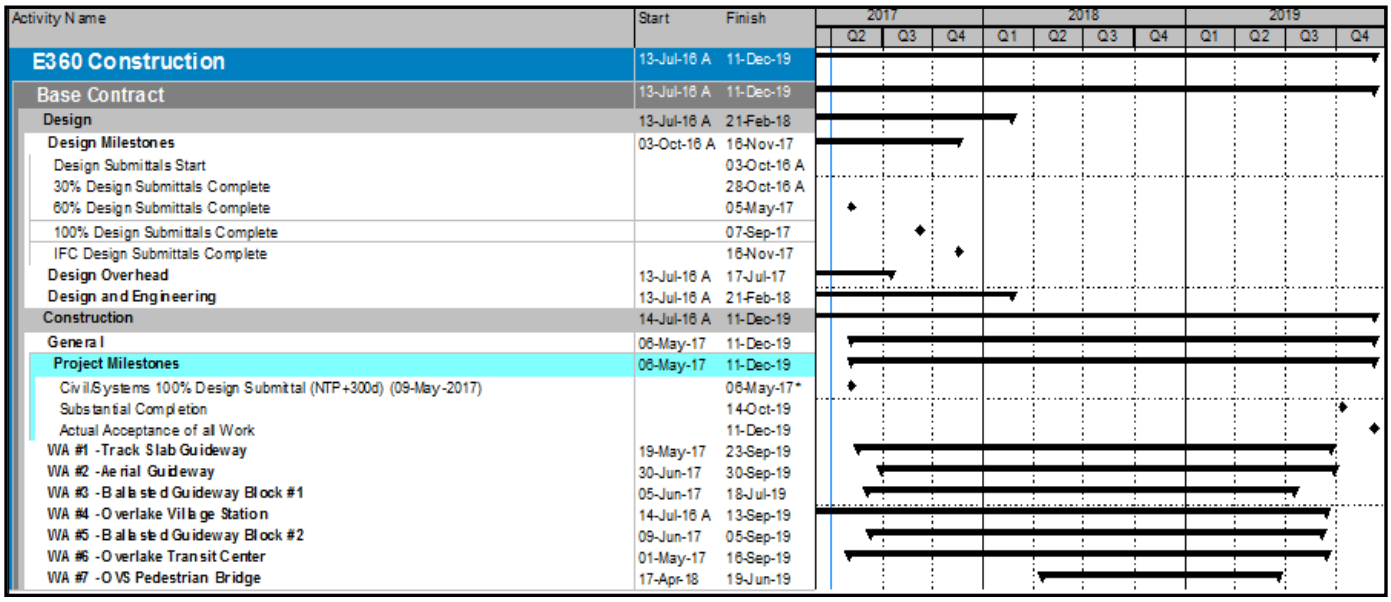
Contract E360 – SR 520 to Redmond Technology Center (RTC)

Current Progress

In March, the Design-Build Contractor (Kiewit-Hoffman East Link Constructors) submitted three 60% design packages including: the guideway substructure, superstructure and walls; and, the Overlake Village Station (OVS) Pedestrian Bridge. Contractor is advancing the 60% designs for the stations at OVS and Overlake Transit Center (OTC). Three 100% design packages were submitted this month for track design, and OVS/guideway civil and site prep, and drainage. ST worked closely with the City of Redmond and the Contractor on outstanding permitting reviews and conditions. The Contractor continues to prepare for early site clearing, demolition and temporary bus loop configuration in May. Contractor worked with Microsoft on design of their temporary facilities.

Schedule Summary

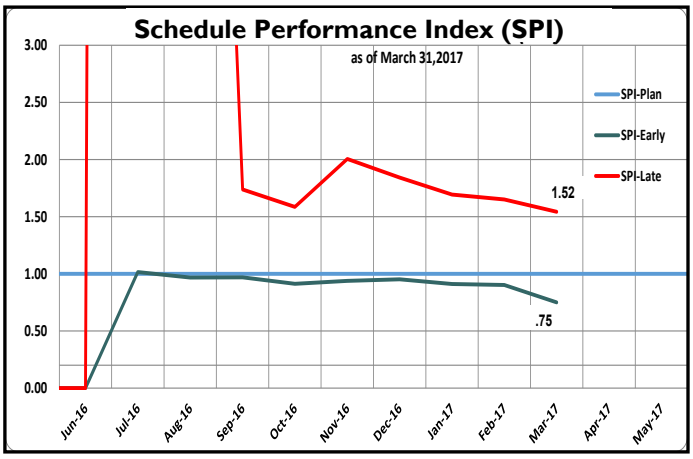
The Contractor is between 60-100% on most design packages. Work is anticipated to start in late Spring or early Summer. The critical path for this contract runs through the aerial guideway and direct fixation trackwork. The Contractor's current projected completion is two months ahead of Sound Transit's baseline milestone.



Activity Name	Start	Finish
E360 Construction	13-Jul-16 A	11-Dec-19
Base Contract	13-Jul-16 A	11-Dec-19
Design	13-Jul-16 A	21-Feb-18
Design Milestones	03-Oct-16 A	16-Nov-17
Design Submittals Start		03-Oct-16 A
30% Design Submittals Complete		28-Oct-16 A
60% Design Submittals Complete		05-May-17
100% Design Submittals Complete		07-Sep-17
IFC Design Submittals Complete		16-Nov-17
Design Overhead	13-Jul-16 A	17-Jul-17
Design and Engineering	13-Jul-16 A	21-Feb-18
Construction	14-Jul-16 A	11-Dec-19
General	06-May-17	11-Dec-19
Project Milestones	06-May-17	11-Dec-19
Civil/Systems 100% Design Submittal (NTP+300d) (09-May-2017)		06-May-17*
Substantial Completion		14-Oct-19
Actual Acceptance of all Work		11-Dec-19
WA #1 -Track Slab Guideway	19-May-17	23-Sep-19
WA #2 -Aerial Guideway	30-Jun-17	30-Sep-19
WA #3 -Ballasted Guideway Block #1	05-Jun-17	18-Jul-19
WA #4 -Overlake Village Station	14-Jul-16 A	13-Sep-19
WA #5 -Ballasted Guideway Block #2	09-Jun-17	05-Sep-19
WA #6 -Overlake Transit Center	01-May-17	16-Sep-19
WA #7 -OVS Pedestrian Bridge	17-Apr-18	19-Jun-19

Schedule Performance Index (SPI)

This period, the SPI early is 0.75 and SPI late is 1.52. Both SPI continue to trend accordingly to the baseline schedule. This trend is supported by design packages continuing to make good progress towards 60% with completion throughout the April/May periods and the majority of 100% design packages following throughout this summer. The early design packages such as site clearing, demolition and temporary bus loop configuration are near 100% and is anticipated to be approved and progress into construction in May.



Next Period's Activities

- 60% design package submittals to continue in April.
- Work progress on 100% design packages
- Submit final civil permit application packages to COR.
- Continue preparation for site access at OVS and OTC for demolition of structures.

Cost Summary

Present Financial Status	Amount
E360 Contractor— Kiewit-Hoffman	
Original Contract Value	$225,336,088
Change Order Value	$500,000
Current Contract Value	$225,836,088
Total Actual Cost (Amount Billed)	$27,637,116.97
Financial Percent Complete:	12.2%
Physical Percent Complete:	12.2%
Authorized Contingency	$22,533,609
Contingency Drawdown	$500,000
Contingency Index*	4.2

Closely Monitored Issues

- Alternative OTC Ped Bridge - Contractor and Microsoft are working on 30% designs for an alternative OTC Ped Bridge. The Parties anticipate delivering the revised bridge as part of the contract.
- Contractor is designing and pricing provision of temporary drainage facilities at OTC. The change may impact schedule.
- Contractor is continuing with design of soil nail wall (wall #14). Ongoing discussion with ST and Microsoft about soil nail walls. High Resilient Fasteners - A design and construction change order is being developed for the high resilient fastener change.
- Contractor has numerous outstanding change proposals to submit. ST and the Contractor continue to discuss allowable mark ups on changes.



Aerial renderings of Overlake Village Station and Pedestrian Bridge

This page left intentionally blank.

Link Light Rail
S. 200th Link Extension

Scope

Limits: South 200th Link Extension consists of 1.6-mile extension of light rail from the SeaTac/Airport Station to South 200th Street.

Alignment: The extension continues in an aerial configuration heading south of the existing SeaTac/Airport Station, east of Air Cargo Road across South 188th Street, and continuing south along the east side of 28th Avenue South to the elevated Angle Lake Station located at South 200th Street. A passenger pickup/drop-off area, bus transfer, plaza, and a minimum 700-space structured park and ride facility will be located at Angle Lake Station. Up to 400 additional spaces may be developed for interim parking while Angle Lake Station is the interim southern terminus of the Link system. Guideway/Station and Parking Garage/Plaza are being delivered under a Design Build contract.

Station: Angle Lake Station is located at South 200th Street.

Systems: Signals, track electrification, and SCADA communications

Budget: $383.2 Million

Schedule: Revenue Service began on September 24, 2016



Map of S. 200th Link Extension.

Key Project Activities

- *Design Build Guideway and Station (S440):* The project team continued working with the Contractor and Operations to finalize remaining work to achieve Acceptance on the contract.
- *Design Build Parking Garage (S445*): Contractor completed punchlist items, and working on two last change items. The project team working toward achieving Acceptance on the contract. The maintenance for garage was transitioned to Facilities, however exceptions of three items are being reviewed. Elevator claim under review.
- *Military Road/South 200th (S446):*) Construction work managed by WSDOT continues.
- *Roadway Improvements (S447):* One City of SeaTac punchlist item remains in dispute.
- *Project-wide*: Continued working toward goal to conclude all punchlist work and close-out by end of 2nd QTR 2017.

Closely Monitored Issues

- *S440*: Completion of Training, O&M Manuals – corrective actions need to be completed.
- *S440, S445, S447*: Continue evaluation/negotiation of Contractor request for change.

Project Cost Summary

The South 200th Link Extension cost is summarized into two cost tables. The current Adopted Budget column reflects the 2016 Adopted Budget. The first table is in accordance with Sound Transit's Work Breakdown Structure (WBS) displaying the cost summary by phase. The second table summarizes project cost by SCC, in accordance to the FTA Standard Cost Categories (SCC) format. Both tables show figures in millions.

Cost Summary by Phase

WBS Phase Elements	Baseline Budget	Current Adopted Budget	Commitment to Date	Incurred to Date	Est. Final Cost (EFC)	Adopted Budget vs. EFC
Administration	$16.1	$17.1	$15.8	$15.8	$17.1	$0.0
Preliminary Engineering	$5.9	$5.7	$5.7	$5.7	$5.7	$0.0
Final Design	$20.0	$9.5	$8.7	$8.7	$8.8	$0.7
Construction Services	$15.1	$17.3	$17.1	$17.0	$17.2	$0.1
3rd Party Agreements	$6.9	$7.1	$5.7	$3.8	$5.0	$2.1
Construction	$275.7	$282.9	$238.0	$234.4	$249.3	$33.7
Vehicles	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0
ROW	$43.5	$43.5	$38.2	$38.0	$38.1	$5.4
Capital Total	**$383.2**	**$383.2**	**$329.1**	**$323.4**	**$341.2**	**$42.0**
Finance Cost	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0
Project Total	**$383.2**	**$383.2**	**$329.1**	**$323.4**	**$341.2**	**$42.0**

The overall project Estimated Final Cost (EFC) for this period reflects $341M, a savings projection of $42M. This period incurred cost increased by approximately $1.1M, of this $0.1M was for construction services, $0.9M for several construction contracts (S440, S445, S446), and the remaining expenditures of $0.1M were related to administrative charges.

The EFC grouped by SCC is also reflecting $341M, a savings projection of $42M. The work remaining can be achieved under the adopted budget.

Cost Summary by SCC

Project Elements by SCC	Baseline Budget	Current Adopted Budget	Commitment to Date	Incurred to Date	Est. Final Cost (EFC)	Adopted Budget vs. EFC
10 Guideway & Track Elements	$144.8	$133.0	$99.9	$100.3	$110.3	$22.7
20 Stations	$46.4	$51.7	$40.0	$40.0	$40.5	$11.3
30 Support Facilities: Yards, Shops	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0
40 Sitework & Special Conditions	$34.6	$55.3	$56.8	$53.3	$58.4	-$3.1
50 Systems	$30.8	$19.9	$20.0	$19.8	$20.1	-$0.1
Construction Subtotal (SCC 10-50)	**$256.6**	**$259.9**	**$216.8**	**$213.3**	**$229.2**	**$30.7**
60 Row, Land, Existing Improvements	$43.5	$43.5	$38.2	$38.0	$38.1	$5.4
70 Vehicles	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0
80 Professional Services	$64.1	$72.5	$74.2	$72.1	$68.2	$4.2
90 Unallocated Contingency	$19.0	$7.3	$0.0	$0.0	$5.7	$1.6
Project Total (SCC 10-90)	**$383.2**	**$383.2**	**$329.1**	**$323.4**	**$341.2**	**$42.0**
100 Finance Cost	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0
Project Total (SCC 10-90)	**$383.2**	**$383.2**	**$329.1**	**$323.4**	**$341.2**	**$42.0**

Cost Contingency Management

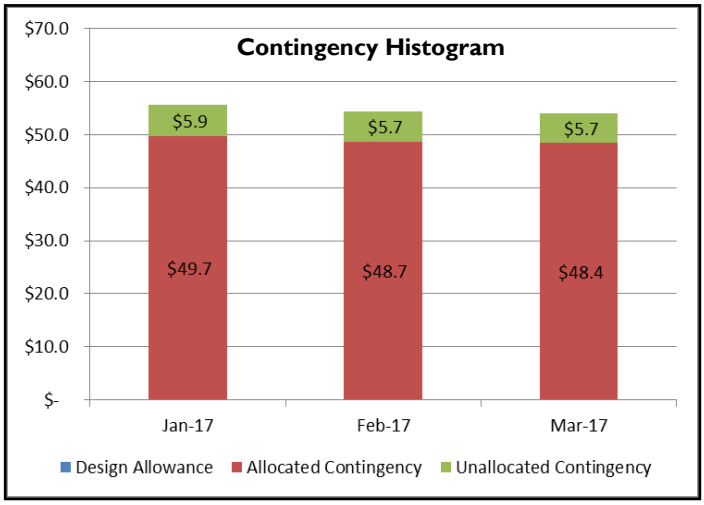
The overall project contingency balance as of this period is $54.1M.

***Design Allowance (DA)*:** N/A

Allocated Contingency (AC): This period the AC shows a current balance of $48.4M, a decrease of $0.3M from the previous period. This change is mostly due to a construction change order for S440 contract and new commitment for follow-on work, both in the Construction phase.

Unallocated Contingency (UAC): This period the UAC balance is $5.7M, same as last period.

Contingency Status	Baseline		Current	
	Amount	% of Total	Amount	% of Work Remaining
Design Allowance	$26.7	7.0%	$ -	0.0%
Allocated Contingency	$29.8	8.0%	$48.4	271.8%
Unallocated Contingency	$19.0	5.0%	$ 5.7	32.0%
Total	**$75.5**	**20.0%**	**$54.1**	**303.9%**



S440 Contract—Design -Build (Guideway and Station)

Close-out

Revenue Service began on September 24, 2016. The S440 Contractor continues to work on closeout and punchlist items.

Close-out Activities

Current Period

- Continued post revenue work arounds, punch list activities and finishing NCRs.
- Continued safety certification.
- Continued change order work and address contract closeout items (training, O&M Manuals, Commissioning Reports and As-Builts).

Next Period

- Complete punch list, closeout materials and work around work.

Closely Monitored Issues

- Need to complete Training and O&M Manuals, Safety Certification, and as-built to transfer maintenance responsibilities to Operations and Facilities.

Cost Summary

Present Financial Status	Amount
S440 Contractor—PCL Civil Contractors, Inc.	
Original Contract Value	$169,000,000
Change Order Value	$6,809,462
Current Contract Value	$175,809,462
Total Actual Cost (Amount Billed)	$174,468,971
Financial Percent Complete:	99%
Physical Percent Complete:	99%
Authorized Contingency	$13,520,000
Contingency Drawdown	$6,809,462
Contingency Index*	2.0

S445 Contract—Design -Build Parking Garage

Close-out

The S445 parking garage is now open to the public. Substantial Completion issued on December 21, 2016.

Close-out Activities

Current Period

- Continued work on Change Orders.

Next Period

- Contract closeout.

Closely Monitored Issues

- Sound Transit is reviewing the claim package from Contractor (for additional time and cost associated with elevator procurement and installation).

Cost Summary

Present Financial Status	Amount
S445 Contractor—Harbor Pacific/Graham	
Original Contract Value	$29,978,000
Change Order Value	$2,604,606
Current Contract Value	$32,582,606
Total Actual Cost (Amount Billed)	$32,485,625
Financial Percent Complete:	99.7%
Physical Percent Complete:	99.7%
Authorized Contingency	$2,798,240
Contingency Drawdown	$2,604,606
Contingency Index*	1.07

S447 Contract – Station Area Roadway Improvements and Surface Parking

Current Progress

S447 Contractor continues punchlist work. Substantial Completion issued December 19, 2016.

Key Activities

Current Period

- Continued punch list work.
- Continued NCR work, one item remains.

Next Period

- Complete last NCR and punchlist item.

Closely Monitored Issues

- None to report.

Cost Summary

Present Financial Status	Amount
S447 Contractor – Johansen Excavating, Inc	
Original Contract Value	$8,426,400
Change Order Value	$2,642,328
Current Contract Value	$11,068,728
Total Actual Cost (Amount Billed)	$11,067,794
Financial Percent Complete:	99%
Physical Percent Complete:	99%
Authorized Contingency	$2,642,640
Contingency Drawdown	$2,642,328
Contingency Index*	1.0

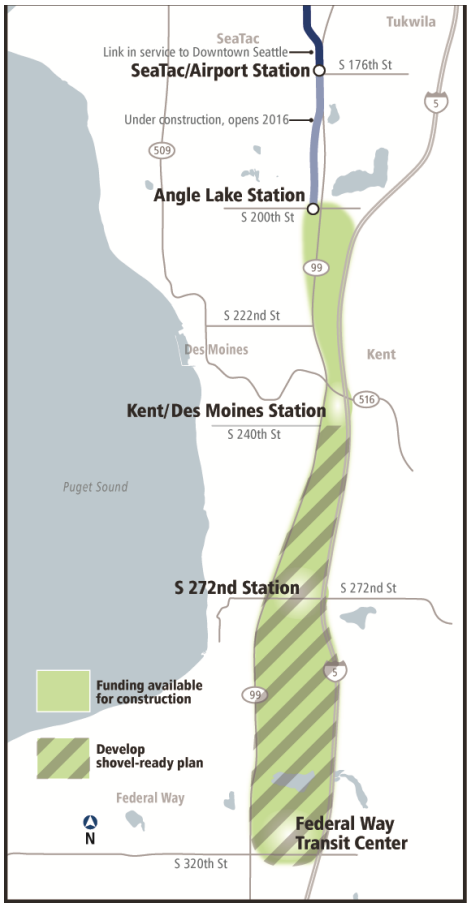
This page left intentionally blank.

Scope

The Federal Way Link Extension includes alternatives analysis, environmental work and conceptual engineering for an extension of Link Light Rail from S. 200th St. in SeaTac to the Federal Way Transit Center. Preliminary engineering will be performed for the segment from S. 200th St. to Kent/Des Moines (in the vicinity of Highline College) and to the Federal Way Transit Center.

Key Project Activities/Issues

- Board identified a Preferred Alternative and approved Phase 3 contract amendment on July 23, 2015.
- Executed Phase 3 contract amendment with HDR for Final EIS and Preliminary Engineering on July 24, 2015.
- ST Board approved contract amendment in February 2016 to extend Preliminary Engineering from Kent/ Des Moines to the Federal Way Transit Center.
- Final EIS was published in November 2016.
- Sound Transit Board has selected the alignment and stations for the project on January 26, 2017.
- Received Record of Decision from FTA in March 2017.
- Entered Final Design (Phase Gate 4) in March 2017.



Map of Federal Way Link Extension.

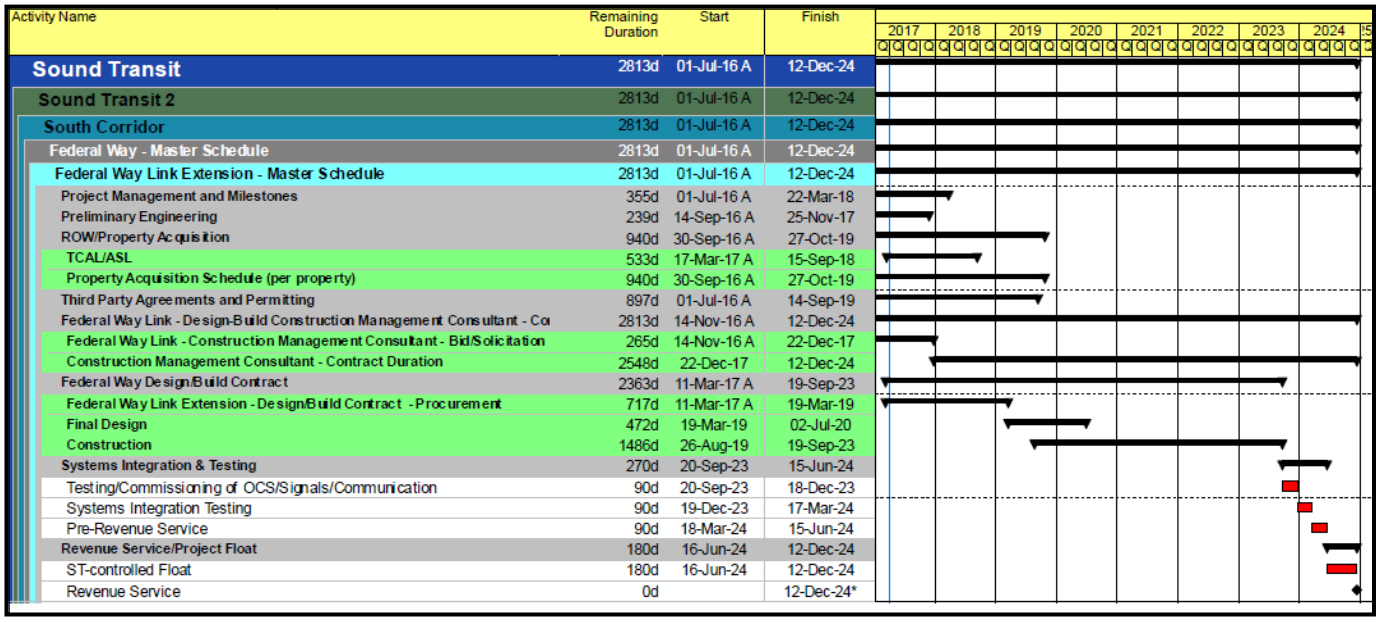
Project Cost Summary

In March 2017, the Sound Transit Board authorized the Federal Way Link Extension Project to advance through Gate 4 (Final Design), increasing the authorized project allocation from $48.8M to $412.6M. Table (below) figures in millions.

Phase	Adopted Budget	Commitment to Date	Incurred to Date	Forecasts and Trends	Estimated Final Cost (EFC)	Budget Versus EFC
Administration	$14.9	$6.4	$6.3	$8.5	$14.9	$0.0
Preliminary Engineering	$40.4	$32.2	$30.8	$8.2	$40.4	$0.0
Final Design	$2.4	$0.0	$0.0	$2.4	$2.4	$0.0
Construction Services	$45.7	$0.0	$0.0	$45.7	$45.7	$0.0
Third Parties	$18.6	$1.2	$0.8	$17.4	$18.6	$0.0
Construction	$25.5	$0.0	$0.0	$25.5	$25.5	$0.0
Right of Way	$265.1	$1.1	$0.9	$264.0	$265.1	$0.0
Total	**$412.6**	**$40.9**	**$38.8**	**$371.7**	**$412.6**	**$0.0**

Project Schedule

The Record of Decision was received this period. Sound Transit will begin acquiring real estate and start procurement for a design-build contractor in the 4th QTR 2017 with construction scheduled to start in 2019. The project schedule is presented below.



Activity Name	Remaining Duration	Start	Finish
Sound Transit	2813d	01-Jul-16 A	12-Dec-24
Sound Transit 2	2813d	01-Jul-16 A	12-Dec-24
South Corridor	2813d	01-Jul-16 A	12-Dec-24
Federal Way - Master Schedule	2813d	01-Jul-16 A	12-Dec-24
Federal Way Link Extension - Master Schedule	2813d	01-Jul-16 A	12-Dec-24
Project Management and Milestones	355d	01-Jul-16 A	22-Mar-18
Preliminary Engineering	239d	14-Sep-16 A	25-Nov-17
ROW/Property Acquisition	940d	30-Sep-16 A	27-Oct-19
TCAL/ASL	533d	17-Mar-17 A	15-Sep-18
Property Acquisition Schedule (per property)	940d	30-Sep-16 A	27-Oct-19
Third Party Agreements and Permitting	897d	01-Jul-16 A	14-Sep-19
Federal Way Link - Design-Build Construction Management Consultant - Co	2813d	14-Nov-16 A	12-Dec-24
Federal Way Link - Construction Management Consultant - Bid/Solicitation	265d	14-Nov-16 A	22-Dec-17
Construction Management Consultant - Contract Duration	2548d	22-Dec-17	12-Dec-24
Federal Way Design/Build Contract	2363d	11-Mar-17 A	19-Sep-23
Federal Way Link Extension - Design/Build Contract - Procurement	717d	11-Mar-17 A	19-Mar-19
Final Design	472d	19-Mar-19	02-Jul-20
Construction	1486d	26-Aug-19	19-Sep-23
Systems Integration & Testing	270d	20-Sep-23	15-Jun-24
Testing/Commissioning of OCS/Signals/Communication	90d	20-Sep-23	18-Dec-23
Systems Integration Testing	90d	19-Dec-23	17-Mar-24
Pre-Revenue Service	90d	18-Mar-24	15-Jun-24
Revenue Service/Project Float	180d	16-Jun-24	12-Dec-24
ST-controlled Float	180d	16-Jun-24	12-Dec-24
Revenue Service	0d		12-Dec-24*

Sound Transit Board Actions

Board motions and resolutions directly related to Federal Way Link Extension is summarized in the table below.

Motion Number	Description	Date
R2017-07	*(1) Authorizing the Federal Way Link Extension project to advance through Gate 4 within Sound Transit's Phase Gate process; (2) increasing the authorized project allocation to date by $363,835,000 from $48,790,000 to $412,625,000; and (3) increasing the 2017 annual budget by $64,637,043 from $6,120,957 to $70,758,000.*	*Mar 23*

Community Outreach

- Scheduled with property owner access for survey work. Updated property owners regarding Board decision on FWLE route, profile and station areas.
- Hosted neighborhood briefings with residents residing near Military Road on Federal Way and in the Jackson Mobile Home Park.
- Met with tenants and property owners of the Silverwood Apartments and other nearly apartments; approx. 20 participants.

Environmental

- None to report.

Phase 3 Preliminary Engineering (PE)

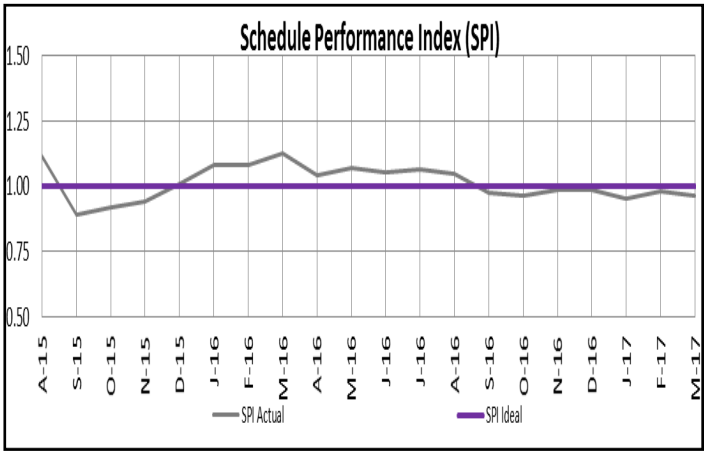
In July 2015, Sound Transit executed an amendment to the professional services agreement with HDR Engineering, Inc. for Phase 3 professional services, including completion of a Final Environmental Impact Statement (FEIS) using the preferred alternative and the other alternatives in the Draft EIS, preliminary engineering of the preferred alternative to Kent/Des Moines and additional engineering as needed to support receipt of environmental approvals. In February 2016, Sound Transit executed an amendment to extend the preliminary engineering effort from Kent/Des Moines to the Federal Way Transit Center.

Phase 3 Key Activities

- Record of Decision (ROD) from FTA received this period (March 2017), moving toward Entry to Engineering.
- Resolution of Pre-Final PE submittal comments by third parties and initiation of development of project requirements associated with RFP development for design-build contract.
- Continued coordination with potentially affected property owners including neighborhood briefings and one-on-one meetings.
- Completion of term sheets with affected jurisdictions documenting concurrence on project improvements, impacts analysis and mitigation and initiation of future agreements.
- Continued discussion with FTA regarding New Starts requirements for Entry to Engineering.

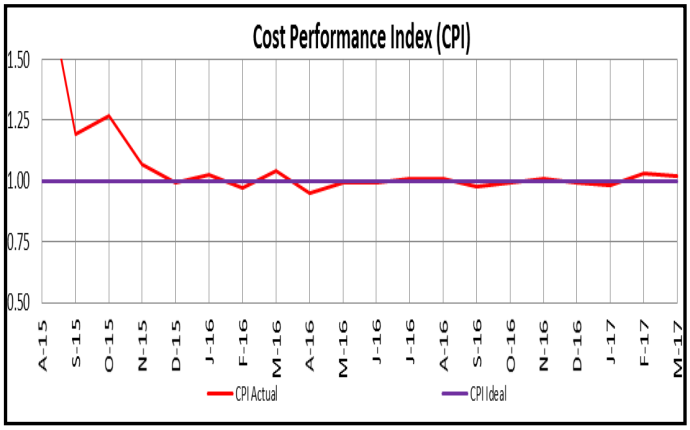
Phase 3 Schedule Performance

The cumulative Schedule Performance Index (SPI) is 0.97 through March 2017, indicating the overall amount of work accomplished is close to planned. Review of the Consultant's draft Phase 3 schedule indicates that the EIS comments response leading to publishing the ROD in March.



Phase 3 Cost Performance

Phase 3 expenditures totaled $18.6M, approximately 95% of the Phase 3 contract total. The Phase 3 percent complete reported at 97%, with an earned value of $19.1M. The cumulative Cost Performance Index (CPI) is 1.02, indicating costs are on track with work accomplished. Based on the current trend, the Phase 3 cost is expected to be on budget.



Phase 3 Performance	Cumulative To Date
Amount Invoiced	$18.6
% Spent	95%
Earned Value	$19.1
% Complete	97%
SPI	**0.97**
CPI	**1.02**

Scope

Limits: City of Tacoma

Alignment: The Tacoma Link Expansion is a 2.4-mile extension of the existing Tacoma Link system from the Theater District in downtown Tacoma to the Hilltop neighborhood. The Expansion will travel at-grade along Stadium Way, North 1st. Street, Division Avenue, and Martin Luther King Jr. Way.

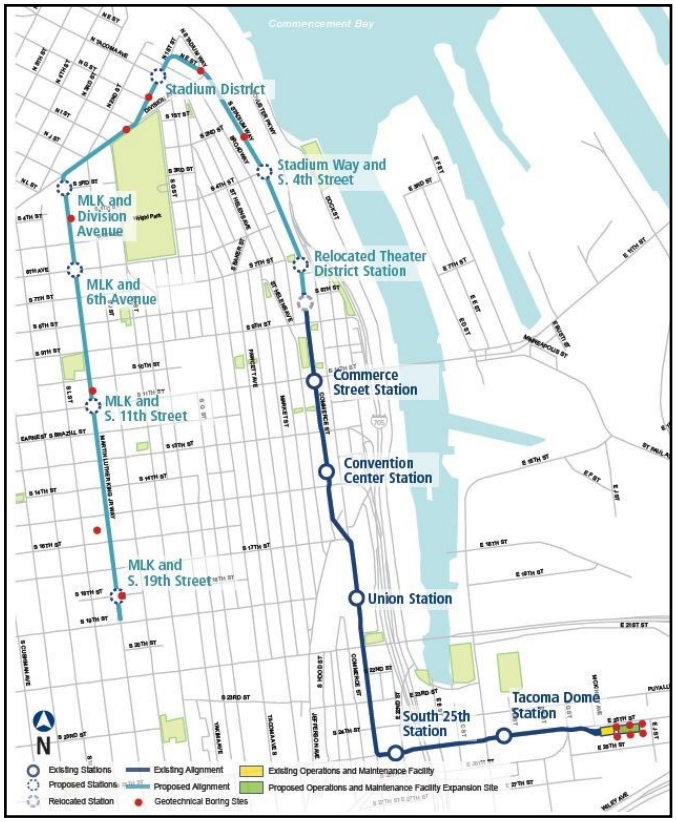
Stations: Relocated Theater District Station, Stadium Way & 4th St, Stadium District, MLK Jr. Way and Division, MLK Jr. Way and 6TH Ave, MLK Jr. Way and S. 11th St., and MLK Jr. Way and S. 19th St.

Systems: Expansion of the Operations and Maintenance Facility; The at-grade alignment will include additional signals, OCS, traction power, and communications infrastructure.

Budget: $34.6 Million Phase Gate 4 budget; excludes Construction (Year of Expenditure Dollars)

Phase: Final Design

Const. Starts: 2018



Map of Tacoma Link Expansion.

Key Project Activities

Final Design

- Staff and third party reviews were completed for 60% mainline design packages. Reviews continued for the 60% OMF design package. The cost estimate was submitted for ST review.

Pre-Construction Services

- Completed negotiations for Phase I (Pre-Construction Services) Construction Management Consultant (CMC) procurement; anticipate contract NTP in April 2017.

Third Parties

- Advancing formal agreements with City of Tacoma.

Right of Way

- Continued early right of way property acquisition activities as they relate to the expansion of the Operations and Maintenance.

Project Cost Summary

The Tacoma Link project cost is summarized below in two cost categories. The first table summarizes costs in accordance with Sound Transit's budget Work Breakdown Structure (WBS). The second table summarizes costs in accordance with the FTA's Standard Cost Categories (SCC) format. The figures in both tables are shown in millions.

Cost Summary by Phase

WBS Phase Elements	Current Adopted Budget	Commitment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget vs. EFC
Administration	$10.6	$3.5	$3.5	$10.6	$0.0
Preliminary Engineering	$5.9	$5.6	$5.6	$5.9	$0.0
Final Design	$10.8	$7.7	$3.5	$10.8	$0.0
Construction Services	$0.8	$0.0	$0.0	$0.8	$0.0
3rd Party Agreements	$1.3	$0.0	$0.0	$1.3	$0.0
Construction	$0.0	$0.0	$0.0	$0.0	$0.0
Vehicles	$2.0	$1.9	$0.2	$2.0	$0.0
ROW	$3.2	$1.6	$1.4	$3.2	$0.0
Total	**$34.6**	**$20.3**	**$14.2**	**$34.6**	**$0.0**

Cost Summary by SCC

Project Elements by SCC	Current Adopted Budget	Commitment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget vs. EFC
10 Guideway & Track	$0.0	$0.0	$0.0	$0.0	$0.0
20 Stations	$0.0	$0.0	$0.0	$0.0	$0.0
30 Support Facilities	$0.0	$0.0	$0.0	$0.0	$0.0
40 Sitework & Special Conditions	$0.0	$0.0	$0.0	$0.0	$0.0
50 Systems	$0.0	$0.0	$0.0	$0.0	$0.0
Construction Subtotal (10 - 50)	$0.0	$0.0	$0.0	$0.0	$0.0
60 Row, Land	$3.2	$1.6	$1.4	$3.2	$0.0
70 Vehicles (non-revenue)	$2.0	$1.9	$0.2	$2.0	$0.0
80 Professional Services	$29.4	$16.8	$12.6	$29.4	$0.0
90 Unallocated Contingency	$0.0	$0.0	$0.0	$0.0	$0.0
Total (10 - 90)	**$34.6**	**$20.3**	**$14.2**	**$34.6**	**$0.0**

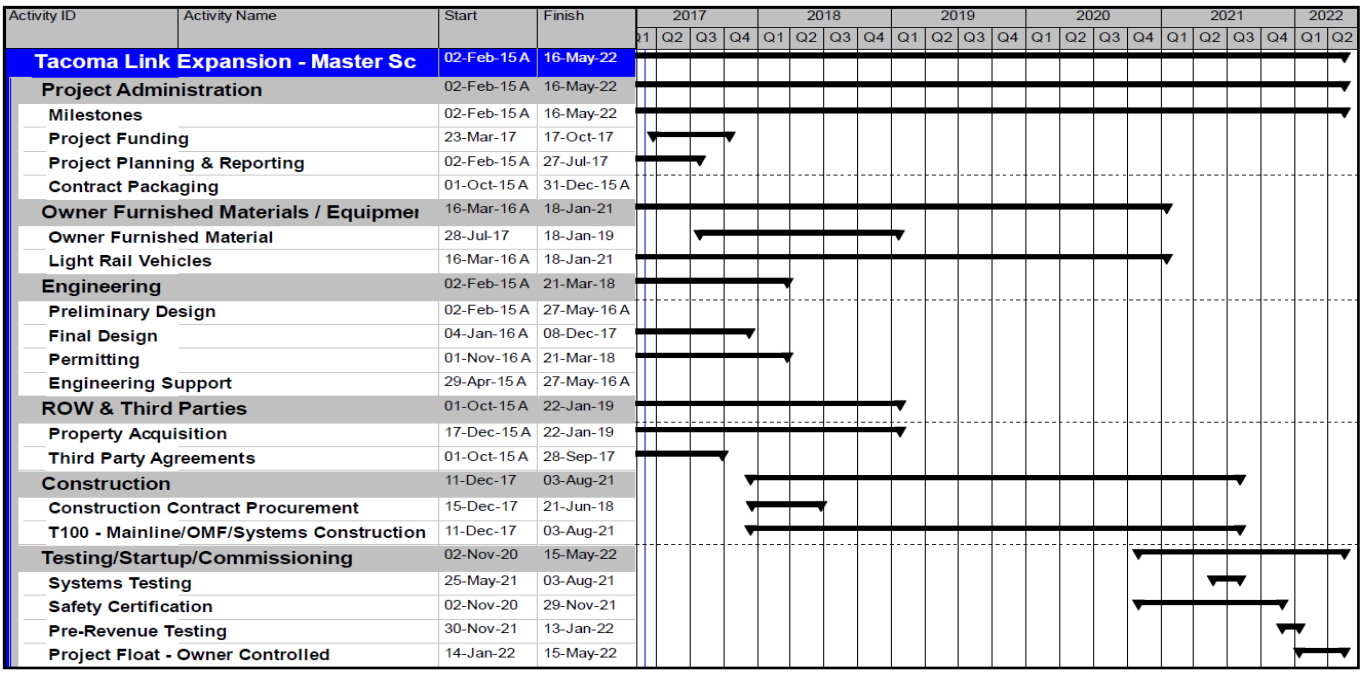
Risk Management

The Tacoma Link Risk and Contingency Management Plan (RCMP) will be completed in 2nd QTR 2017. This establishes a risk management and oversight process for identifying, assessing, and monitoring risks and develops risk reduction/mitigation plans. It will provide a tool for the project team to proactively manage risks on the project. It is Sound Transit's responsibility for continuously monitoring project status and associated risks, taking mitigation actions for known risks identified on the risk register, and effectively responding in a timely manner to those risks that pose a significant impact to the project cost or schedule forecast.

The Tacoma Link Quarterly Risk Review Meeting was held in March 2017. The baseline risk assessment will be held in May 2017.

Project Schedule

The 60% design submittals for T100 Mainline Construction was submitted on February 17 and were reviewed by ST and its stakeholders. HDR submitted the 60% Cost Estimate and Construction Schedule in mid-March. Due to the number of questions received and changes to the technical specifications for the automated vehicle locating system, the LRV proposal due date was moved back mid-May 2017. This change does not have an impact on the anticipated NTP date of early December 2017. ST closed on the OMF site at the end of March. An early works contract for demolition of the existing warehouse will be procured prior to the end of 2017. Selection of the CMC firm and negotiations were completed for Phase I Preconstruction Services.. The ST Board approved of the Right-of-Use and MOU for funding in March.



Activity ID	Activity Name	Start	Finish
	Tacoma Link Expansion - Master Sc	02-Feb-15 A	16-May-22
	Project Administration	02-Feb-15 A	16-May-22
	Milestones	02-Feb-15 A	16-May-22
	Project Funding	23-Mar-17	17-Oct-17
	Project Planning & Reporting	02-Feb-15 A	27-Jul-17
	Contract Packaging	01-Oct-15 A	31-Dec-15 A
	Owner Furnished Materials / Equipmer	16-Mar-16 A	18-Jan-21
	Owner Furnished Material	28-Jul-17	18-Jan-19
	Light Rail Vehicles	16-Mar-16 A	18-Jan-21
	Engineering	02-Feb-15 A	21-Mar-18
	Preliminary Design	02-Feb-15 A	27-May-16 A
	Final Design	04-Jan-16 A	08-Dec-17
	Permitting	01-Nov-16 A	21-Mar-18
	Engineering Support	29-Apr-15 A	27-May-16 A
	ROW & Third Parties	01-Oct-15 A	22-Jan-19
	Property Acquisition	17-Dec-15 A	22-Jan-19
	Third Party Agreements	01-Oct-15 A	28-Sep-17
	Construction	11-Dec-17	03-Aug-21
	Construction Contract Procurement	15-Dec-17	21-Jun-18
	T100 - Mainline/OMF/Systems Construction	11-Dec-17	03-Aug-21
	Testing/Startup/Commissioning	02-Nov-20	15-May-22
	Systems Testing	25-May-21	03-Aug-21
	Safety Certification	02-Nov-20	29-Nov-21
	Pre-Revenue Testing	30-Nov-21	13-Jan-22
	Project Float - Owner Controlled	14-Jan-22	15-May-22

Sound Transit Board Actions

Motion Number	Description	Date
M2017-32	*Authorizing the chief executive officer to execute an amendment to the Tacoma Link Right of Use Agreement with the City of Tacoma for the Tacoma Link Expansion.*	*Mar 23*
M2017-33	*Authorizing the chief executive officer to execute a Funding Agreement with the City of Tacoma for the Tacoma Link*	*Mar 23*
R2017-08	*Authorizing the chief executive officer to acquire certain real property interests, including acquisition by condemnation to the extent authorized by law, and to reimburse eligible relocation and reestablishment expenses incurred by affected owners and tenants as necessary for the Tacoma Link Expansion.*	*Mar 23*

Community Outreach

- Met with a community group "Sound Outreach" to provide an overview of and update on the project.
- Presented a project update to the New Tacoma Neighborhood Council.
- Attended the Hilltop Business District meeting to promote the April Public Open House.
- Contacted 15 business owners regarding upcoming vibration testing.
- Sent out mailers for upcoming April 13th Open House to the TLE neighborhood residents and businesses.

Final Design

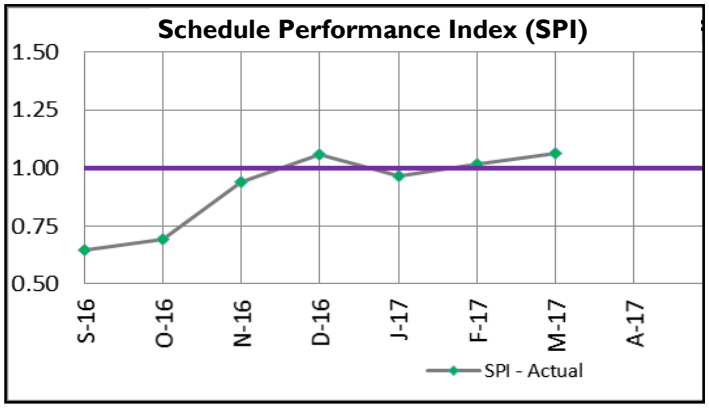
In July 2016 Sound Transit executed a contract with HDR Engineering, Inc. to provide civil and systems final design services for Tacoma Link.

Final Design Key Activities

- Submitted the 60% mainline and OMF design packages for ST and stakeholder review. Review of the OMF package is scheduled to be completed in early April.
- Submitted the 60% construction cost and schedule estimates to ST for review.
- Continued coordination meetings with the City of Tacoma regarding TPSS locations.
- Continued work order development with the City of Tacoma for construction services as identified in the Construction Services Agreement.
- Issued remaining addenda, including responses to Requests for Information, to interested LRV proposers.

Final Design Schedule Performance

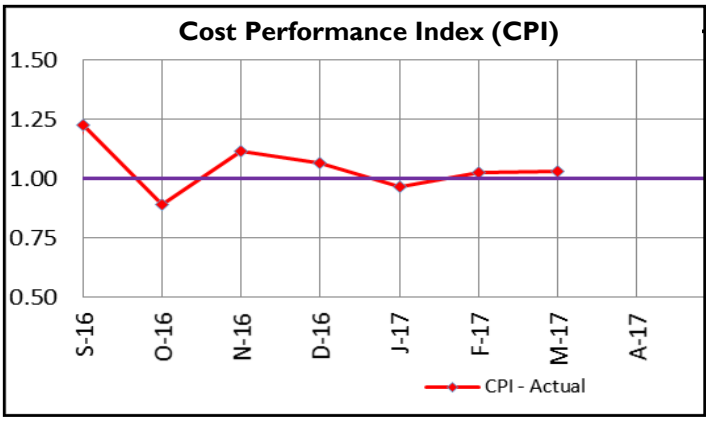
The cumulative Schedule Performance Index (SPI) for the Final Design contract is 1.06 through March, indicating that the project Earned Value is slightly ahead of the original plan.



Final Design Cost Performance

Contract expenditures through March totaled $3.4M, approximately 45% of the Final Design contract total.

The Final Design contract percent complete reported at the end of March is 45%, with an earned value of $3.5M. The cumulative Cost Performance Index (CPI) through March is 1.03; indicating slightly lower expenditures than planned for work performed.



Phase 3 Performance	Cumulative To-Date
Amount Invoiced	$3.4M
% Spent	45%
Earned Value	$3.5M
% Complete	45%
SPI	1.06
CPI	1.03

Link Light Rail
Link Operations and Maintenance Facility: East

Project Summary

Scope: The Link Operations and Maintenance Facility: East (OMFE) project, located in the City of Bellevue, includes a 160,000 sq. ft. operations and maintenance facility that will maintain, store and deploy and expanded light rail fleet, including seven LRV service bays, a LRV wash facility and storage for up to 96 LRVs.

Budget: $449.2 Million (Baselined July 2016)

Schedule: Project completion December 2020

Phase: Proceed to Construction

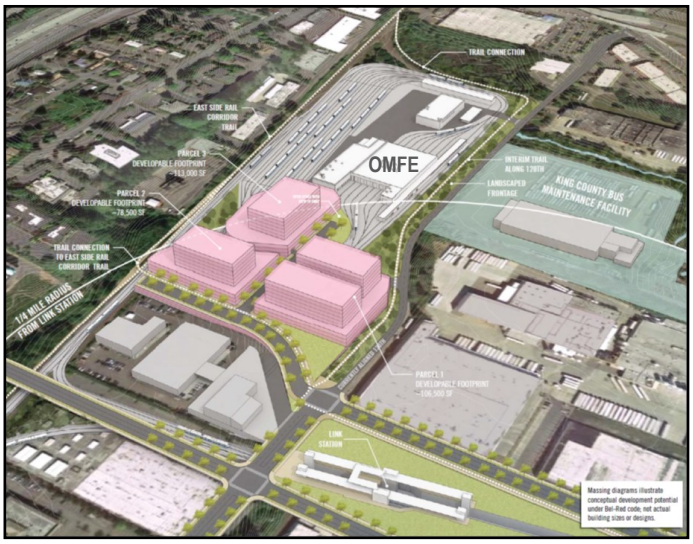


Illustration of OMFE site

Key Activities

- Continue negotiations with City of Bellevue (COB) regarding acquisition of spur property for construction staging.
- Concept Validation/negotiation phase continued with highest ranking firm Hensel Phelps. ST Board action anticipated May 2017 followed by NTP June 2017.
- Incorporate Value Management review observations into negotiation plan with Hensel Phelps.
- Design Build PM contract anticipate ST Board in April and NTP in May 2017.
- King County Sewer replacement agreement has been finalized.
- The critical areas permit application has been submitted and under review by COB.
- Revised draft PMP submitted to PMOC for further review.

Closely Monitored Issues

- Property acquisition schedule is on critical path. Condemnation process timeline closely followed and focused on relocations. Possession and Use of all properties are being pursued.

Project Cost Summary

The Adopted Capital Budget for the OMFE is $449.2M. The OMFE cost is summarized into two cost tables. The first table is in accordance with Sound Transit's Work Breakdown Structure (WBS) displaying the cost summary by phase. The second table summarizes project cost by SCC, in accordance to the FTA Standard Cost Categories (SCC) format. Both tables show figures in millions.

Cost Summary by Phase

WBS Phase Elements	Baseline Budget	Current Adopted Budget	Commit-ment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget vs. EFC
Administration	$17.6	$17.6	$2.5	$2.5	$17.6	$0.0
Preliminary Engineering	$9.5	$9.5	$9.0	$8.5	$9.5	$0.0
Final Design	$0.4	$0.4	$0.0	$0.0	$0.4	$0.0
Construction Services	$19.3	$19.3	$0.0	$0.0	$19.3	$0.0
3rd Party Agreements	$3.1	$3.1	$0.0	$0.0	$3.1	$0.0
Construction	$264.8	$264.8	$0.0	$0.0	$264.8	$0.0
ROW	$134.5	$134.5	$39.4	$38.4	$134.5	$0.0
Total	**$449.2**	**$449.2**	**$50.9**	**$49.5**	**$449.2**	**$0.0**

The Estimated Final Cost (EFC) for this reporting period is $449.2M, which is equal to the current project budget. This period approximately $14.8M was incurred, of which $14.7M was ROW acquisitions and relocation, and remaining balances of $0.1M from Administration and PE expenditures.

The Estimated Final Cost (EFC) reflects the current adopted budget values due to project baselined this period. ROW and Professional Services categories continue to show the only activity at this stage of the project.

Cost Summary by SCC

Project Elements by Phase	Baseline Budget	Current Adopted Budget	Commit-ment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget vs. EFC
10 Guideway & Track Elements	$2.6	$2.6	$0.0	$0.0	$2.6	$0.0
20 Stations	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0
30 Support Facilities: Yards, Shops	$140.1	$140.1	$0.0	$0.0	$140.1	$0.0
40 Sitework & Special Conditions	$43.6	$43.6	$0.0	$0.0	$43.6	$0.0
50 Systems	$43.0	$43.0	$0.0	$0.0	$43.0	$0.0
Construction Subtotal (SCC 10-50)	**$229.3**	**$229.3**	**$0.0**	**$0.0**	**$229.3**	**$0.0**
60 Row, Land, Improvements	$134.5	$134.5	$39.4	$38.4	$134.5	$0.0
70 Vehicles	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0
80 Professional Services	$71.6	$71.6	$11.5	$11.1	$71.6	$0.0
90 Unallocated Contingency	$13.8	$13.8	$0.0	$0.0	$13.8	$0.0
Project Total (SCC 10-90)	**$449.2**	**$449.2**	**$50.9**	**$49.5**	**$449.2**	**$0.0**

Cost Contingency Management

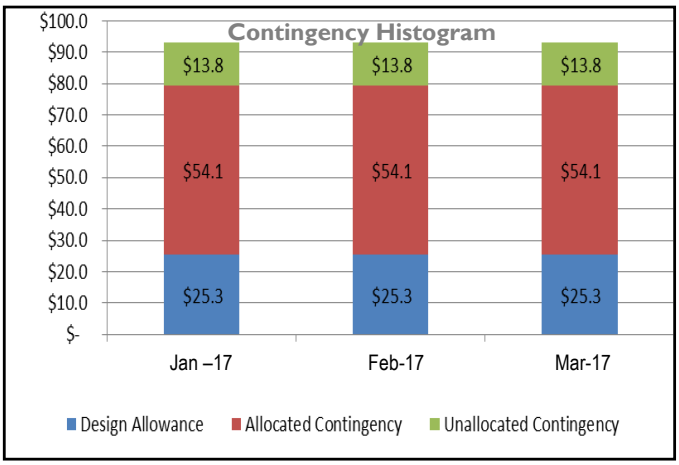
Compared to the Baseline amount of $93.2M, the current Total Contingency remains the same at $93.2M.

Design Allowance (DA) – The baseline DA of $25.3M reflects no changes this period.

Allocated Contingency (AC) – Compared to the baseline amount of $54.1M, AC reflects no changes this period.

Unallocated Contingency (UAC) – The baseline UAC amount of $13.8M reflects no changes this period.

Contingency Status	Baseline		Current	
	Amount	% of Total	Amount	% of Work Remaining
Design Allowance	$ 25.3	5.6%	$ 25.3	6.1%
Allocated Contingency	$ 54.1	12.0%	$ 54.1	13.1%
Unallocated Contingency	$ 13.8	3.1%	$ 13.8	3.3%
Total	**$ 93.2**	**20.7%**	**$ 93.2**	**22.5%**



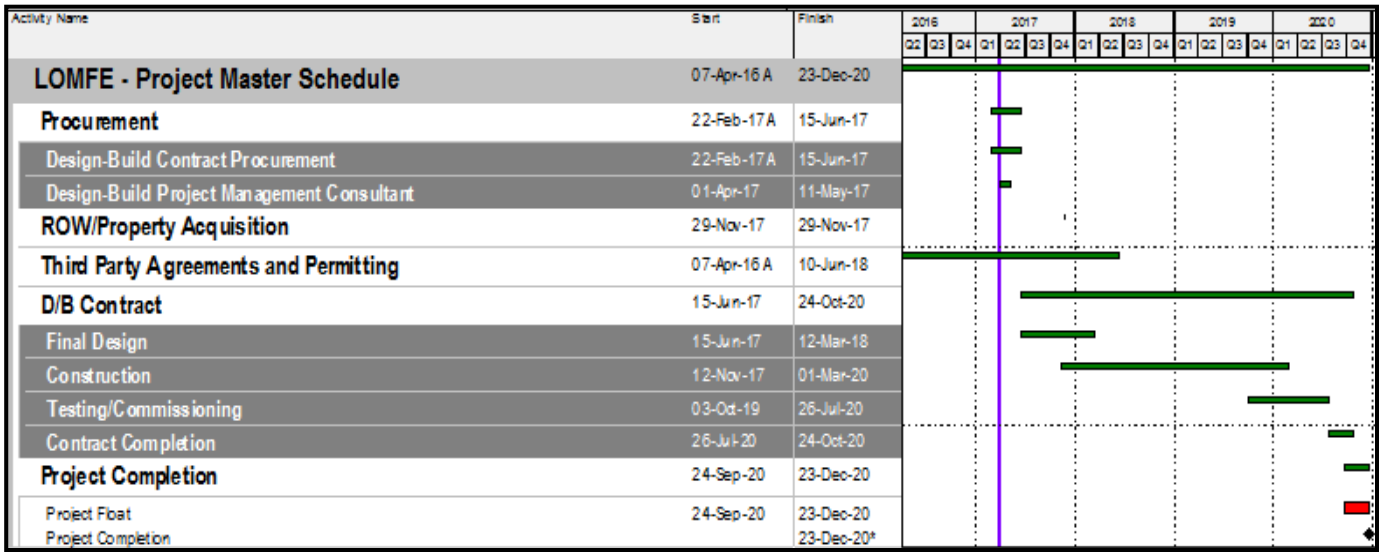
Risk Management

The Project has a Risk Register and is actively managing risks. The following are the top five risks:

- Volatility in property values around the Bel-Red area.
- Third Party requesting additional site improvements not included in the agreements.
- Authorities Having Jurisdiction not fully communicating all of their requirements to Sound Transit.
- Interface point of E335/OMFE and ST3 leads to change in track, systems, signals and utilities.
- Third Party not satisfied with progress on MDP causing reconsideration for use of a development agreement process and late permits.

Project Schedule

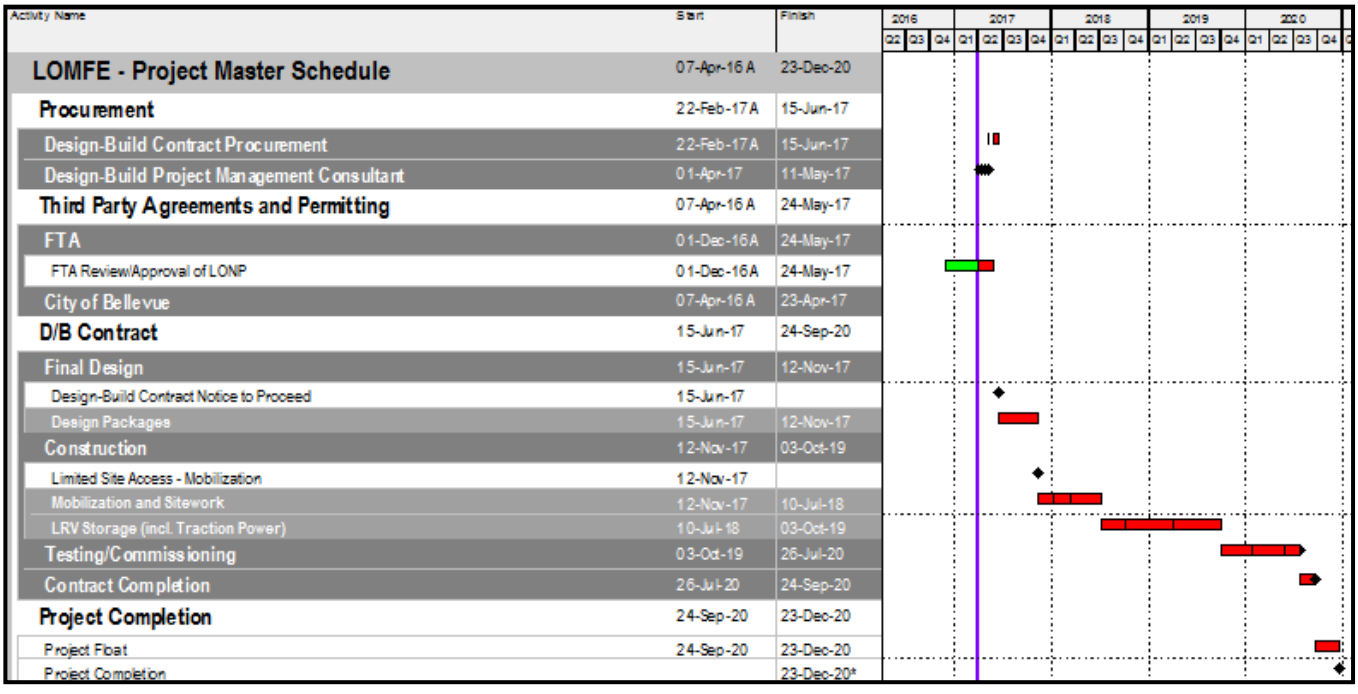
The project schedule is presented below and reflected in the baseline. Concept validation with Hensel Phelps began this period and will continue through April. Property acquisitions are trending ahead of schedule to be completed in late 2017, in time for construction to start as planned.



Activity Name	Start	Finish
LOMFE - Project Master Schedule	07-Apr-16 A	23-Dec-20
Procurement	22-Feb-17 A	15-Jun-17
Design-Build Contract Procurement	22-Feb-17 A	15-Jun-17
Design-Build Project Management Consultant	01-Apr-17	11-May-17
ROW/Property Acquisition	29-Nov-17	29-Nov-17
Third Party Agreements and Permitting	07-Apr-16 A	10-Jun-18
D/B Contract	15-Jun-17	24-Oct-20
Final Design	15-Jun-17	12-Mar-18
Construction	12-Nov-17	01-Mar-20
Testing/Commissioning	03-Oct-19	26-Jul-20
Contract Completion	26-Jul-20	24-Oct-20
Project Completion	24-Sep-20	23-Dec-20
Project Float	24-Sep-20	23-Dec-20
Project Completion		23-Dec-20*

Critical Path Analysis

The project critical path this period continues to follow the Letter of No Prejudice (LONP) to allow work to progress ahead of the Full Funding Grant Agreement for Lynnwood Link Extension. The contract will not be awarded without receipt of the LONP. The current conceptual construction plan shows a critical path through the foundation design and subsequently the construction of the LRV storage yard. The critical path of construction is subject to change upon award of the DB contract.



Activity Name	Start	Finish
LOMFE - Project Master Schedule	07-Apr-16 A	23-Dec-20
Procurement	22-Feb-17 A	15-Jun-17
Design-Build Contract Procurement	22-Feb-17 A	15-Jun-17
Design-Build Project Management Consultant	01-Apr-17	11-May-17
Third Party Agreements and Permitting	07-Apr-16 A	24-May-17
FTA	01-Dec-16 A	24-May-17
FTA Review/Approval of LONP	01-Dec-16 A	24-May-17
City of Bellevue	07-Apr-16 A	23-Apr-17
D/B Contract	15-Jun-17	24-Sep-20
Final Design	15-Jun-17	12-Nov-17
Design-Build Contract Notice to Proceed	15-Jun-17	
Design Packages	15-Jun-17	12-Nov-17
Construction	12-Nov-17	03-Oct-19
Limited Site Access - Mobilization	12-Nov-17	
Mobilization and Sitework	12-Nov-17	10-Jul-18
LRV Storage (incl. Traction Power)	10-Jul-18	03-Oct-19
Testing/Commissioning	03-Oct-19	26-Jul-20
Contract Completion	26-Jul-20	24-Sep-20
Project Completion	24-Sep-20	23-Dec-20
Project Float	24-Sep-20	23-Dec-20
Project Completion		23-Dec-20*

Right of Way

Filed condemnations and requested Possession and Use of all properties. ST responded to second counter offer for one property. One parcel granted P & U. One parcel vacated. The right - of-way program status of the OMFE as of this period is summarized below:

Link OMFE Property Acquisition Status								
	Board Approved	Offers Made/In Negotiations	Signed Agreements	Possession and Use	Admin Settlement	Closings to date	Relocations Required	Parcels Vacated
Total*	11	9	0	2	0	1	30	2

**Number totals may differ from other reports due to the timing of reporting periods.*

Sound Transit Board Actions

Board motions and resolutions directly related to OMFE is summarized in the table below.

Motion Number	Description	Date
	None to report.	

Environmental

- None to report.

Community Outreach

- OMFE public website updates in progress.

Phase 2 Preliminary Engineering and FEIS

Sound Transit executed a professional services agreement with Huitt-Zollars, Inc. (HZ) for civil engineering and architectural design services and issued NTP for Phase 2, Preliminary Engineering and FEIS for the Link Operations and Maintenance Facility: East (OMFE) in September 2014.

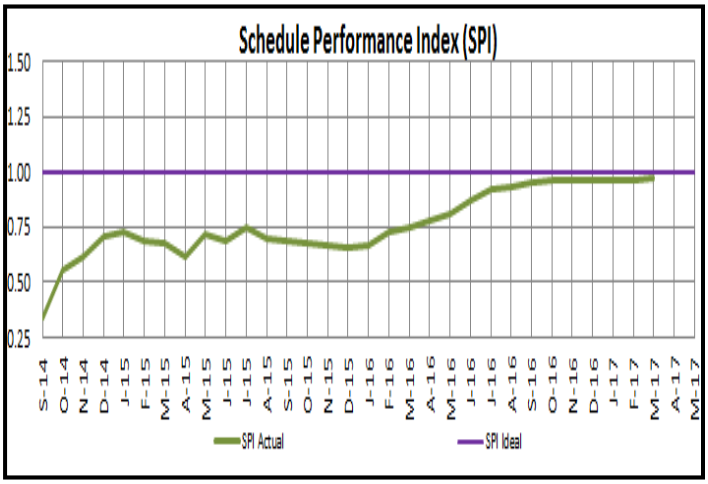
Phase 2 Key Activities

- Continued working on draft agreement with King County regarding mainline sewer replacement .
- JARPA approval received, awaiting City of Bellevue review of CAR prior to implementing fee in lieu program.
- Finalized conformed RFP documents.
- Final design build M200 proposals received and evaluated. Announcement of highest scoring team imminent.
- Received last Phase 2 environmental report from cooperating properties.
- Coordinated permit support services payment concurrence with the City of Bellevue.

Phase 2 Schedule Performance

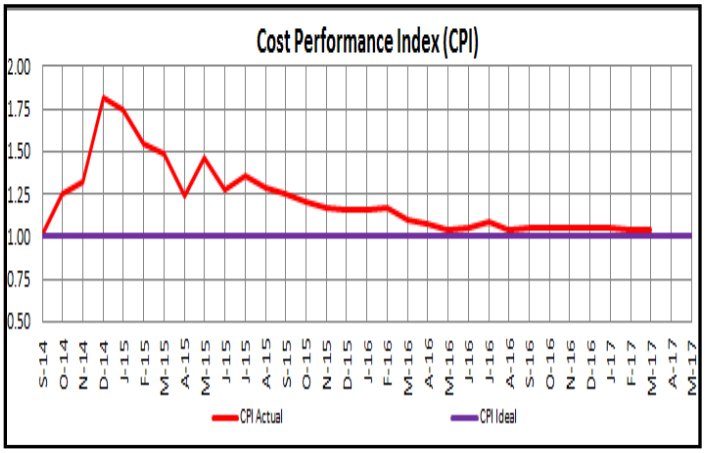
The cumulative Schedule Performance Index (SPI) trends at 0.97 through March 2017, which means that cumulative work accomplished is slightly less than work planned.

Change Order 008 incorporated Phase 2A of the project in February 2016. The remaining Phase 2 scope has been re-negotiated to reflect the recent Design Build Project Delivery decision. The consultant has been directed to focus efforts on delivering Design Build bridging documents to help inform the procurement of the final design build team starting February 2016 as Phase 2A.



Phase 2 Cost Performance

$4.9M of the total Phase 2 contract amount, approximately 93%, was spent through March 2017. The Phase 2 percent complete is 96%, with an earned value of $5.2M. The cumulative Cost Performance Index (CPI) is 1.03, meaning that expenditures are slightly less than the earned value.



Phase 2 Performance	Cumulative To Date
Amount Invoiced	$4.9M
% Spent	93%
Earned Value	$5.2M
% Complete	96%
SPI	**0.97**
CPI	**1.03**

This page left blank intentionally.

Project Summary

Scope: Design, manufacturing, assembly, inspection, testing and delivery of 122 low floor light rail vehicles for service requirements of Northgate, East Link and Lynnwood Link Extensions.

Budget: $733 Million (Baseline September 2015)

Schedule: Project completion 3rd QTR 2024



Graphic rendering of the new light rail vehicles.

Key Activities

- Review and approval of Contract Delivery Requirement List (CDRL) submittal prior to the start of manufacturing.
- Review and preparation of the feasibility of execution of the contractual options.

Project Cost Summary

The ST2 LRV Fleet Expansion project cost is summarized below by two cost categorizations. In the first table, the cost is summarized in accordance with Sound Transit's Work Breakdown Structure (WBS) at the Phase Level. The second table summarizes the project in accordance with the FTA's Standard Cost Category (SCC) format. Both tables below are shown in millions. TIFIA is financing 28% of this project.

To date, approximately $9.9M has been expended. This period LTK reviewed engineering submittals, vehicle test plan and quality assurance plan. Siemen's Industry prepared engineering submittals. Negotiation for the excising of option LRV was completed. The approval for the change order is pending ST Board approval.

Cost Summary by Phase

Project Phase	Baseline Budget	Current Adopted Budget	Commit-ment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget Vs. EFC
Administration	$4.7	$4.7	$0.3	$0.3	$4.7	$0.0
Construction Services	$14.1	$14.1	$4.8	$1.9	$14.1	$0.0
Vehicles	$714.2	$714.2	$524.0	$7.7	$714.2	$0.0
Total	**$733.0**	**$733.0**	**$529.1**	**$9.9**	**$733.0**	**$0.0**

Cost Summary by SCC

SCC Element	Baseline Budget	Current Adopted Budget	Commit-ment to Date	Incurred to Date	Estimated Final Cost (EFC)	Adopted Budget Vs. EFC
Construction Subtotal (SCC 10-50)	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0
60 ROW, Land, Existing Improvements	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0
70 Vehicles	$696.7	$696.7	$528.7	$9.6	$696.7	$0.0
80 Professional Services	$4.5	$4.5	$0.3	$0.3	$4.5	$0.0
90 Contingency	$31.7	$31.7	$0.0	$0.0	$31.7	$0.0
Capital Total (SCC 10-90)	**$733.0**	**$733.0**	**$529.1**	**$9.9**	**$733.0**	$0.0

Cost Contingency Management

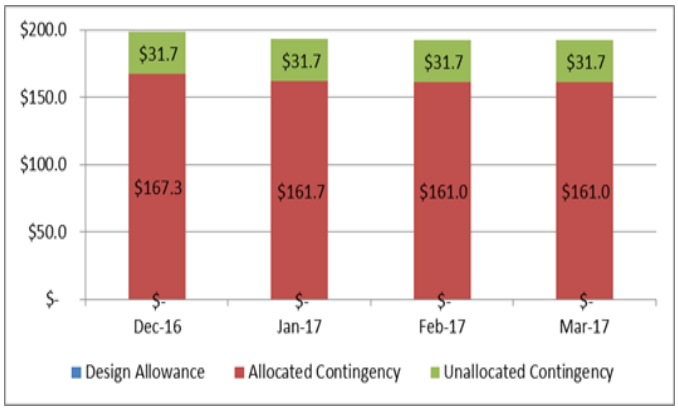
The project's baseline budget includes a total of $78.0M of Total Contingency. The project's Total Contingency is currently at $192.7M or about 37.1% of remaining work in the project.

Design Allowance – This project contains $0 design allowance.

Allocated Contingency – The project baseline's allocated contingency was $46.3M. Due to a successful vehicle procurement and a lower than anticipated contract value the allocated contingency was realized at $167.3M in October 2016. The current available allocated contingency is $161M.

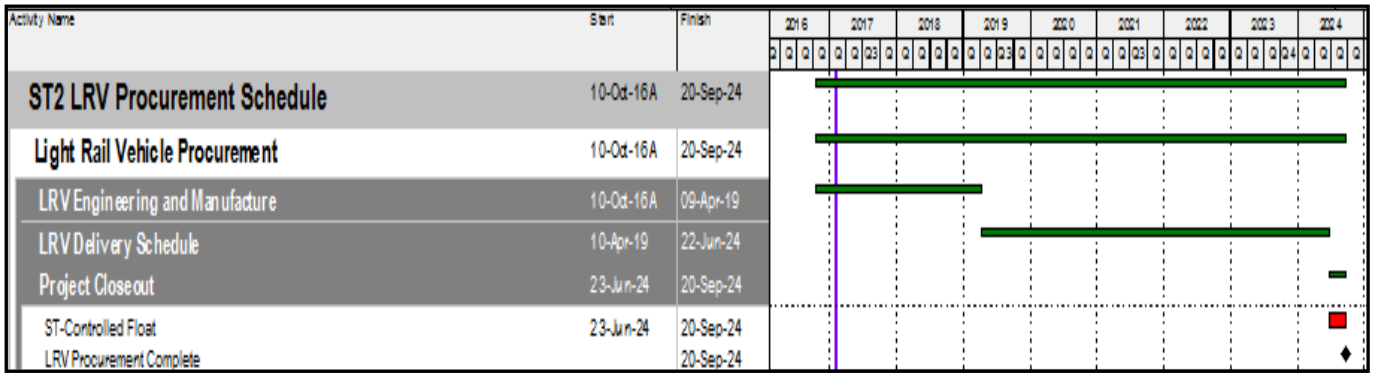
Unallocated Contingency – The project baseline's unallocated contingency remained unchanged at $31.7M.

Contingency Status	Baseline		Current	
	Amount	% of Total	Amount	% of Work Remaining
Design Allowance	$ -	0.0%	$ -	0.0%
Allocated Contingency	$ 46.3	6.4%	$ 161.0	31.0%
Unallocated Contingency	$ 31.7	4.3%	$ 31.7	6.1%
Total	**$ 78.0**	**10.7%**	**$ 192.7**	**37.1%**



Project Schedule

The ST2 LRV Expansion project schedule is shown below.



Sound Transit Board Actions

Motion Number	Description	Date
	None to report.	

Project Staffing – Link Light Rail Program – FEBRUARY 2017

Staffing variance reported in the following section is relative to the April 2016 Agency Staffing Plan (Version 6). During this period, the following positions were filled to support the Link capital program:

Recruiting is on-going for the following design, engineering and construction management staff.

Position	Project Assignment	Planned Hire Date
Sr. Project Manager	Sounder Capital Program	2016 Attrition*
Civil Engineer	Right-of-Way	2016 Attrition*
Civil Engineer	East Link Extension	2016 Attrition*
Design Technology Specialist	Civil and Structural Engineering	2016 Attrition*
Design Technology Specialist	Civil and Structural Engineering	2016 Attrition
Electrical Engineer	Systems Engin. and Integration	2016 Attrition*
Principal Architect	Federal Way Link Extension	2016 Attrition*
Corridor Design Manager	Systems	2016 Attrition*
Deputy Director	Construction Management	2016 Attrition*
Construction Contracts Claims Sp.	Construction Management	2016 Attrition*
Sr. Systems Engineer	LRV Procurement and Testing	2016 Attrition
Sr. Real Property Agent	Appraisals	2016 Attrition
Sr. Project Manager	Operations and Maintenance Facility: East	2016 Reclass
Executive Project Director	Executive Director's Office	2016 Reclass
Construction Manager	Systems	2017 Attrition
Sr. Systems Engineer	Operations Technology	January 2017
Sr. Systems Engineer	Operations Technology	January 2017
Deputy Director	East Link Extension	February 2017
Executive Project Director	Federal Way Link Extension	February 2017
Project Coordinator	Operations and Maintenance Facility: East	February 2017
Civil Engineer	Non-Corridor Projects – Operations Supt.	February 2017
Project Control Supervisor	Tac./Fed. Way –Tac Link Extension	February 2017
Construction Manager	Systems	February 2017
Construction Manager	Lynnwood Link Extension	February 2017
Assistant Manager	Property Management	February 2017
Project Manager	Public Art (ST*art)*	February 2017

**Attrition resulting from internal promotion*

Project Staffing – Link Light Rail Program continued

There were 594.1 (94% of plan) consultant and internal staff full time equivalents (FTE) participating in the on-going planning, design and construction of Link light rail extensions in February. Staffing was within 10% of plan for the Tacoma, Lynnwood, East, and Northgate Link Extensions. Staffing for the Federal Way Link Extension was 69% of plan while staffing for the Link Operations & Maintenance Facility: East was 40% of plan.

Staffing variance to plan for the month and year to date summary by project follows.

	February 2017 Staffing Plan Variance Summary											
	Consultant Staff				Sound Transit Staff				Total (AVG. YTD)			
	FTE		Variance		FTE		Variance		FTE		Variance	
Project	Planned	Actual	FTE	% of Plan	Planned	Actual	FTE	% of Plan	Planned	Actual	FTE	% of Plan
Federal Way Link Extension	42.0	37.9	-4.1	90%	37.9	17.0	-20.9	45%	79.9	54.4	-25.6	68%
Northgate Link Extension	71.1	88.4	17.3	124%	44.2	37.9	-6.3	86%	115.3	119.1	3.8	103%
OMF East	6.6	1.9	-4.7	29%	16.7	7.5	-9.2	45%	23.3	7.9	-15.4	34%
East Link Extension	66.8	70.8	4.1	106%	72.1	60.0	-12.0	83%	138.8	146.2	7.4	105%
Lynnwood Link Extension	178.0	187.8	9.8	105%	51.9	39.5	-12.4	76%	229.9	231.0	1.1	100%
Tcoma Link Expansion	24.0	31.5	7.5	131%	18.1	13.9	-4.2	77%	42.1	44.8	2.7	107%
Total	388.5	418.3	29.9	108%	240.9	175.8	-65.1	73%	629.3	603.3	-26.0	96%

Federal Way Link Extension Staffing

Total Internal and External Staffing – Federal Way Link Extension

There were 54.8 (68% of plan) consultant and internal staff on the Federal Way Link Extension in February including 37.9 consultant FTE (90% of plan) and 17 internal FTE (45% of plan). Average year-to-date (YTD) internal staffing (17.2 FTE) is 53% (20.3 FTE) below plan; YTD consultant staffing (36.5 FTE) is 13% (5.5 FTE) below plan. All consultant staff is supporting preparation of bridging documents that will be included in the solicitation for a Design-Build (DB) contractor to be issued later this year. All consultant staff was supporting preliminary engineering.

	Labor Hours Jan- Feb. 2017				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	25,612.8	17,371.3	-8,241.5	67.8%	12,806.4	8,762.4	-4,044.0	68.4%
Monthly Average	12,806.4	8,685.7	-4,120.8					
Monthly FTE	80.0	54.3	-25.8		80.0	54.8	-25.3	

Operations & Maintenance Facility: East Staffing

Total Internal and External Staffing – OMFE

There were 7.5 internal and 1.9 consultant FTE participating in the Operations & Maintenance Facility: East project in February. This is 48% (3 FTE) above January staffing and 40% of plan Average year-to-date staffing (7.9 FTE/mo.) is 66% (15.9 FTE) below plan. All consultant staff are participating in concept validation of the selected DB contractor.

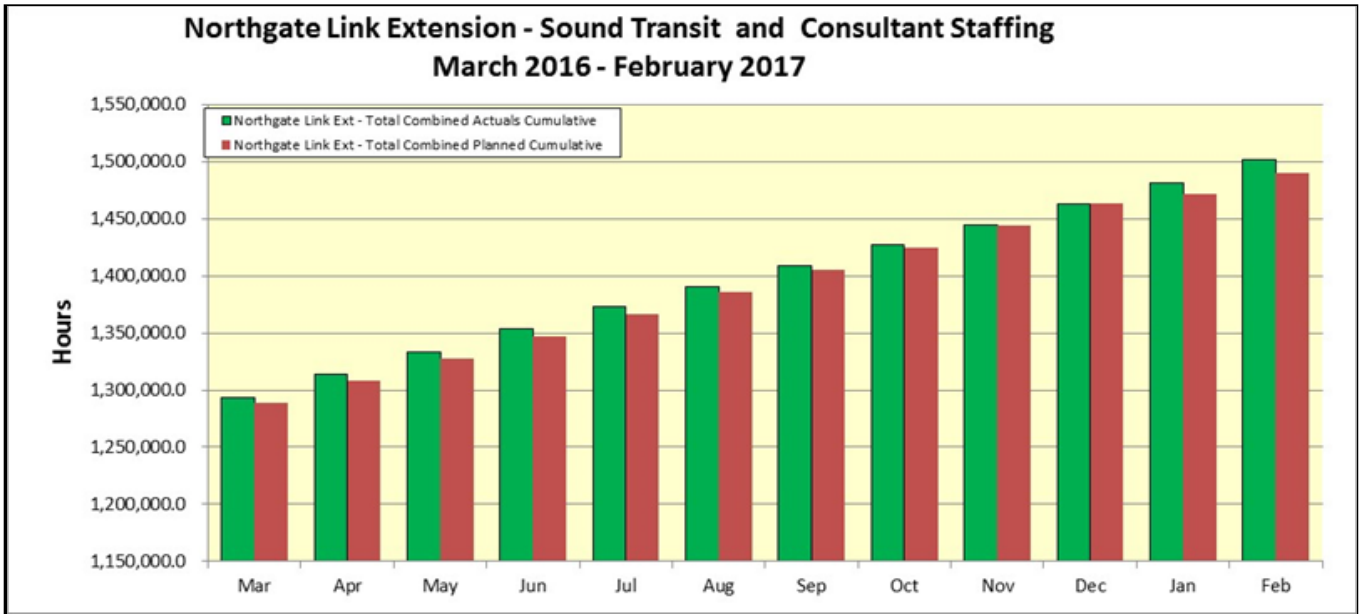
	Labor Hours to Date				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	7,452.8	2,513.4	-4,939.4	33.7%	3,726.4	1,500.0	-2,226.4	40.3%
Monthly Average	3,726.4	1,256.7	-2,469.7					
Monthly FTE	23.3	7.9	-15.4		23.3	9.4	-13.9	

This page left blank intentionally.

Northgate Link Extension Staffing

Total Internal and External Staffing – Northgate Link Extension

During February, 37.9 internal and 88.4 consultant FTE were assigned to the Northgate Link Extension. This is 13% (14.5 FTE) above January staffing and is 10% (11.0 FTE) above plan. Average year-to-date staffing (119.1 FTE/mo.) is 3% (3.8 FTE/mo.) above plan; cumulatively since January 2011 average monthly staffing is trending with plan.

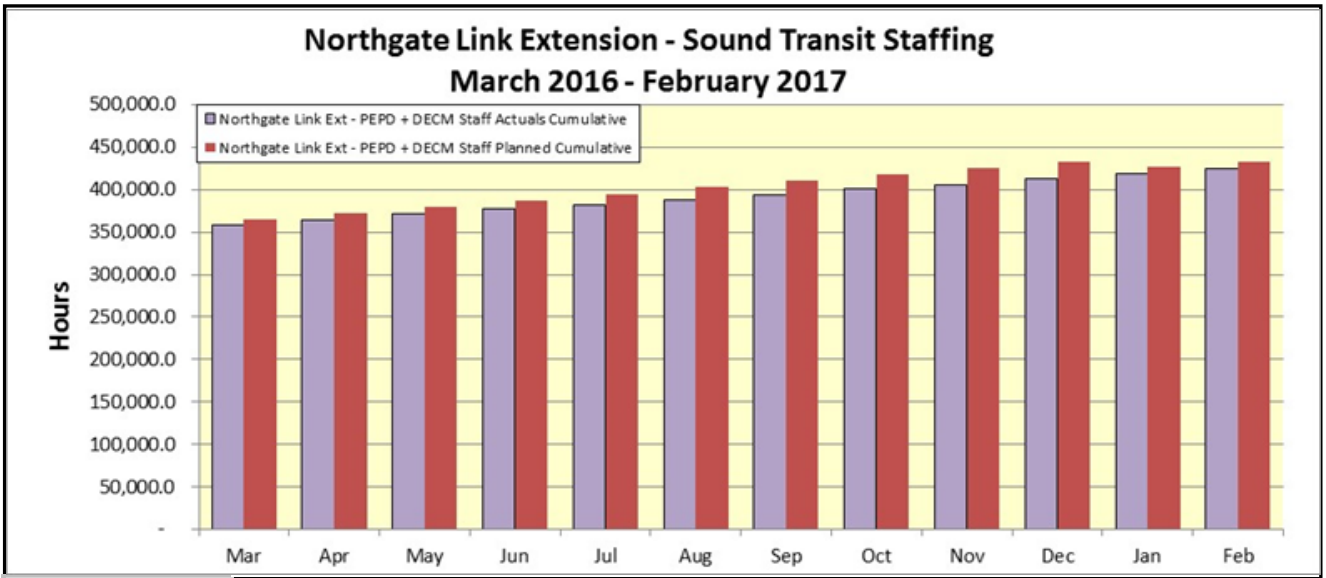


	Labor Hours Jan 2011 - Feb 2017				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	1,489,690.4	1,501,489.7	11,799.3	100.8%	18,444.8	20,211.1	1,766.3	109.6%
Monthly Average	20,131.0	20,290.4	159.5					
Monthly FTE	125.8	126.8	1.0		115.3	126.3	11.0	

Northgate Link Extension Staffing

Internal Resource Commitments to Northgate Link Extension

Internal staffing in February (37.9 FTE) decreased 10% (3.7 FTE) from January and was 14% (6.3 FTE) below plan. Cumulatively, since January 2011, average monthly internal staffing (35.9 FTE/mo.) is 1.9% (0.7 FTE/mo.) below plan.

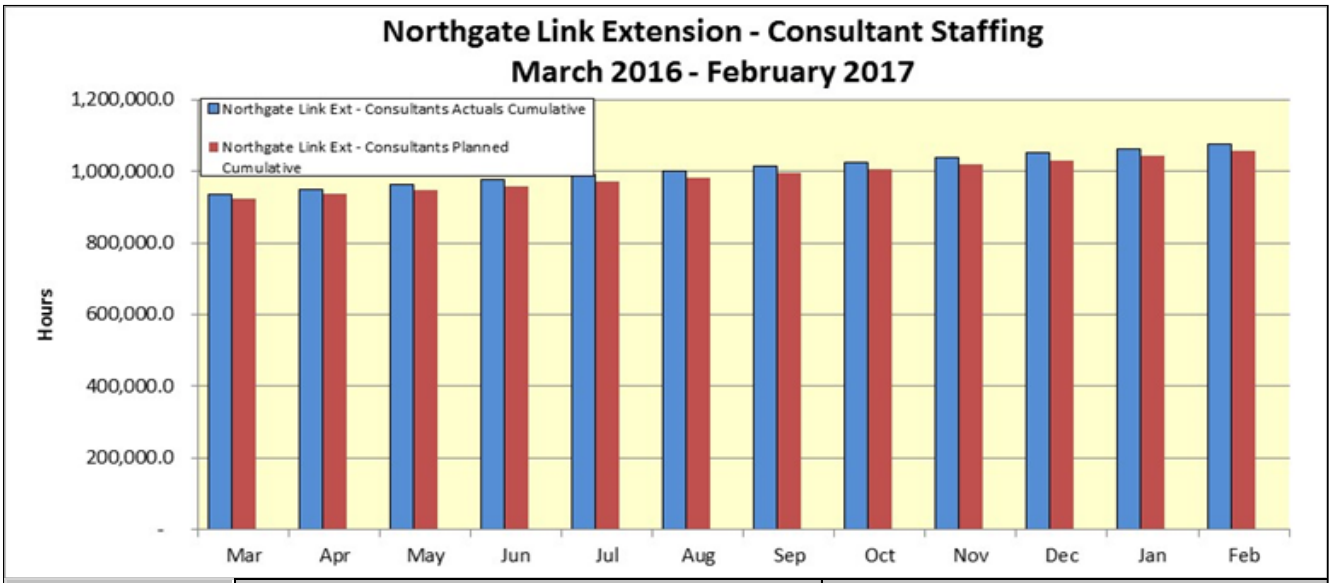


	Labor Hours Jan 2011 - Feb 2017				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	433,227.3	425,152.3	-8,075.0	98.1%	7,068.8	6,062.6	-1,006.2	85.8%
Monthly Average	5,854.4	5,745.3	-109.1					
Monthly FTE	36.6	35.9	-0.7		44.2	37.9	-6.3	

Northgate Link Extension Staffing

Consultant Resource Commitments to Northgate Link Extension

In February, 88.4 consultant FTE were assigned to the Northgate Link Extension; consultant staffing was 26% (18.2 FTE) above January staffing and was 24% (17.3 FTE) above plan. Cumulatively, since January 2011, average monthly consultant utilization (90.9 FTE/mo.) is trending 2% (1.7 FTE/mo.) above plan.



	Labor Hours Jan 2011 - Feb 2017				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	1,056,463.1	1,076,337.4	19,874.3	101.9%	11,376.0	14,148.5	2,772.5	124.4%
Monthly Average	14,276.5	14,545.1	268.6					
Monthly FTE	89.2	90.9	1.7		71.1	88.4	17.3	

Consultant utilization by discipline follows:

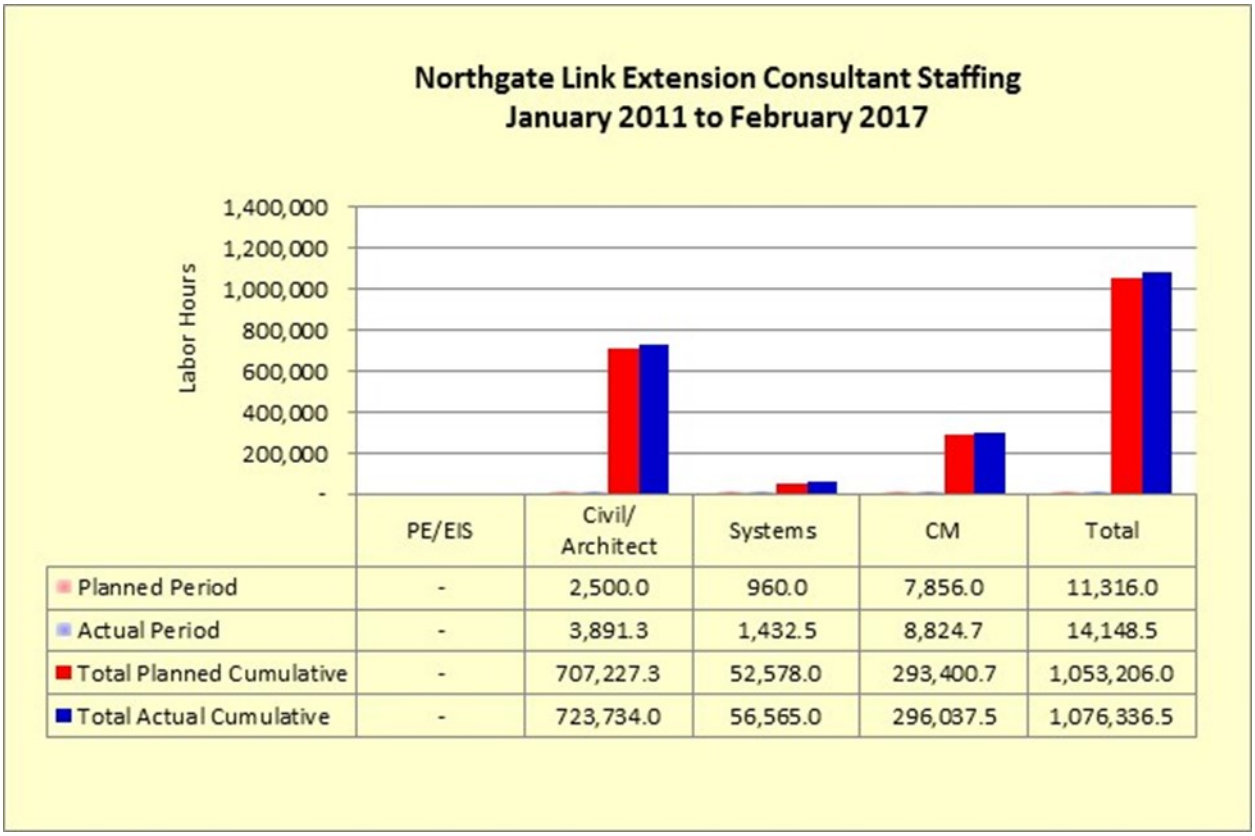
- Civil/architecture consultant utilization in February (24.3 FTE) was 5.4% (1.3 FTE) above January staffing and was 52% (8.3 FTE) above plan. Cumulatively since January 2011, average civil engineering/architecture consultant staffing of 61.1 FTE/mo. is trending within 2% of plan.

- Systems consultant utilization (9 FTE) was 7% (0.6 FTE) above January staffing and was 49% (3 FTE) above plan. Since January 2011, average monthly Systems consultant staffing (4.8 FTE/mo.) is trending with plan.

- CM consultant staffing in February (55.2 FTE) was 42% (16.3 FTE) above January staffing and was 12% (6.1 FTE) above plan. Cumulatively, since January 2011, average monthly CM consultant staffing (25 FTE/mo.) is trending with plan.

Northgate Link Extension Staffing

Consultant Resource Commitments to Northgate Link Extension



	PE/EIS	Civil/ Architect	Systems	CM	Total
Planned Period	-	2,500.0	960.0	7,856.0	11,316.0
Actual Period	-	3,891.3	1,432.5	8,824.7	14,148.5
Total Planned Cumulative	-	707,227.3	52,578.0	293,400.7	1,053,206.0
Total Actual Cumulative	-	723,734.0	56,565.0	296,037.5	1,076,336.5

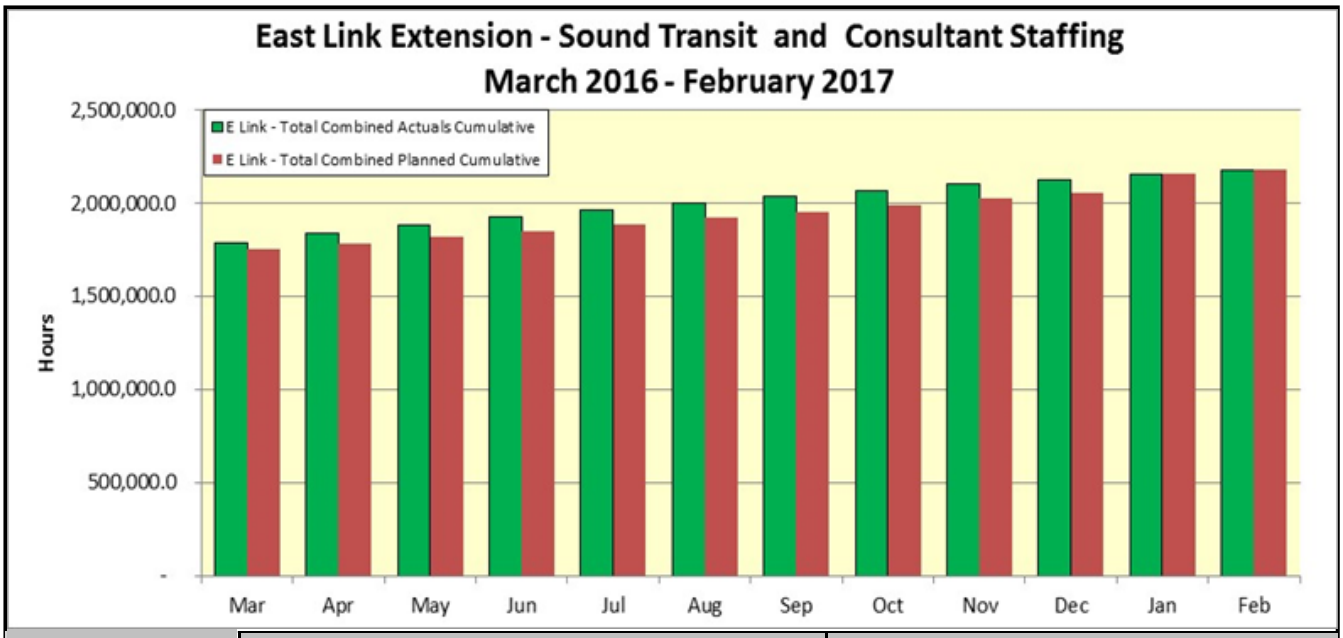
Consultant utilization by discipline follows:

- Civil/architecture consultant utilization January (23.1 FTE) was 17.3% (3.4 FTE) above December staffing and was 44% (7.1 FTE) above plan. Cumulatively since January 2011, average civil engineering/architecture consultant staffing of 61.6 FTE/mo. is trending within 2% of plan.
- Systems consultant utilization (8.4 FTE) was 46% (2.6 FTE) above December staffing and was 39% (2.4 FTE) above plan. Since January 2011, average monthly Systems consultant staffing (4.7 FTE/mo.) is trending with plan.
- CM consultant staffing in January (38.8 FTE) was 24% (12 FTE) below December staffing and was 21% (10.3 FTE) below plan. Cumulatively, since January 2011, average monthly CM consultant staffing (24.6 FTE/mo.) is trending with plan.

East Link Extension Staffing

Total Internal and External Staffing – East ink Extension

Staffing for the East Link Extension in February (130.8 FTE) decreased 19% (30.8 FTE) from January and was 6% (8 FTE) below plan. Average year-to-date staffing of 146.2 FTE/mo. is 5% (7.4 FTE/mo.) above plan. Cumulatively since January 2011, average monthly staffing of 183.8 FTE/mo. is trending with plan.

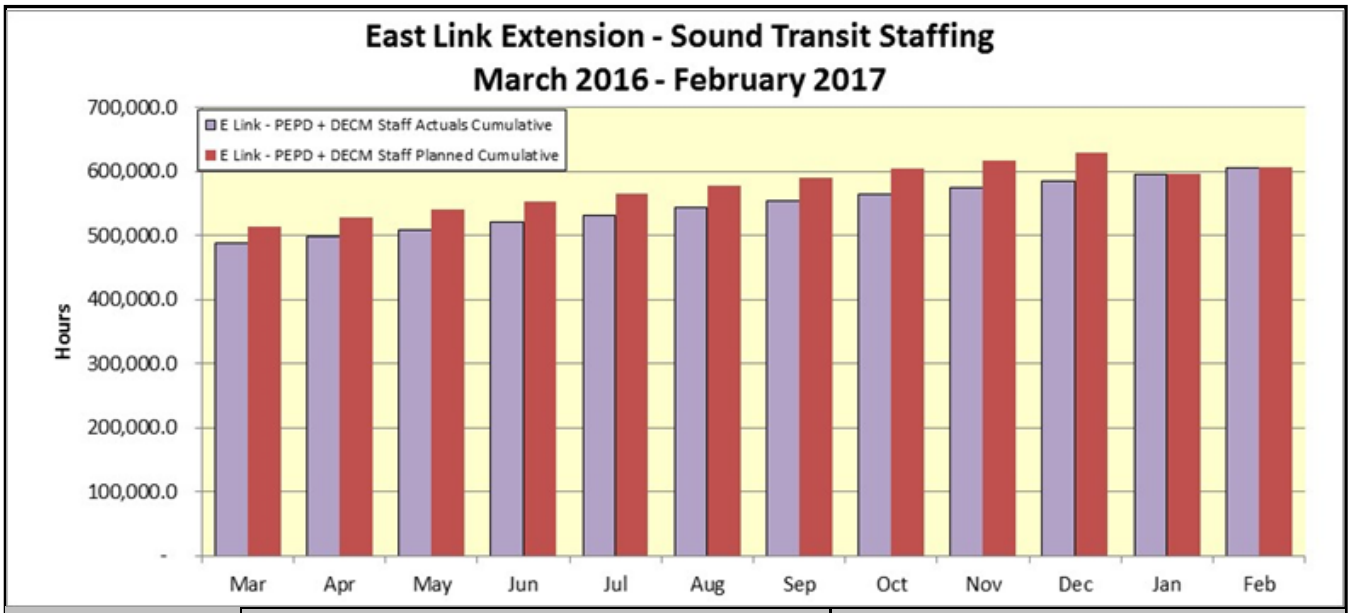


	Labor Hours Jan 2011 - Feb 2017				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	2,182,641.3	2,176,592.0	-6,049.3	99.7%	22,208.0	20,931.0	-1,277.0	94.2%
Monthly Average	29,495.2	29,413.4	-81.7					
Monthly FTE	184.3	183.8	-0.5		138.8	130.8	-8.0	

East Link Extension Staffing

Internal Resource Commitments to East Link Extension

During February internal staffing for the East Link Extension (60 FTE) was 14% (8.4 FTE) below January staffing and was 17% (12 FTE) below plan. Cumulatively, since January 2011, average monthly internal staffing (51.2 FTE/mo.) is trending with plan.

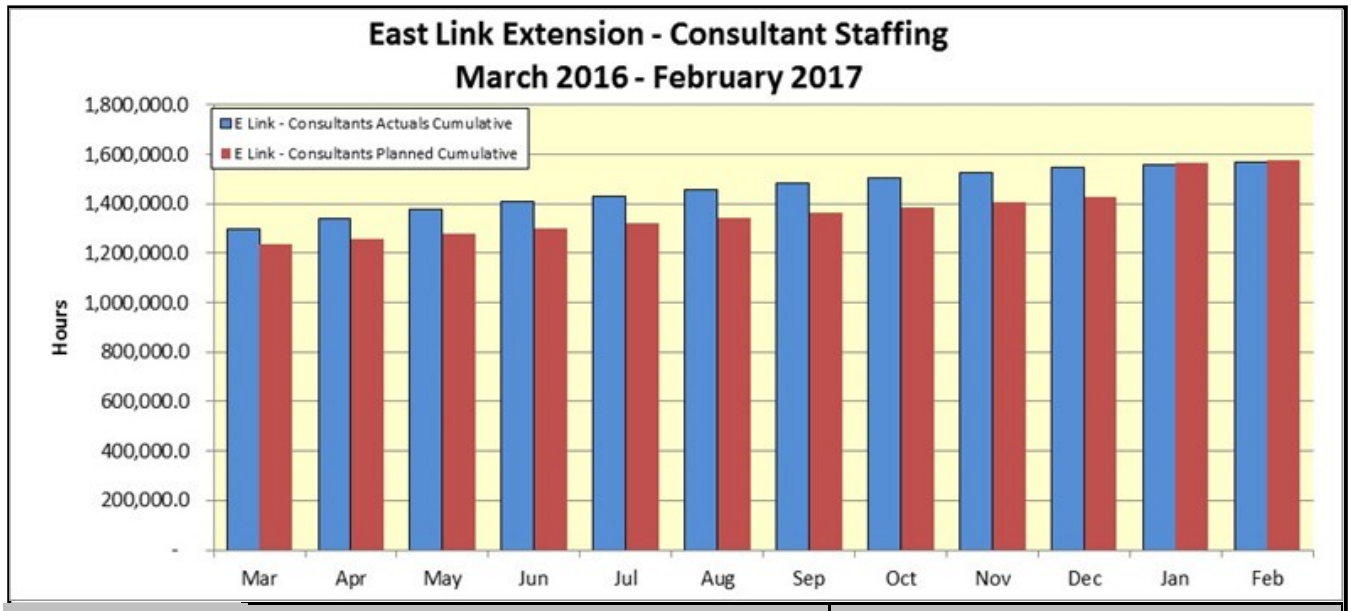


	Labor Hours Jan 2011 - Feb 2017				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	607,590.6	605,662.9	-1,927.7	99.7%	11,528.0	9,600.3	-1,927.7	83.3%
Monthly Average	8,210.7	8,184.6	-26.0					
Monthly FTE	51.3	51.2	-0.2		72.1	60.0	-12.0	

East Link Extension Staffing

Consultant Resource Commitments to East Link Extension

There were 70.8 consultant FTE assigned to the East Link Extension during February. Consultant staffing decreased 24% (22.4 FTE) from January and was 6% (4.1 FTE) above plan. Cumulatively since January 2011, average monthly consultant staffing (132.7 FTE/mo.) is trending with plan.

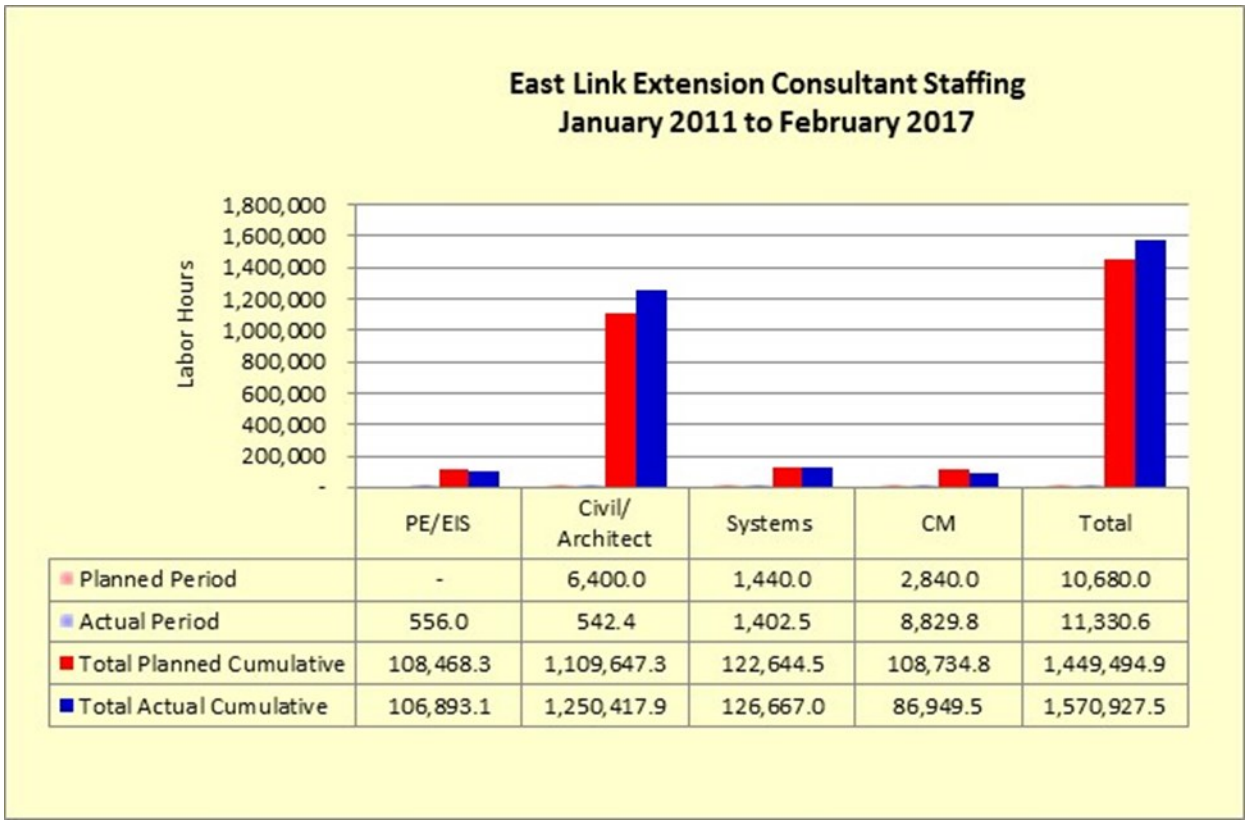


	Labor Hours Jan 2011 - Feb 2017				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	1,575,050.7	1,570,929.1	-4,121.6	99.7%	10,680.0	11,330.7	650.7	106.1%
Monthly Average	21,284.5	21,228.8	-55.7					
Monthly FTE	133.0	132.7	-0.3		66.8	70.8	4.1	

East Link Staffing

Consultant Resource Commitments to East Link, continued

In February construction management consultant staffing (55.2 FTE) was 78% of consultant staffing and was 25.6% (11.2 FTE) above January construction management staffing. Civil engineering consultant staffing (3.4 FTE) comprised 5% of consultant staffing and was 90% (31.5 FTE) below January staffing and 92% (36.6 FTE) below plan. Systems consultant staffing (8.8 FTE) was 3% (0.2 FTE) below plan and was 35% (4.8 FTE) below January staffing. There were 3.5 consultant FTE performing Preliminary Engineering (PE).

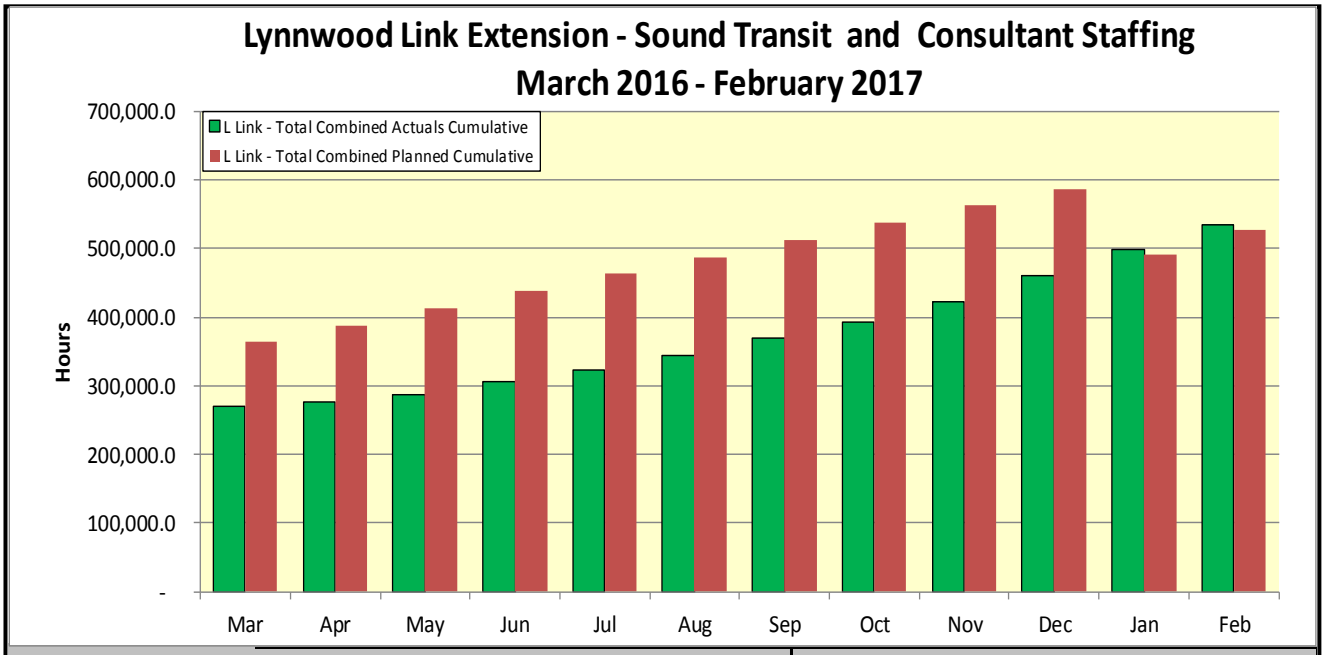


	PE/EIS	Civil/ Architect	Systems	CM	Total
Planned Period	-	6,400.0	1,440.0	2,840.0	10,680.0
Actual Period	556.0	542.4	1,402.5	8,829.8	11,330.6
Total Planned Cumulative	108,468.3	1,109,647.3	122,644.5	108,734.8	1,449,494.9
Total Actual Cumulative	106,893.1	1,250,417.9	126,667.0	86,949.5	1,570,927.5

Lynnwood Link Extension Staffing

Total Internal and External Staffing – S. 200th Link Extension

During February, Lynnwood Link Extension staffing was 227.3 FTE (1% below plan) and included 187.8 consultant FTE (5% above plan) and 39.5 internal FTE (76% of plan). Average year-to-date staffing of 231 FTE/mo is trending with plan; cumulatively since January 2013 staffing is averaging 66.8 FTE/mo. and is also trending with plan.

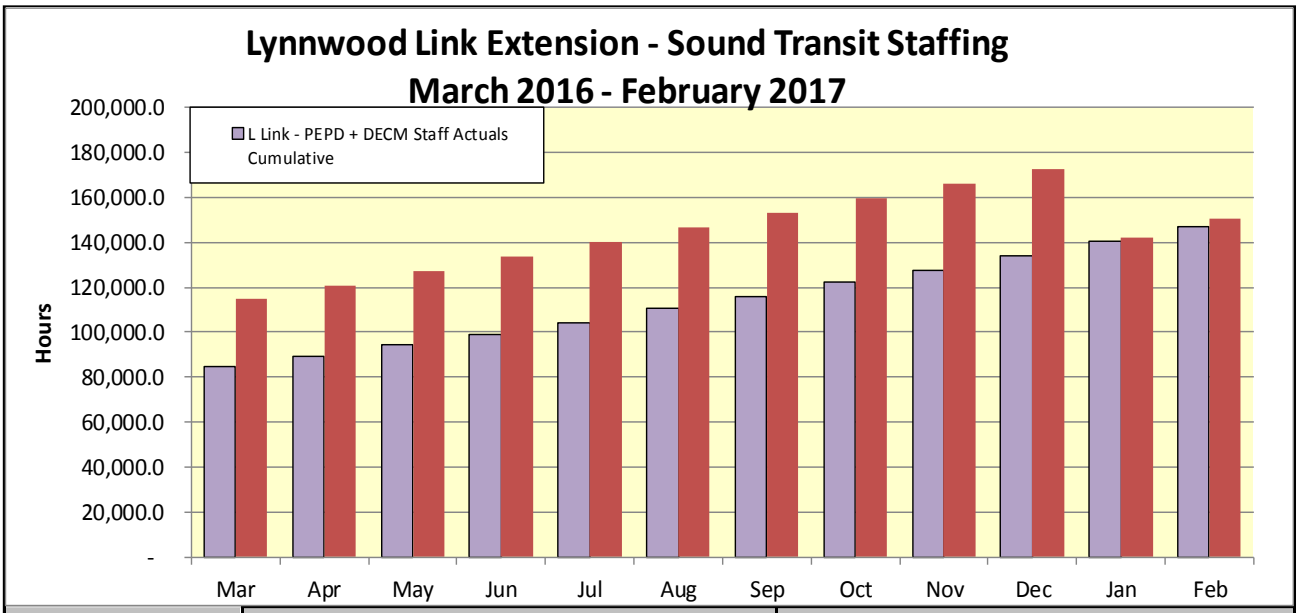


	Labor Hours Jan 2013 - Feb 2017				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	527,012.1	534,351.6	7,339.5	101.4%	36,787.2	36,364.0	-423.2	98.8%
Monthly Average	10,540.2	10,687.0	146.8					
Monthly FTE	65.9	66.8	0.9		229.9	227.3	-2.6	

Lynnwood Link Extension Staffing

Internal Staffing—Lynnwood Link

During February internal staffing for the Lynnwood Link Extension (39.5 FTE) was 10% (3.9 FTE) below January staffing and was 24% (12.4 FTE) below plan. Cumulatively, since January 2013, average monthly internal staffing (18.4 FTE/mo.) is trending with plan.

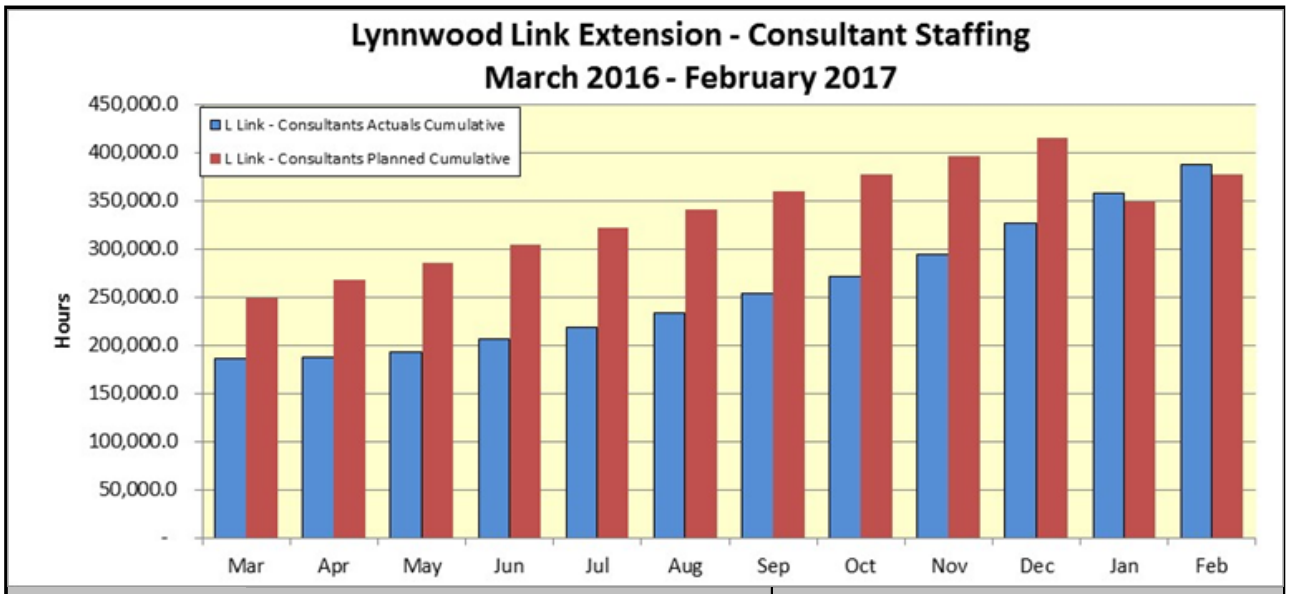


	Labor Hours Jan 2013 - Feb 2017				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	150,368.4	147,019.2	-3,349.2	97.8%	8,307.2	6,321.6	-1,985.6	76.1%
Monthly Average	3,007.4	2,940.4	-67.0					
Monthly FTE	18.8	18.4	-0.4		51.9	39.5	-12.4	

Lynnwood Link Expansion

Consultant Staffing—Lynnwood Link

There were 187.8 consultant FTE assigned to the Lynnwood Link Extension during February. Consultant staffing increased 37% (51.8 FTE) from January and was 5.5% (9.8 FTE) above plan. Cumulatively since January 2013, average monthly consultant staffing (48.4 FTE/mo.) is trending 2.8% (1.3 FTE/mo.) above plan.

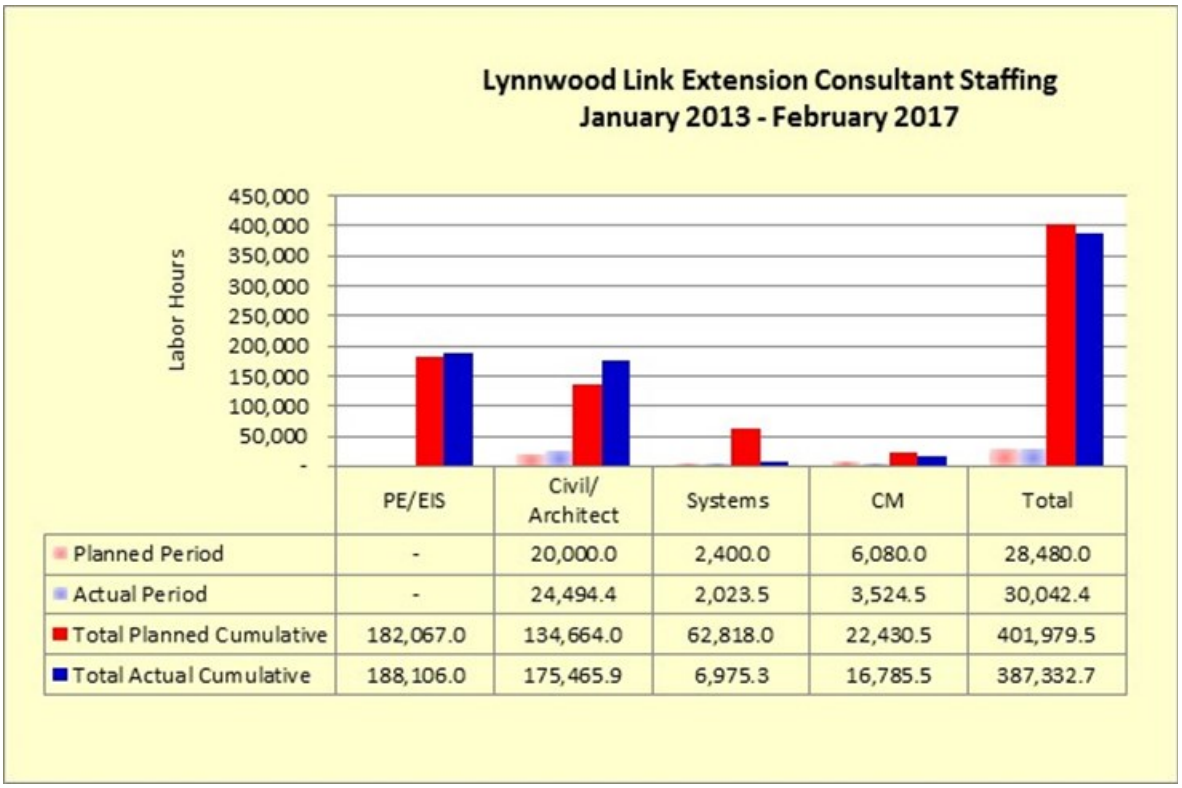


	Labor Hours Jan 2013 - Feb 2017				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	376,643.7	387,332.4	10,688.7	102.8%	28,480.0	30,042.4	1,562.4	105.5%
Monthly Average	7,532.9	7,746.6	213.8					
Monthly FTE	47.1	48.4	1.3		178.0	187.8	9.8	

Lynnwood Link Extension Staffing

Consultant Resource Commitments to Lynnwood Link

Sixty-seven percent of consultant staffing (187.8 FTE) were civil engineering consultants. Additional consultant staffing (22 FTE) provided construction management and systems design (12.6 FTE) support.

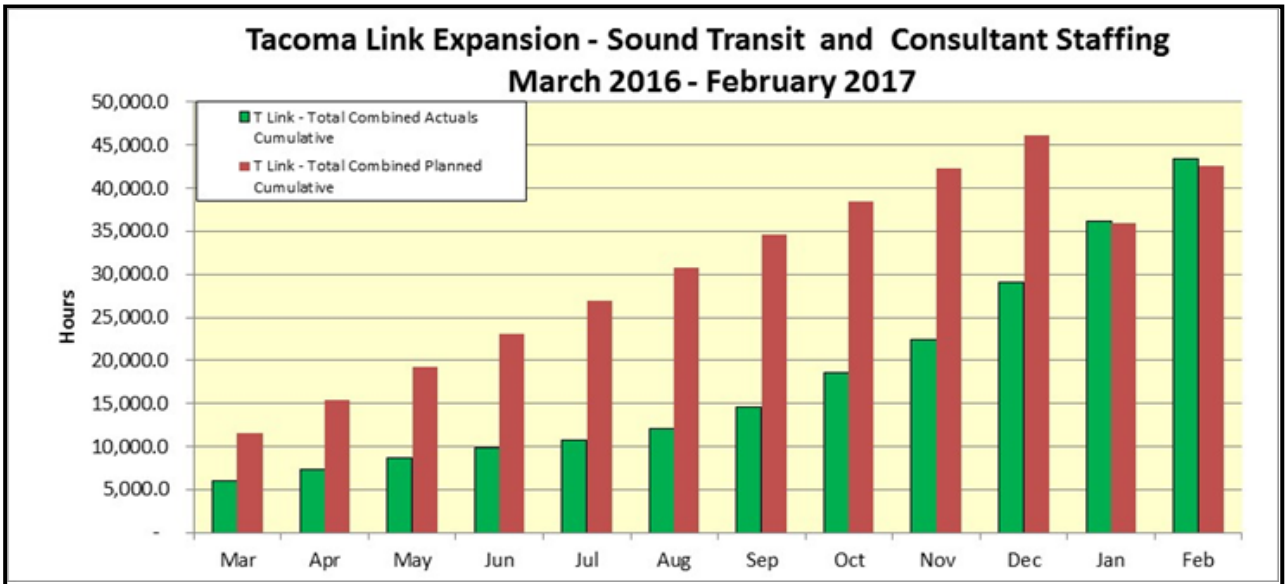


	PE/EIS	Civil/ Architect	Systems	CM	Total
Planned Period	-	20,000.0	2,400.0	6,080.0	28,480.0
Actual Period	-	24,494.4	2,023.5	3,524.5	30,042.4
Total Planned Cumulative	182,067.0	134,664.0	62,818.0	22,430.5	401,979.5
Total Actual Cumulative	188,106.0	175,465.9	6,975.3	16,785.5	387,332.7

Tacoma Link Expansion

Total Internal and Consultant Staffing – Tacoma Link Expansion

During February, there were 45.4 FTE (108% of plan) assigned to the Tacoma Link Expansion including 31.5 consultant FTE (31% above plan) and 13.9 internal FTE (77% of plan). Average year-to-date staffing (44.8 FTE/mo.) is 7% above plan; cumulatively since January 2016 staffing is trending with plan.

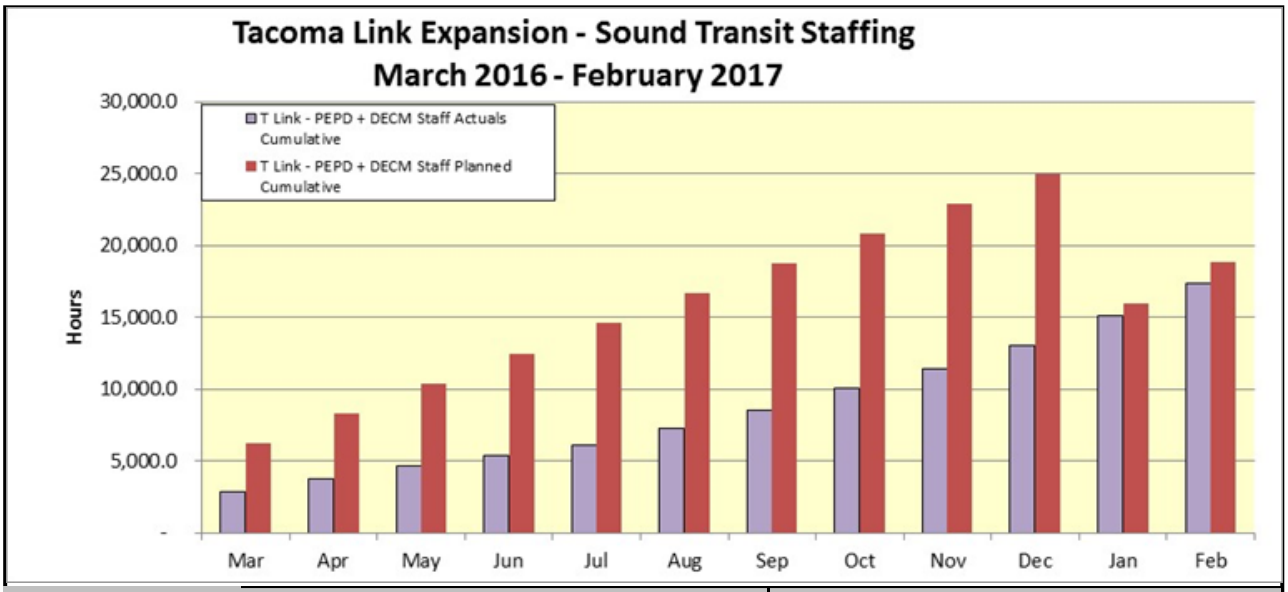


	Labor Hours Jan 2016				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	42,615.5	43,489.0	873.6	102.0%	6,732.8	7,265.4	532.6	107.9%
Monthly Average	3,044.0	3,106.4	62.4					
Monthly FTE	19.0	19.4	0.4		42.1	45.4	3.3	

Tacoma Link Expansion

Internal – Tacoma Link Expansion

During February, internal staffing for the Tacoma Link Extension (13.9 FTE) was 9% (1.2 FTE) above January staffing and was 23% (4.2 FTE) below plan. Cumulatively, since January 2016, average monthly internal staffing (7.7 FTE/ mo.) is trending 8% below plan.

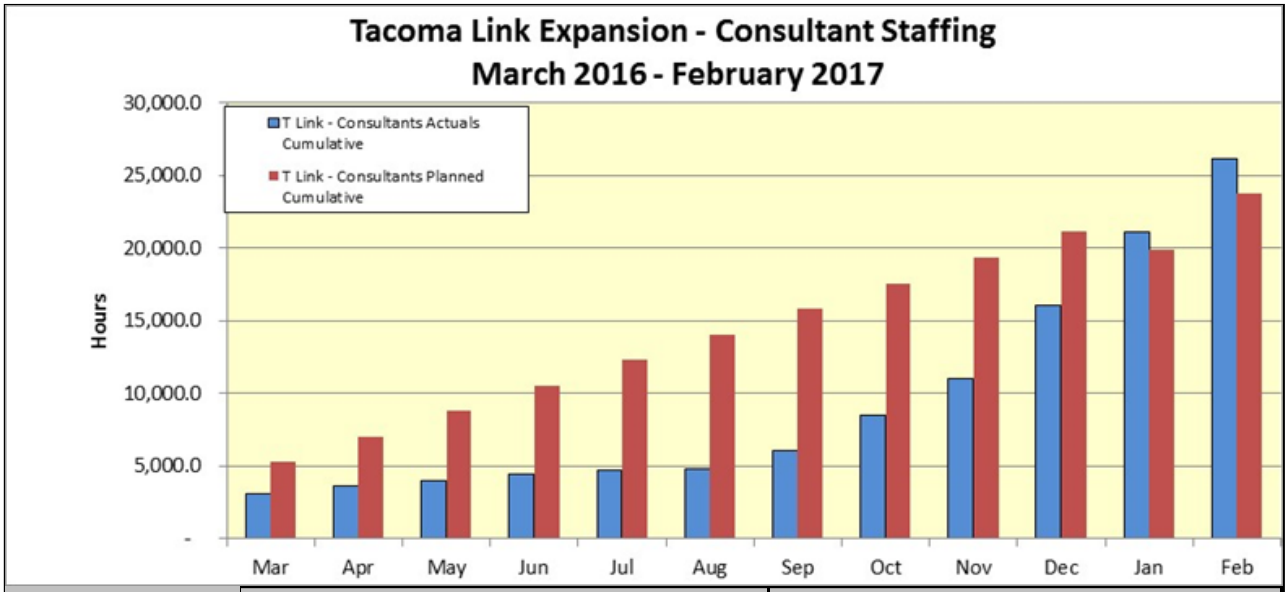


	Labor Hours Jan 2016				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	18,878.6	17,346.4	-1,532.2	91.9%	2,892.8	2,222.5	-670.3	76.8%
Monthly Average	1,348.5	1,239.0	-109.4					
Monthly FTE	8.4	7.7	-0.7		18.1	13.9	-4.2	

Tacoma Link Expansion

Consultant Resource Commitments – Tacoma Link Expansion

Consistent with January staffing, there were 31.5 consultant FTE assigned to the Tacoma Link Extension during February. Consultant staffing was 31% (7.5 FTE) above plan. Cumulatively since January 2016, average monthly consultant staffing (8.8 FTE/mo.) is trending 17% (8.8 FTE/mo.) above plan. All consultant staff assigned to the project (31 FTE) were supporting civil engineering.



	Labor Hours Jan 2016				Current Period Feb 2017			
	Total Planned	Total Actual	Total Variance	Percent Expended	Period Planned	Period Actual	Period Variance	Percent Expended
No. of Hours	23,736.9	43,489.0	19,752.1	183.2%	3,840.0	5,042.9	1,202.9	131.3%
Monthly Average	1,695.5	3,106.4	1,410.9					
Monthly FTE	10.6	19.4	8.8		24.0	31.5	7.5	

ACRONYMS

AA	Alternative Analysis
APE	Area of Potential Impact
BCE	Baseline Cost Estimate
BCWS	Budgeted Cost of Work
BIM	Building Information Modeling
BNSF	Burlington Northern Santa Fe Railway
CCB	Change Control Board
CDF	Controlled Density Fill
CHS	Capitol Hill Station
CM	Construction Management
CMU	Concrete Masonry Unit
CO	Change Order
CPI	Cost Performance Index
CPM	Critical Path Method
DAHP	Department of Archaeology & History Preservation
DART	Days Away, Restricted or Modified
DB	Design -Build
DECM	Design, Engineering and Construction Management
DEIS	Draft Environmental Impact Statement
DPD	Seattle Department of Planning and Development
DSC	Differing Site Conditions
DSDC	Design Support During Construction
DSTT	Downtown Seattle Transit Tunnel
EFC	Estimated Final Cost
EMI	Electro Magnetic Interference
FD	Final Design
FHWA	Federal Highway Administration
FSEIS	Final Supplemental Environmental Impact Statement
FFGA	Full Funding Grant Agreement
FTA	Federal Transit Administration
FTE	Full Time Employee
GC/CM	General Contractor /Construction Management
HVAC	Heating, Ventilation and Air Conditioning
ICD	Integration Control Document
IRT	Independent Review Team
IWP	Industrial Waste Permit
JA	Jacobs Associates
JARPA	Joint Aquatic Resource Permit Application
KCM	King County Metro
LNTP	Limited Notice to Proceed

ACRONYMS, continued

LRRP	Light Rail Review Panel
LRT	Light Rail Transit
LRV	Light Rail Vehicle
LTK	LTK Engineering Services
MACC	Maximum Allowable Construction Cost
MDA	Major Discharge Authorization
MLK	Martin Luther King, Jr. Way
MOA	Memorandum of Agreement
MOS	Minimum Operable Segment
MOU	Memorandum of Understanding
MPPCV	Major Public Project Construction Variance
MRB	Material Review Board
MTP	Montlake Triangle Project
MUP	Master Use Permit
NB	Northbound
NCR	Notification of Change Report
NCTP	North Corridor Transit Partners
NEPA	National Environmental Policy Act
NOAA	National Oceanic and Atmospheric Administration
NTP	Notice to Proceed
OCS	Overhead Catenary System
OMF	Operations and Maintenance Facility
OMSF	Operations and Maintenance Satellite Facility
PE	Preliminary Engineering
PEP	Project Execution Plan
PEPD	Planning, Environment and Project Development
PMOC	Project Management Oversight Consultant
PSST	Pine Street Stub Tunnel
QA	Quality Assurance
QC	Quality Control
QTR	Quarter
RE	Resident Engineer
RFC	Request for Change
RFD	Request for Deviation
RFI	Request for Information
RFP	Request for Proposal
RFQ	Request for Qualifications
RIR	Recordable Injury Rates

ACRONYMS, continued

RMP	Risk Management Plan
ROD	Record of Decision
ROW	Right of Way
SB	Southbound
SCADA	Supervisory Central and Data Acquisition
SCC	Standard Cost Categories
SCL	Seattle City Light
SDEIS	Supplemental Draft Environmental Impact Statement
SEPA	State Environmental Policy Act
SIP	Street Improvement Permitting
SPI	Schedule Performance Index
SR	State Route
ST	Sound Transit
START	Seattle Tunnel and Rail Team
SWI	Stacy & Witbeck, Inc.
TBM	Tunnel Boring Machine
TCE	Temporary Construction Easement
TE	Traction Electrification
TFK	Traylor Frontier Kemper Joint Venture
TOD	Transit Oriented Development
TVM	Ticket Vending Machine
UAC	Unallocated Contingency
U-Link	University Link project
USFWS	U.S. Fish and Wildlife Service
UW	University Of Washington
UST	Underground Storage Tank
UWS	University of Washington Station
VE	Value Engineering
VECP	Value Engineering Cost Proposal
WBS	Work Breakdown Structure
WSDOT	Washington Department of Transportation